

BT501-HEALTH BIOTECHNOLOGY

1-What is biotechnology all about?

Biotechnology

- bio - life Technology - the application, or harnessing of science for a specific purpose
- technology, process or practice - modifies or harnesses any living organism or system to be useful to any human purpose

Classical Biotech vs Modern

- Classical Biotech –since ages ,Growing better crops, Breeding animals
- Cheese and yogurt production – use of microorganisms
- Mendel’s work – Laying foundation of modern Biotech
- Major discoveries : DNA structure, Ligase, Restriction enzyme
- Birth of rDNA technology
- Cut Gene from one organism and paste into others

Applications of Biotech

- AgriBiotech –: Pest-resistant, Herbicide resistant, Abiotic stress tolerant crops
- Environmental Biotech: Bioremediation, Bacteria and Fishes as Biosensors

- Health Biotech: biotherapeutics (insulin, growth hormones) Diagnostics tools (PCR, FISH, microarray)

The field of biotechnology has been in use for ages in various forms, which include the growing of better crops (agricultural biotechnology) and animal breeding (animal biotechnology). Similarly, the use of biotechnology has been around for thousands of years, especially the application of microorganisms in the production of cheese and yogurt (food biotechnology). In addition, the tools of biotechnology have been implied in animal husbandry, to develop pest-resistant crops, bioremediation (environmental biotechnology), as well as in bioethanol production. But the most promising application of biotechnology is found to be in the medical field by generations of biotherapeutics (insulin, growth hormones) and diagnostics tools (PCR, FISH, micro-array technique). Before we discuss various applications of medical biotechnology, let us briefly go through the historical aspects of biotechnology and this information will make you to understand the field better.

It all began with the discovery made by Sir Alexander Fleming in the year 1918 where he observed that the mold *Penicillium* inhibited the growth of human skin disease-causing bacteria called *Staphylococcus aureus*. The discovery by Sir Alexander Fleming lead to the making of antibiotics that we use today. These antibiotics are highly recommended and extensively used medicinally for bacterial infections. These antibiotics are basically substances produced by microorganisms that normally inhibit the growth of other microorganisms. Later on, antibiotics became widely available as a drug for treating microbial infections in human beings, especially with the development of penicillin as the most used antibiotic. Currently, a variety of

microorganisms have been used to generate thousand liters of antibiotic drugs by using advanced biotechnology tools.

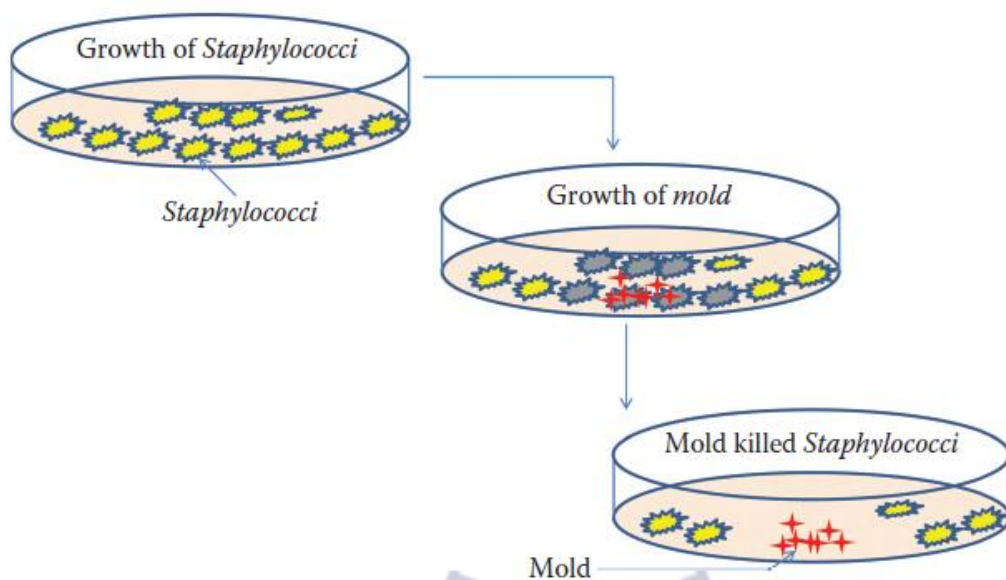
The field of biotechnology has taken a leap with the discovery of the double-helix structure of the deoxyribose nucleic acid (DNA) molecule and the credit goes to one research publication titled “This structure has novel features which are of considerable biological interest” authored by James Watson and Francis Crick in 1953, which claimed to discover the structure of the human DNA helix, the molecule that carries genetic information from one generation to the other. Nine years later, in the year 1962, they shared the Nobel Prize with Wilkins for cracking one of the most important of all biological puzzles. This discovery has led to the birth of genetic mapping, manipulation, and genetic engineering-type fields.

Surprisingly, with the help of genetic engineering, the gene of interest can be cut and inserted into the genome of other living organisms (microorganisms and viruses) and this process of gene insertion and manipulation is called recombinant DNA technology. Over the past few years recombinant DNA technology has been extensively employed to generate therapeutic products (insulin and growth hormones) for treating human diseases. With a rapid increase in the number of patients worldwide, there has been a tremendous scope to identify and create medicine for various human diseases. Thankfully with the availability of biotechnology tools, now it is possible to develop therapy for various diseases.

2-Medical products developed by using biotechnology tools:

Antibiotics

Penicillin is one of the earliest discovered antibiotics which is basically derived from molds such as *Penicillium*. Penicillin was discovered by Scottish scientist and Nobel laureate Sir Alexander Fleming in 1928. It all started with a basic experiment where Sir Fleming noticed a Petri dish containing *Staphylococcus* plate culture which he had mistakenly left open was contaminated by blue-green mold, which had formed a visible growth. But surprisingly he also found that there was a circle of inhibited bacterial growth around the mold. Later on, Sir Fleming hypothesized that the mold was releasing a substance that was preventing the growth and could contain a substance with antibiotic properties. In order to prove his hypothesis, he grew a pure culture and discovered the first antibiotic substance from the *Penicillium* mold, known as *Penicillium notatum*. Sir Alexander showed that *P. notatum* when grown in the appropriate substrate caused the release of chemical substances and these chemical substances were named as antibiotics. He later on named that chemical substance as penicillin. Soon after this discovery, penicillin was considered to be the most effective drug against bacteria (Gram-positive bacteria), and not at all effective against Gram-negative bacteria. The discovery of penicillin marks the beginning of the antibiotic production and so far more than 200 different types of antibiotics have been produced.



The pioneering experiment conducted by Dr. Alexander Fleming resulted in the discovery of the penicillin drug, an antibiotic drug.

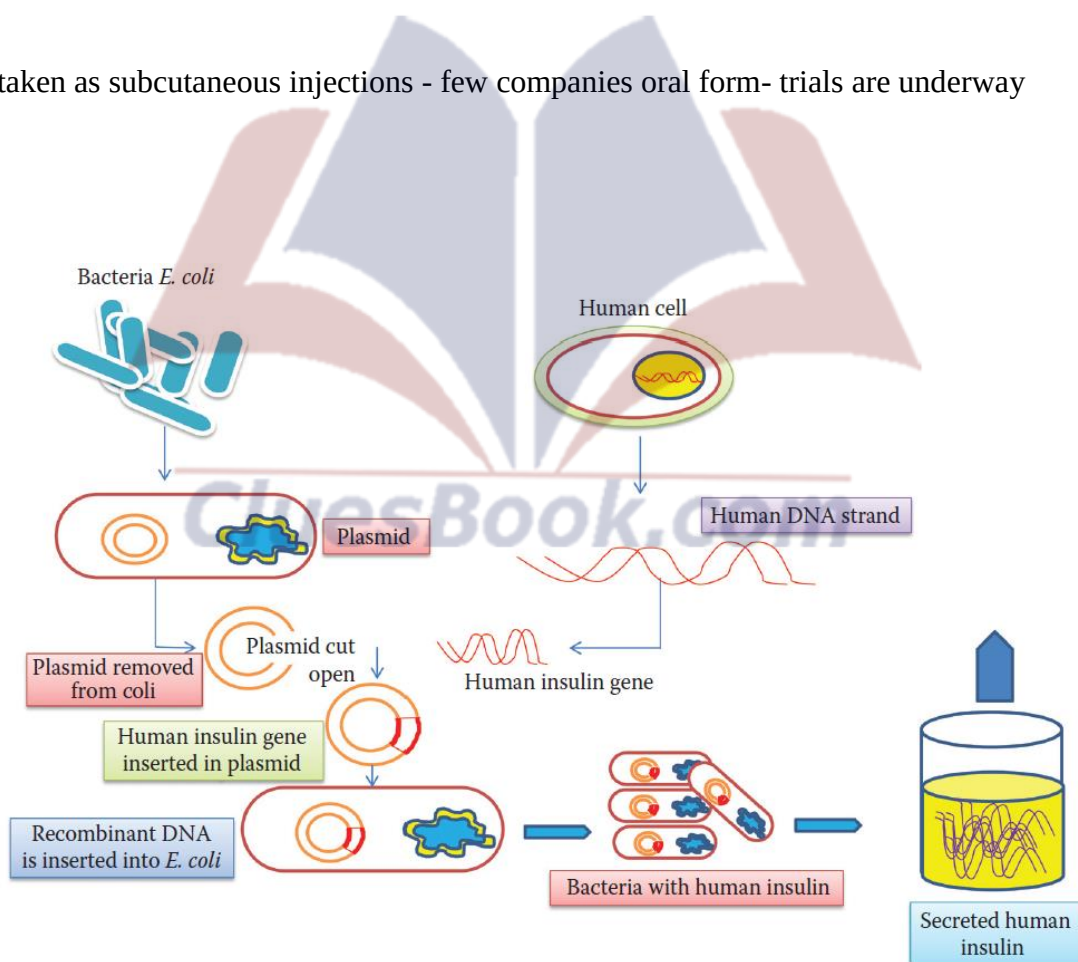
3-Recombinant insulin

Another discovery which was made in the medical field was the recombinant DNA technology, which has completely revolutionized disease treatments especially in the Type 1 diabetes mellitus, where insulin-producing cells become dysfunctional and do not produce sufficient amount of insulin to regulate blood-sugar level. In a healthy human being, insulin regulates glucose metabolism in the body. In diabetic conditions, the level of insulin decreases which causes the elevation of the blood sugar level and this clinical condition is known as diabetes mellitus. One of the best treatments of diabetes mellitus is insulin injections, where insulin is injected into the patient's body and the insulin regulates the blood-sugar level. The insulin is either synthesized chemically or produced by recombinant DNA technology. These synthesized

insulins are used medically to treat patients with Type 1 diabetes mellitus, whereas patients with Type 2 diabetes mellitus are insulin resistant.

Insulin Production -Advances

- Recently, human insulin gene into plants - safflower plant as bioreactors
- insulin analogues- chemically synthesized
- major hurdles - delivery of insulin - cannot be taken orally- improper absorption - biological activity lost.
- taken as subcutaneous injections - few companies oral form- trials are underway



Making of human insulin by recombinant DNA technology.

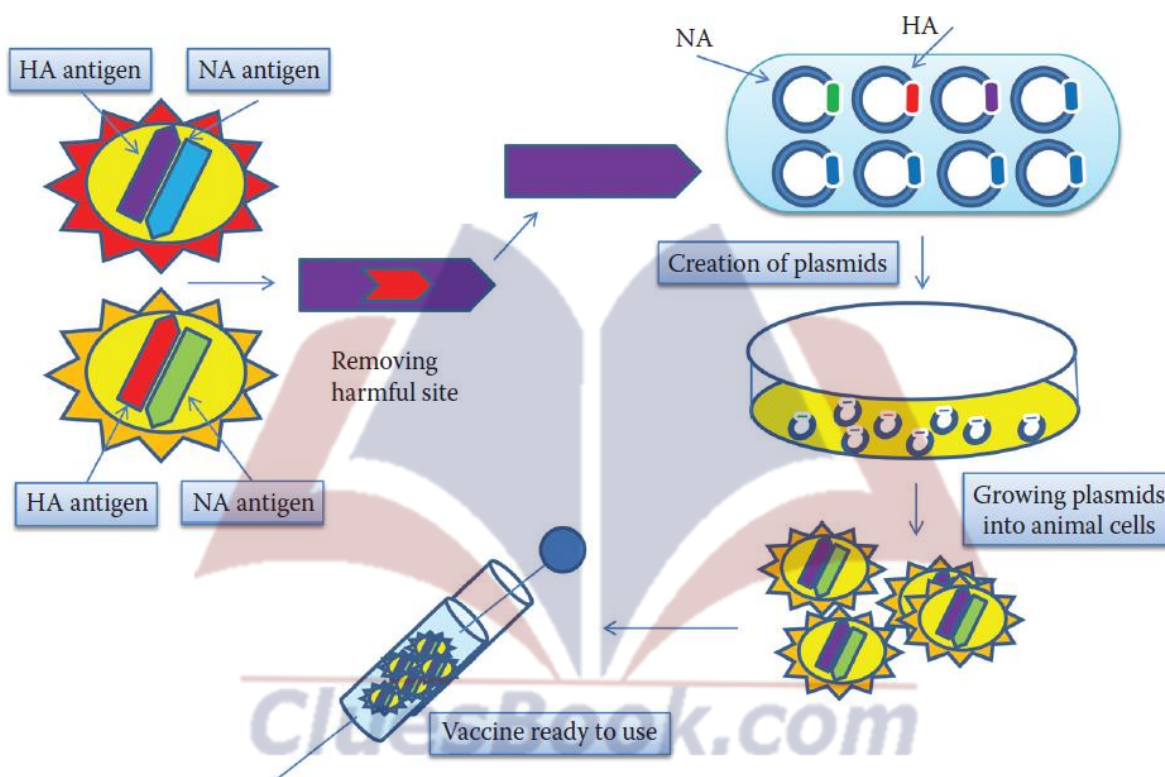
4-Vaccines

The vaccine is basically a biological preparation that improves the human immune defense system to fight diseases. The vaccines normally contain an agent that diligently resembles a disease-causing bacteria, and is frequently produced from debilitating or killed forms of the microorganisms. Upon injection into the human body, the vaccine can stimulate the body's immune system to recognize the agent as a foreign body, and recognize it, so that the immune system can recall and destroy or kill any of these microorganisms that later infect the human body. In a simpler way, the vaccine basically trains the human body to defend or fight or kill the microorganisms. The vaccines are classified based on its application; the vaccines that are used to prevent the effects of an upcoming infection by any natural or wild pathogen are called prophylactic, and the vaccines which are used against cancer are called therapeutic vaccines.

The story of vaccine development dates back to the seventeenth century when Edward Jenner found a milkmaid infected with smallpox, and a few days later he took pus from the hand of the milkmaid with cowpox and inoculated an eight-year-old boy with it. After six weeks he found that the boy did not contract smallpox. After a few years, Sir Louis Pasteur adapted Jenner's idea of developing a rabies vaccine and, subsequently, vaccine development progress started and became a matter of national concern. After that obligatory vaccination laws were passed in various countries around the globe. During the twentieth century, we have seen an introduction of several successful vaccines against diseases such as diphtheria, measles, mumps, and rubella. Interestingly, the major achievement was done in the 1950s when the polio vaccine was made and clinically used; also vaccines were synthesized to eradicate smallpox.

Successful vaccines

- Vaccines against diseases such as diphtheria, Anthrax, measles, mumps, and rubella, polio, smallpox, Hepatitis A, Hepatitis B, Flu etc



5-Monoclonal antibodies

Another milestone achieved in the beginning of the twentieth century was the making of monoclonal antibodies, which was proposed by Paul Ehrlich. He suggested that drug compounds can be accurately, precisely, and specifically delivered along with monoclonal antibodies. Monoclonal antibodies are mono-specific antibodies as they are made by identical cells that are all clones of a distinctive parent cell. Currently, it is possible to produce monoclonal antibodies

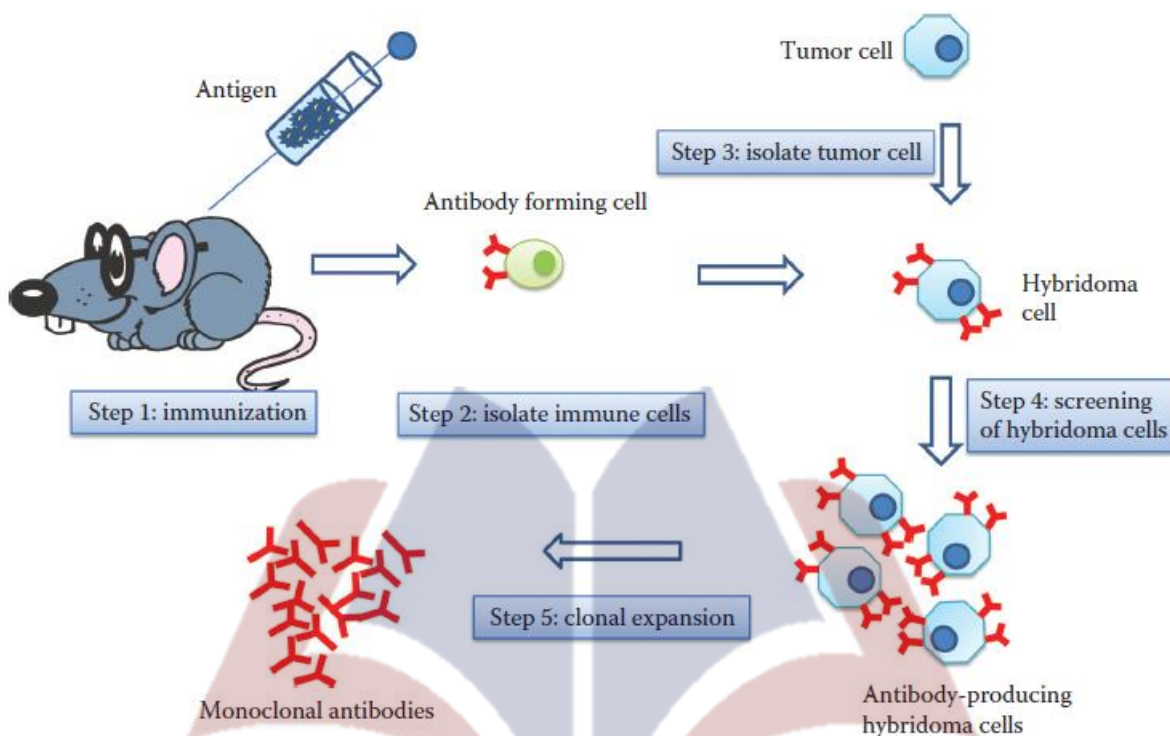
that specifically bind to that antigen and have a variety of applications, especially to detect or purify that substance (antigen), and has become an important tool in diagnostics and biomedical research.

In diagnostics and biomedical research, monoclonal antibodies are very useful tools. First, they are extremely specific; that is, each antibody binds to the specific site of the antigen. Second, some antibodies, once activated by the occurrence of a disease, continue to confer resistance against that disease; classic examples are the antibodies to the childhood diseases chickenpox and measles. Another important application of monoclonal antibodies is to develop vaccines against various diseases. As you are aware, a vaccine is made from bacteria or viruses either killed or inactivated. Upon introduction into the human body this vaccine stimulates the production of antibodies against the antigens to fight back the diseases. The production of monoclonal antibodies involves human and mouse hybrid cells and this technology is known as hybridoma technology. During monoclonal antibody production, tumor cells are merged with mammalian cells that produce an antibody against a particular antigen. The result of these merged tumors with mammalian cells is called hybridoma, which can frequently produce antibodies. These antibodies are called monoclonal antibodies because they come from only one type of cell; whereas, the antibodies that are produced by conventional methods contain many kinds of cells and are called polyclonal antibodies.

Applications

- Diagnostic applications: e.g Quantification of blood hormones etc
- Therapeutic Applications: blocking the function of targeted molecule Protein Purification, Killing cancer cells

- Targeted Drug delivery
- Protein Purification



6-Bioengineered tissues

Over the last decade, tissue engineering became the most fascinating medical field, especially in body-parts reconstruction or cosmetic surgery areas. The basic concept about tissue engineering is to make human tissues through in vitro methods under controlled laboratory conditions and the making of such tissues is called bioengineered tissues. The definition of tissue engineering covers a broad range of applications in the healthcare field, however, in practice the term tissue engineering closely relates to repair or replacing portions of, or whole, human body tissues which include neurons, cardiomyocytes, bone, and cartilage. The bioengineered tissues are constructed by integrating certain mechanical and structural properties for proper functioning in

the human body, so one can say that bioengineering is basically the use of a combination of cells, engineering, and biocompatible materials, and is finally suitable to improve or replace biological functions. The bioengineered field was once categorized as a sub-field of biomaterials, but having grown in scope and application it is considered as a field in its own right. Tissue engineering is an emerging multidisciplinary field involving biology, medicine, and engineering that is likely to revolutionize the ways we improve the health of millions of people worldwide by repairing, maintaining, or enhancing tissue and organ function. The tissue engineering can also have diagnostic applications where the human cells or tissues are used to test drug metabolism and drug uptake, toxicity, and pathogenicity.

In vitro cell culture techniques have successfully cultured and constructed the human cells outside of the body. These outside body constructed cells are often referred to as bioengineered cells. There are various applications of bioengineered cells in clinical conditions.

The term regenerative medicine is frequently used synonymously with cell and tissue engineering, though in regenerative medicine, cell-based therapy is considered to be a main branch of regenerative medicine.

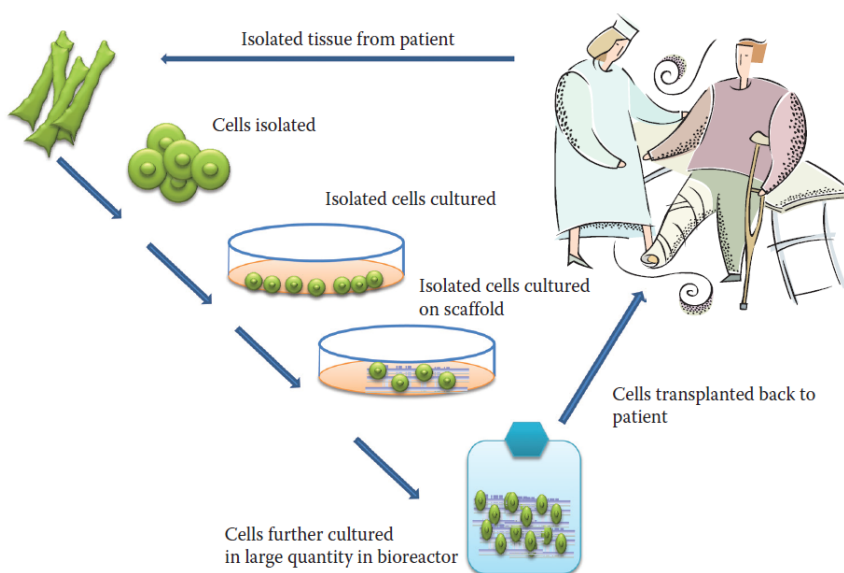
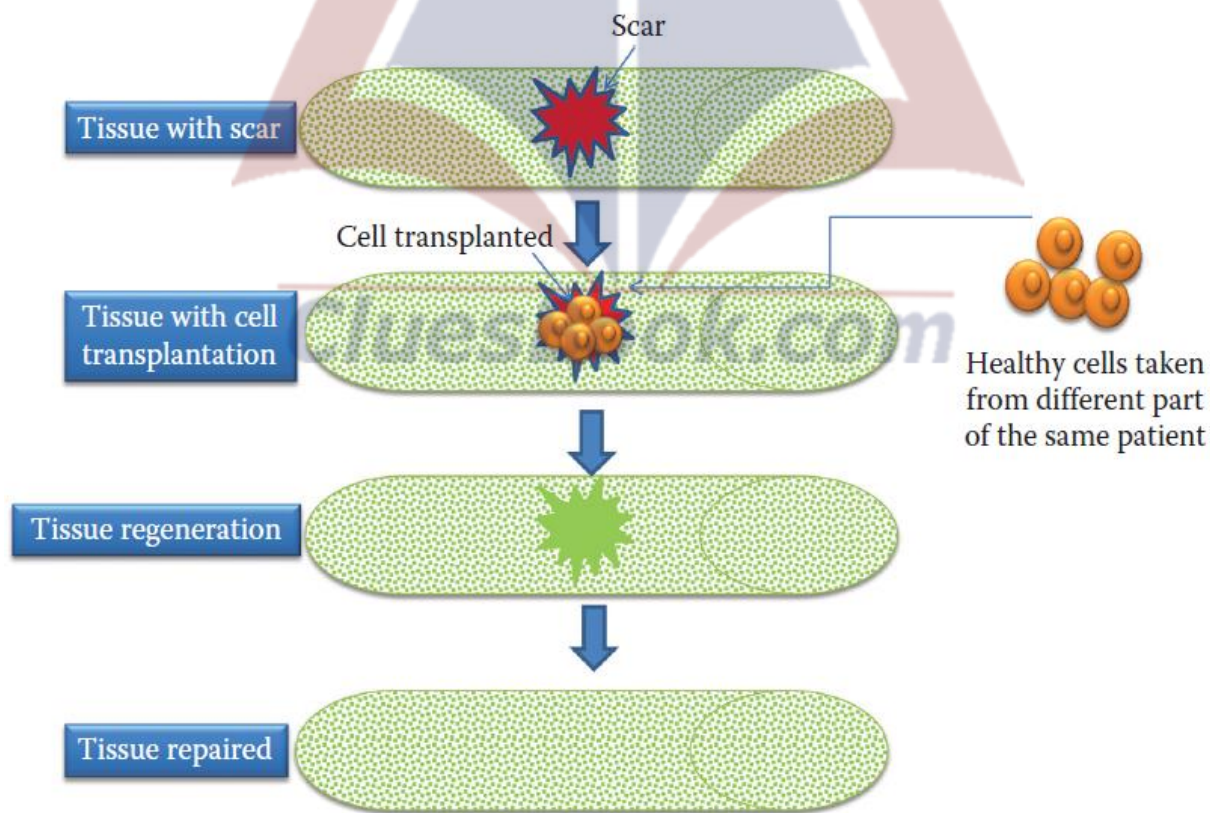


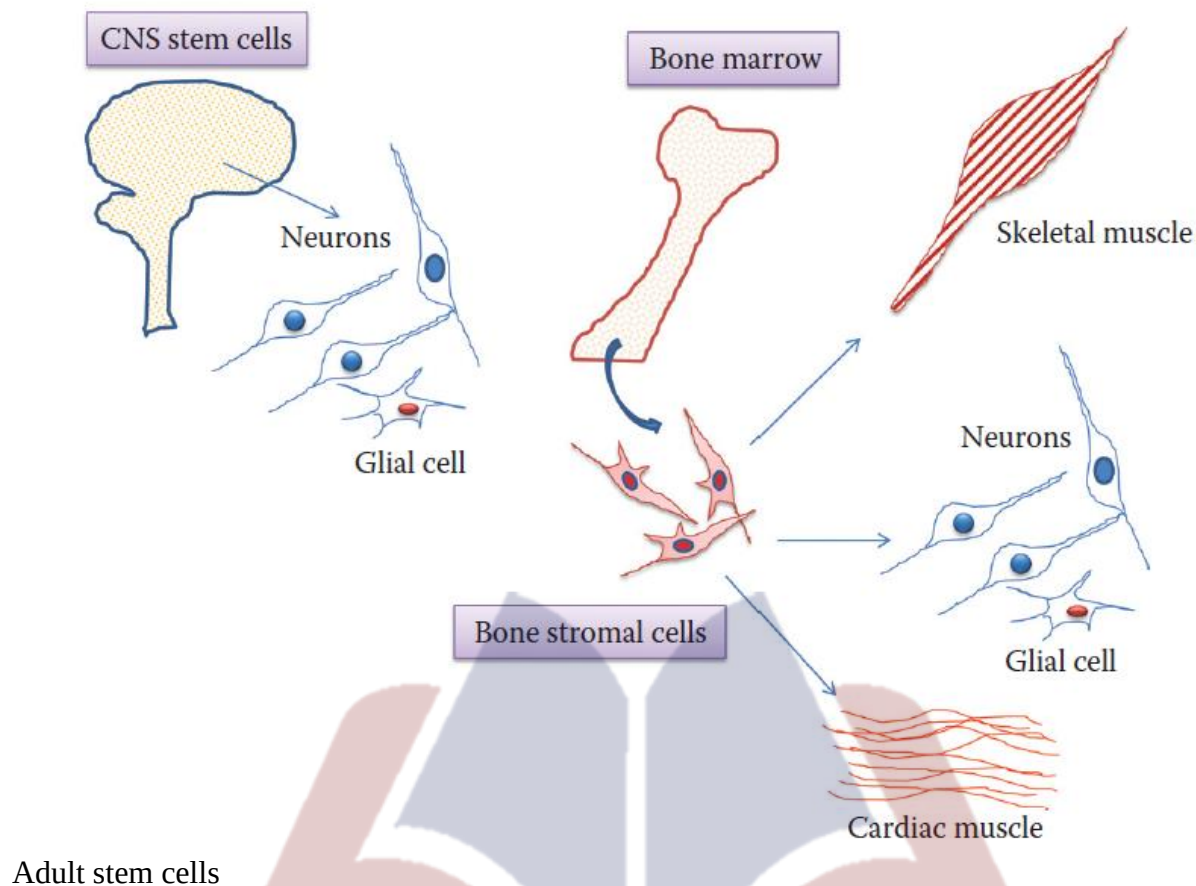
Figure. Cell and tissue engineering



7-Adult stem cell therapy

Over the last few years, there has been tremendous interest in adult stem cells for the autologous cell transplantation as these specialized cells have the potential ability to repair or restore the dysfunctional cells or tissues in the human body afflicted with various diseases that include blood cancer and neurological disorders. The stem cells are generally found in all multicellular organisms and they are characterized by the ability to renew themselves through cell division and differentiate into a varied range of specialized cell types. Moreover, mammalian stem cells may be broadly classified into two major types: adult stem cells and embryonic stem cells (ESCs) that are isolated from the inner cell mass of blastocysts.

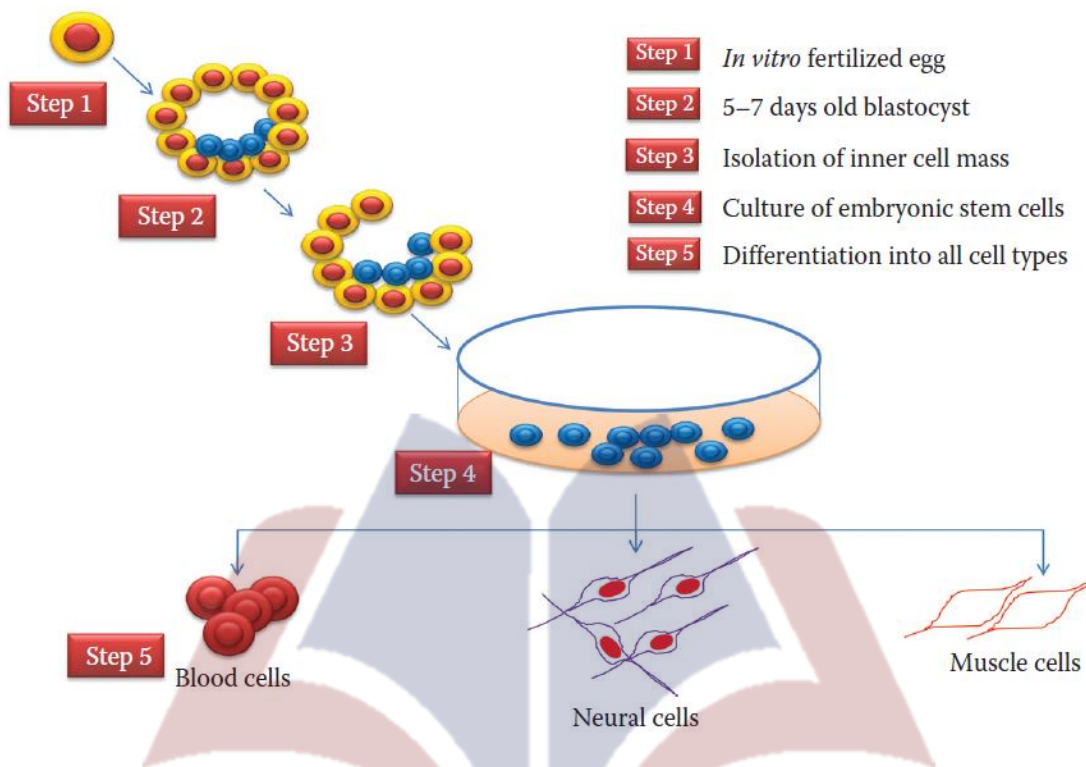
One of the main roles of adult stem cells is that they remain in an undifferentiated state in the human body and multiply by cell division to replenish dying cells and restore damaged tissues and organs. The adult stem cells are also known as somatic stem cells; these stem cells can be found in both juvenile and adult ages. Moreover, adult stem cells are specialized cells that have the capability to divide and generate all cell types of the organ from which they originate, and can also possibly regenerate the entire organ. Unlike ESCs, the use of adult stem cells is not contentious as they are derived from adult source or tissue rather than by killing human embryos. Moreover, adult stem cells have been in use for many years predominantly in the treatment of cancer (such as leukemia and related bone and blood cancers) employing bone marrow transplants. Interestingly, the majority of the government funding especially in the United States is confined to adult stem cell-based research. Among adult stem cells, hematopoietic stem cells (HSCs) and mesenchymal stem cells (MSCs) are found to be the most successful stem cells because they can be clinically used in patients. These stem cells can be directly injected or placed at the site of repair or injected through vascular delivery.



8-Emerging trends: Embryonic stem cells

Over the last few years ESCs have been a highly discussed and debated scientific topic at all levels which include public life, scientific forums, ethical, legal, and political platforms. Before we discuss the ethical or legal issues associated with ESCs, we will first learn what ESCs are all about. The ESCs are basically pluripotent stem cells isolated from the inner cell mass of the blastocyst, which is an early-stage embryo. Moreover, human embryos reach the blastocyst stage 4 to 5 days post fertilization, at which time they consist of about 50–150 cells. In addition to this, ESCs are adept at propagating themselves for an indefinite period under controlled culture situations. The culture and propagation of ESCs through in vitro methods is considered to be the

best technology to culture human cells in large quantities and to be engaged as valuable tools for both research and cell-based therapy.



Embryonic stem cells

One of the best applications of ESCs is to provide a large supply of required cells to be used as cell-based therapy and tissue replacement in patients who are suffering from degenerative diseases. Moreover, a number of diseases that could possibly be treated by ESCs are cancers, diabetes, Parkinson's disease, Alzheimer's disease, stroke, blindness, and spinal cord injuries. In contrast to its various therapeutic applications, the issues related to ESCs need to be resolved before making them useful for human use. These issues are the problem of immune rejection associated with allogeneic stem cell transplantation. These problems can be solved by using adult stem cells in autologous transplantation. Besides, cell-based therapy, the ESCs can be used as diagnostic tools to study an early human development and genetic disease, as well as in vitro

drug testing. Other issues related to ESCs transplantation is that there is a possibility that transplanted stem cells could form tumors and have the possibility of becoming cancerous if cell division continues uncontrollably. Contrary to this, supporters of ESCs research argue that such research should be continued because the resultant research findings will have significant medical potential. The recent development of induced pluripotent stems cells (iPSc) has created tremendous interest among scientists, as adult stem cells can be converted into pluripotent stem cells and there is no need to kill the human embryos.

In spite of all controversies over ethical use of human ESCs, on January 23, 2009, Geron Corporation, USA, has received an approval from US Food and Drug Administration (FDA) to test ESCs in humans. The Phase 1 clinical trial for transplantation of a human-ES-derived cell population into spinal cord-injured individuals became the first human ES cell human trial. The Phase 1 clinical study was conducted on eight to ten paraplegic patients who have had their injuries no longer than two weeks before the trial began, and stem cells were injected before the formation of scar in the tissue. Nevertheless, the researchers are stressing that the injections are not expected to fully cure the patients and restore all mobility. The success of such clinical trials goes back to the experiments conducted by University of California, USA, and which was supported by Geron Corporation of Menlo Park (Biotechnology Company), California. The results of an animal study showed that there was functional improvement in locomotive recovery in spinal cord-injured rats after stem cell transplantation. Interestingly, the first trial is mainly for testing the safety of cell-based transplantation. However, the trial had been put on hold in August 2009 due to apprehensions made by the FDA regarding a small number of microscopic tumors found in several treated rat models, but the hold has been lifted since.

9-Human genome project

The Human Genome Project (HGP) was an international collaborative research project with a primary aim to decode the human DNA genome. The main objective of the HGP was to map about 20–25 k genes of the human genome from both a physical and functional perspective. The initial task of the project was to identify the full set of genetic instructions enclosed inside human DNA. The project began with the end of several years of work supported by the United States Department of Energy (DoE). It has been reported that the \$3-billion project was founded in 1990 by the DoE and the US National Institutes of Health. In addition to the United States, other countries also took part in the HGP, including the United Kingdom, France, Germany, Japan, China, and India. The team of scientists worked hard over the years and drafted a proposal for the genome project and the final sequencing of the human genome was done in 2006. Although the objective of the HGP is to understand the genetic organization of the human genome, the project has worked with other non-mammalian organisms.

Significance of the genome project

The reason for starting the HGP was to understand the role of genes in the development of diseases. There are reports that suggest more than 3000 genetic disorders known to be caused by genetic mutations. With the current treatment modalities, these genetic disorders cannot be cured without knowing the real cause. And to know the cause we have to know the role of genes in the diseases. Recent efforts have been made to find out the causes of cancer with genetic tools, but not much success was achieved. It has been recommended that information gained from the HGP would help to understand the genetic cause of the devastating illnesses, which include Parkinson's disease, schizophrenia, or Alzheimer's disease.

The genetic mutations play a role in many genetic diseases that include heart disease, diabetes, immune system disorders, and congenital defects. These diseases are thought to be the consequence from complex collaborations between genes and environmental factors. When genes for diseases have been identified, researchers can study how environmental factors, food, drugs, or pollutants can influence those genes. The location of the gene is important in identifying the type of protein which is produced by a particular gene. It has been reported that understanding the mechanism of a genetic disease is an important step in curing genetic diseases. It has been suggested that one day it may be possible to treat genetic diseases by gene therapy. Besides therapeutic applications, the information gained from the HGP can help to know the reason for the pluripotency nature of ESCs and how these ESCs can be differentiated into many different specialized cells such as muscle cell, neural cells, or hepatocytes.

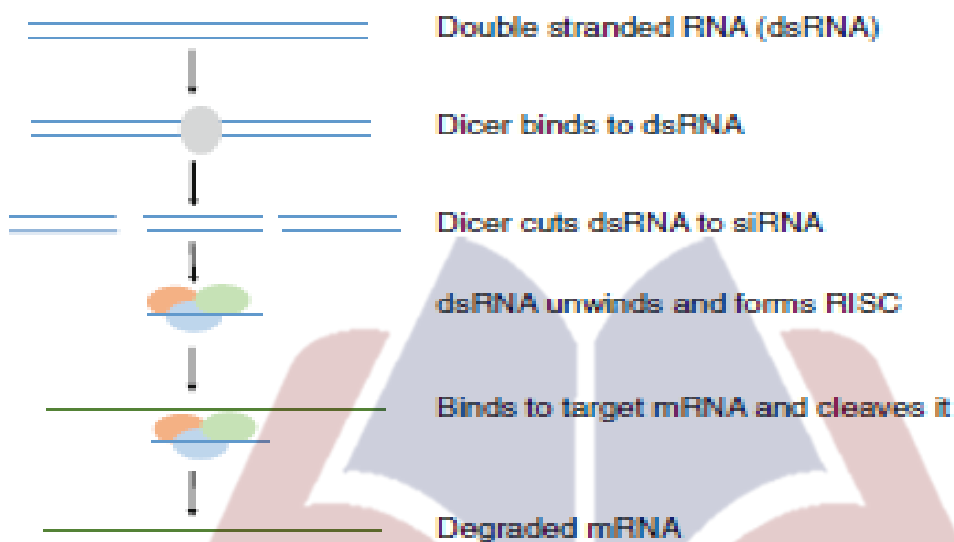
10-RNA interference technology

The RNA interference (RNAi) is a natural process that cells dictate to turn down or go to a silent state; such activity cell is regulated by a specific gene known as RNAi. The RNAi technology provides excellent tools to study the role specific gene in the development, causes, and progression of particular diseases, and has taken the biomedical community by storm. It has been reported that RNAi was first noticed in the plant *Petunia* when plant biologists tried to develop the purple color of the flower by introducing a pigment-producing gene. Surprisingly, the gene which was introduced to intensify color in the plants was found to suppress the color. In the end the resulting flowers developed white patches or became completely white. Surprisingly, a few years later another group of researchers observed a similar kind of gene-silencing effect in *Caenorhabditis elegans*. Later on, it has been reported that the gene-silencing effect is caused by

the presence of double-stranded RNA, and this double-stranded RNA is normally not found in healthy cells.

RNAi technology has been in use in various fields of biotechnology, particularly in the food plant engineering that produces lower levels of natural toxins. No plant that uses RNAi technology has yet passed the experimental stage; however, research work has been known to effectively reduce the allergen levels in tomato plants and also is known to cut the cancer-causing agents in tobacco plants. Also, other plant characters that have also been bioengineered include the production of nonnarcotic natural products by using the opium plant, development of resistance to common plant viruses, and protection of plants with dietary antioxidants. Interestingly, bioengineered plants such as *FlavrSavr* tomatoes and two cultivars of *ring-spot*-resistant papaya were originally developed by employing antisense technology. One of the pronounced applications of RNAi technology is its application in medicine. Over the past few years, efforts have been made to understand RNAi's role in normal and diseased cells, and also to use the RNAi technology for use in medical therapies. It has been suggested that human disease progression can be blocked by using RNAi-based therapies to turn down the activity of genes. Cancer, for example, is frequently caused by over-excited genes in the cells, and retarding their activities could stop the disease progression. Over the last few years, several pharmaceutical companies are using RNAi-based therapies to treat for various forms of cancer. In addition, viral infections can also be treated using RNAi-based therapies and this can be done by reducing the activity of key viral genes. It has been shown in the laboratory that human cells have successfully stopped the growth of HIV, polio, hepatitis C, and other viruses in human cell culture and RNAi-based therapies against HIV are under clinical trial stages. Moreover, the importance of RNAi technology has some beneficial effects in finding out the cause of the

disease. Using RNAi technology, the activity of a particular gene can be knocked down which will help us to understand its role in the disease development and progression. For many years, researchers have been studying how proteins regulate gene activity. Now with the help of RNAi technology, it would be possible to discover the role of proteins in gene regulation.

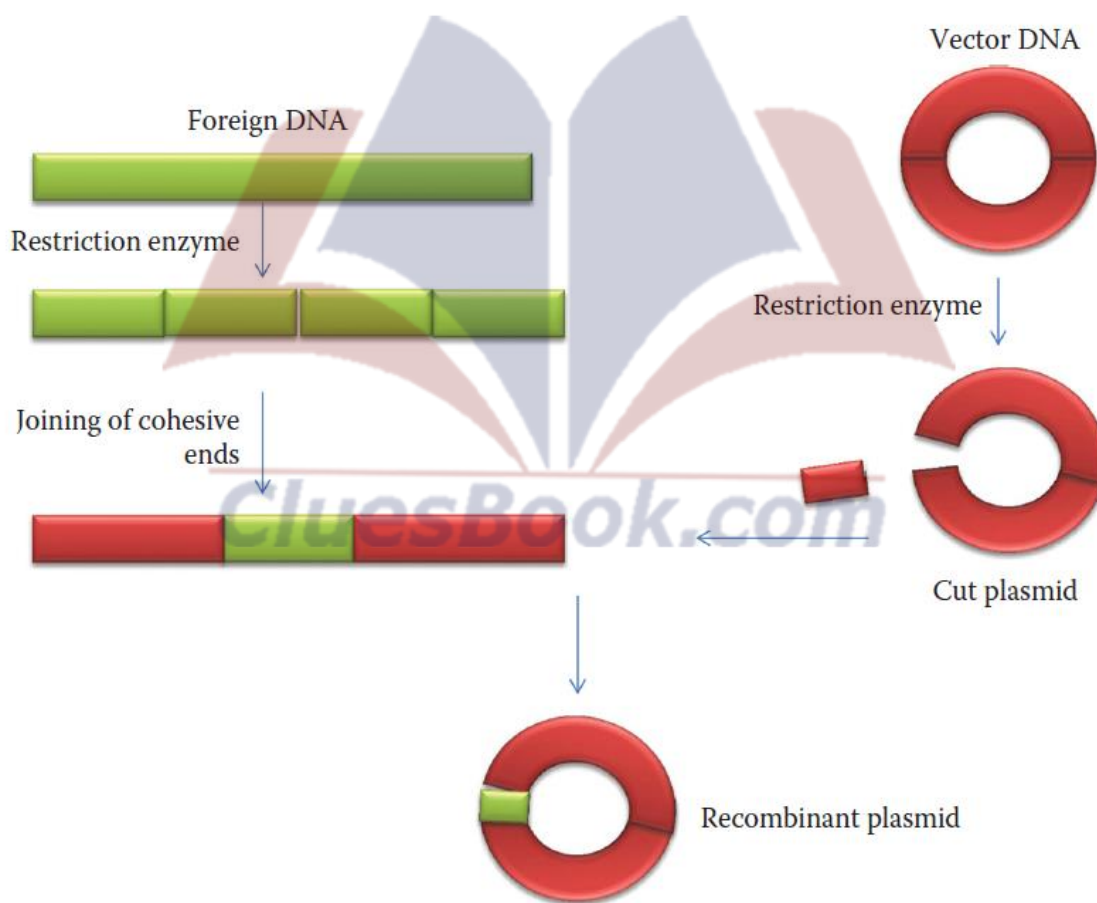


RNAi method for gene silencing. Dicer binds to dsRNA and cleaves it to small fragments. The fragments unwind and form RISC complex. The RISC complex binds to target mRNA and cleaves its sequence specifically.

11-Recombinant DNA technology

The process of r-DNA technology begins with the isolation of a gene of interest, which is then inserted into a vector; these vectors are further cloned into multiple copies. A vector is basically a piece of DNA that is capable of independent growth; bacterial plasmids and viral phages are the commonly used vectors. When the gene of interest (foreign DNA) is integrated into the plasmid or phage, this process is generally referred to as r-DNA. The next step in r-DNA technology is to introduce the vector containing the foreign DNA into host cells so that the cells

can express the desirable proteins. In order to get sufficient amounts of protein, the vectors must be cloned to produce large quantities of the DNA. Once the vector is isolated in large quantities, it can be introduced into the desired host cells, which include mammalian, yeast, or special bacterial cells. Finally, the host cells can synthesize the foreign protein from the r-DNA. When the cells are grown in vast quantities in the bioreactor or fermenter, the recombinant protein can be isolated in large amounts and the entire process is commonly referred as r-DNA technology. There are three different methods available through which r-DNA products are being developed: (1) transformation, (2) phage introduction, and (3) nonbacterial transformation.



Steps of constructing a recombinant DNA

The making of r-DNA has been briefly described below in a stepwise manner.

- Isolation of DNA
- Digestion of DNA
- Ligation of Two DNA Fragments
- Transferring Recombinant Vector into Host Cells (Transformation)
- Selection of the Transformed Cells

Goals of recombinant DNA technology

- To isolate and characterize a gene
- To make desired alterations in one or more isolated genes
- To return altered genes to living cells
- Artificially synthesize new gene
- Alternating the genome of an organism
- Understanding the hereditary diseases and their cure
- Improving human genome

Techniques used in rDNA technology

- Gel electrophoresis
- Cloning libraries
- Restriction enzyme mapping
- PCR
- Nucleic Acid Hybridization
- DNA Microarrays

Applications of rDNA technology

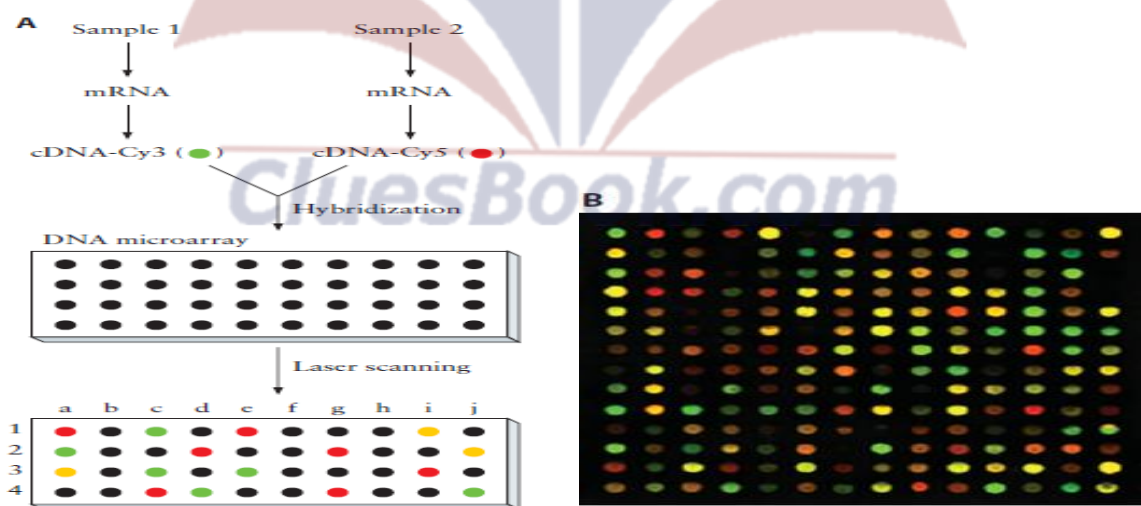
- Agriculture: growing crops of your choice (GM food), pesticide resistant crops, fruits with attractive colors, all being grown in artificial conditions

- Pharmacology: artificial insulin production, drug delivery to target sites
- Medicine: gene therapy, antiviral therapy, vaccination, synthesizing clotting factors
- Other uses: fluorescent fishes, glowing plants etc

12-Biochips

With the advent of information technology, it is possible to store biological information in a chip format which is called as a biochip. A microarray, also known as a DNA chip or biochip, is basically a collection of microscopic DNA spots attached to a solid surface (see picture). DNA microarray enables investigators to analyze expression of several genes in a single reaction. Typically it uses a solid surface such as a glass slide or a silicon film known as chip.

Gene expression profiling with a DNA microarray



(A) mRNA is extracted from two samples (sample 1 and sample 2), and during reverse transcription, the first cDNA strands are labeled with the fluorescent dyes Cy3 and Cy5,

respectively. The cDNA samples are mixed and hybridized to an ordered array of either gene sequences or gene-specific oligonucleotides. After the hybridization reaction, each probe cell is scanned for both fluorescent dyes and the separate emissions are recorded. Probe cells that produce only a green or red emission represent genes that are transcribed only in sample 1 or 2, respectively; yellow emissions indicate genes that are active in both samples; and the absence of emissions (black) represents genes that are not transcribed in either sample. **(B)** Fluorescence image of a DNA microarray hybridized with Cy3- and Cy5-labeled cDNA

Applications

DNA microarray has increasing number of applications:

1-Expression analysis : In this method, the DNA of an organism is spotted on a chip and hybridized to cDNA obtained from mRNA of the same organism. The intensity signifies the abundance of a particular mRNA in the sample, signifying the expression level of the gene under a given condition. This is the most commonly used technique.

2-Mutation analysis : The genomic DNA is spotted on a chip and hybridized to DNA isolated from different individuals. Single nucleotide polymorphism can be identified by this method. It is important to analyze variation in a particular gene.

3- Comparative genomic hybridization: Increase or decrease of chromosomal fragments associated with a disease state. Prenatal chromosomal aberrations can be studied.

13-Gene therapy

To treat genetic disorders, the default or dysfunctional gene can either be replaced with a new gene or repaired with gene therapy. In brief, the gene therapy involves correcting defects in genes and in this process a normal/healthy gene is introduced to replace a defective gene. A person carrying the defect in a specific gene might develop disease, which may be inheritable. In gene therapy, nucleic acid is the therapeutic agent.

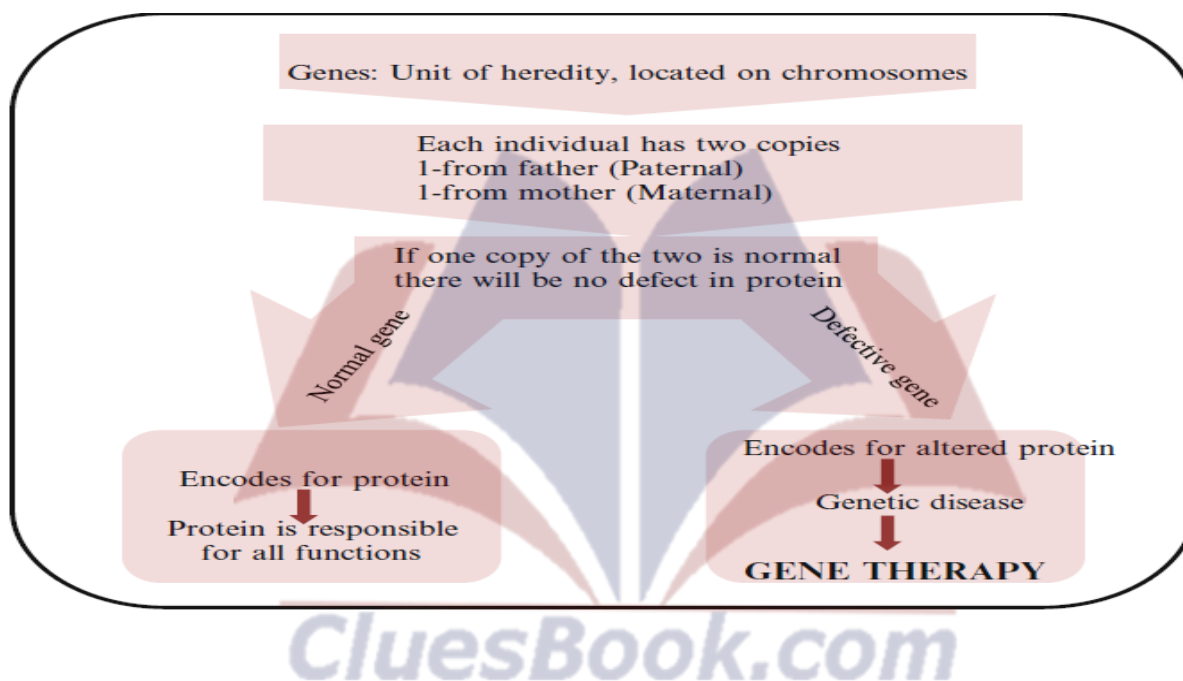
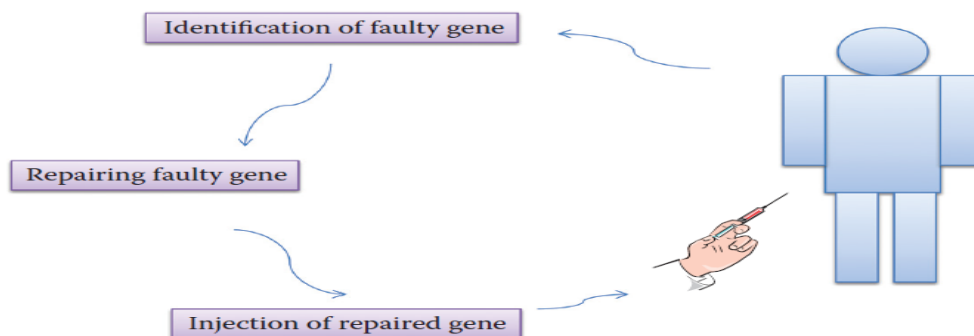


Fig. The function of normal gene and effect of defective gene, which causes disease. The individuals with defective gene are the candidate for gene therapy.

Gene therapy is still under development stages and various clinical trials are underway around the world. Its success will be based on the successful cure of genetic diseases by gene therapy.

The therapy is achieved by using a viral vector or gene delivery vehicles. The steps involved in the therapy in which the gene of interest enters the target cell via viral vector and begins expression are referred to as transduction.



The gene therapy can be classified into two major types: somatic gene therapy and germline gene therapy.

Somatic gene therapy

By using somatic gene therapy, somatic cells can be treated by inserting a vector loaded with the correct gene into a person's body. The somatic cells are cells that form in the body and cannot produce progenies. Gene therapy, in its present stage only treats somatic cells in humans.

There are two types of somatic gene therapy, *ex vivo* and *in vivo*.

Ex vivo somatic gene therapy: In *ex vivo* gene therapy genes or cells are modified outside the body and then transplanted back into the body. **In vivo gene therapy,** cells are modified or treated within the patient's body.

Germline Therapy

In the germline therapy, the modification is done in germ cells that are sperms or eggs, producing a permanent and genetically transferable modification. The functional gene is introduced into either of them, which then integrate in the selected germ cell, and subsequently the therapeutic gene is present in each cell of the body. The fertilized egg may be used to insert the therapeutic

gene and reimplanted into the mother. If successful, the functional gene is present and expressed in all cells of the resulting individual. The technique of making permanent changes might prove effective in the genetic diseases, but due to ethical and technical reasons, their use has been prohibited in humans.

Importance of vectors in gene therapy

Vectors should be undetected by the body's immune system to avoid immune rejection. Most viruses attack their hosts to insert their genetic material into the DNA of the host and this DNA contains instructions to produce viruses in large numbers. Considering these capabilities, scientist thought of using these viruses as a vehicle to deliver genetic materials to the host cells in order to treat genetic diseases. Currently the most common vector is used as a vehicle virus that has been genetically altered to carry normal human DNA. We have described various types of viral vectors which are being used in gene therapy and these viruses are found to be different in their mechanisms of action.

To make gene therapy effective, there are two classes of viruses—retroviruses and adenoviruses—found to be critical in gene delivery.

Recommendation to resolve the Legal and ethical issues

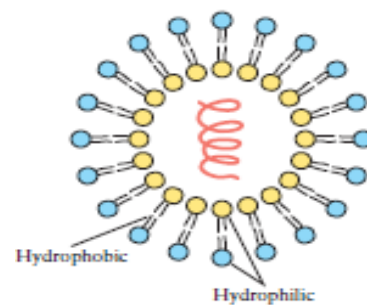
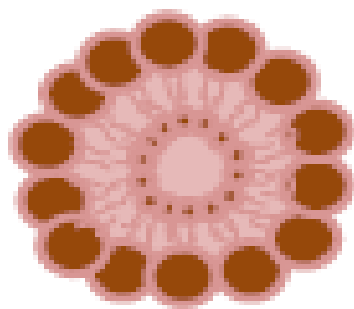
Nevertheless, there are some ethical concerns which have been raised in different parts of the world, and some ethical organizations have released recommendations to resolve the gene therapy issues and these recommendations include (a) establishing a national ethics body in each country to look at somatic gene therapy; (b) supporting somatic gene therapy research that follows the recommendations; (c) asking researchers, organizations, and governments to listen and respond to public concerns about gene therapy research; and (d) asking that research follows quality and safety controls.

14-Liposome-Based Drug Delivery

Liposomes are artificially prepared vesicles made of lipid bilayer. Liposomes can be filled with drugs, and used to deliver drugs for cancer and other diseases.

Liposomes have liquid core surrounded by phospholipid bilayer. Liposomes can be formed by sphingomyelins or lecithins and synthetic lipids as dimyristoyl, distearoyl, dipalmitoyl, and dioleoyl. They are 10 nm to several-micrometer spherical nanostructured molecules with defined shape and size. Liposomes are formed by an external phase having double phospholipid membranes and an aqueous internal phase with the capability to encapsulate both hydrophobic and hydrophilic compounds. Their advantages are biocompatibility, low toxicity, non-immunogenicity, and improved drug efficacy. They fuse to cell membrane and deliver the drug. They are used as carriers of antifungal, antibacterial, and cancer drugs:

- Liposome is used to deliver anticancer drug doxorubicin. Doxorubicin can induce myocardial toxicity. However when delivered in a liposomal preparation, the off-target toxicity is significantly reduced.
- Liposomes consisting of phosphatidylcholine, phosphatidylethanolamine, oleic acid, and cholesteryl hemisuccinate were developed by encapsulating a purified recombinant T4 endonuclease V . When delivered to the skin, they tend to improve DNA repair.



Structure of liposomes. They have liquid core surrounded by phospholipid layer. They are biocompatible, nontoxic, and non-immunogenic and used for packaging of drugs such as amphotericin B, ampicillin, polymyxin B, ciprofloxacin, anticancerous (daunorubicin and doxorubicin), insulin, and DNA. Used for drug delivery in fungal and bacterial infections, cancer cells, and diabetes.

15-Bionanotechnology

Bionanotechnology is a term that refers to the intersection of nanotechnology and biology. This discipline helps to indicate the merger of biological research in various fields of nanotechnology. Nanomaterial is the material used to make or construct biomaterials with nanomorphological features and these biomaterials can be used for various applications. These nanomaterials are basically smaller than 1/10th of a micrometer in at least one dimension; however, this term is occasionally used for materials smaller than 1 μm .

Application of nanomaterials in medicine

Therapeutic delivery: It has been reported that drug needs to be transported well in the body to easily reach the target site and there are times some drugs are not able to reach to target sites and get lost in the process. Recent efforts have been made to improve drug bioavailability, and nanomaterials play an important role in it. The word “bioavailability” denotes the presence of drug molecules where they are needed in the body and perform normal body function.

Nanoparticles can also be used as contrastagents, to take ultrasound and magnetic resonance imaging (MRI) pictures. Additionally, polymer-based nanoparticles can be designed to improve the pharmacological and therapeutical properties of drugs.

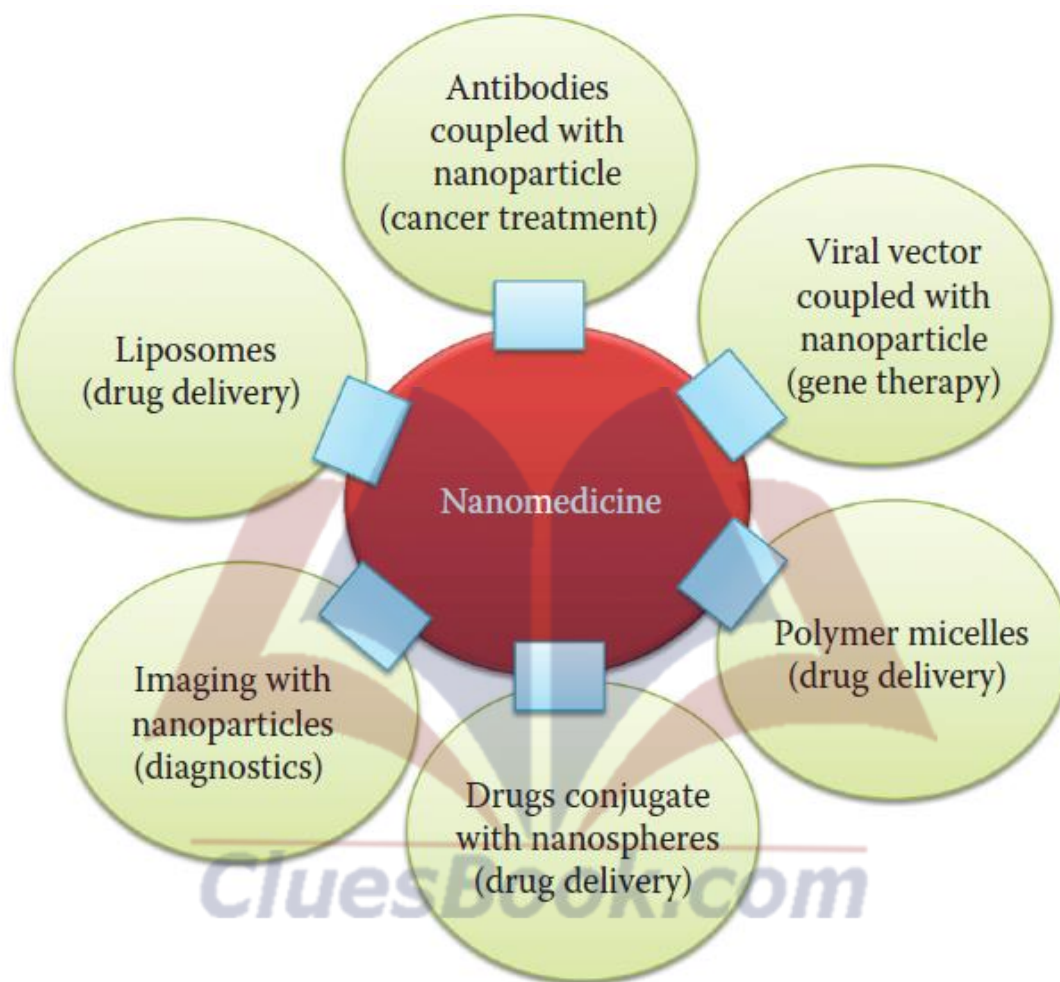


Figure- Applications of nanotechnology in medicine

The concepts that are enhanced through nanobiology include: nanodevices, nanoparticles, and nanoscale that occurs within the discipline of nanotechnology. Moreover, this technical method in biology allows scientists to imagine and create systems that can be used for biological research. Biologically inspired nanotechnology uses biological systems as the inspiration for technologies not yet created. The most important objectives that are frequently found in

nanobiology involve applying nanotools to relevant medical or biological problems and refining these applications. Moreover, developing new tools for the medical and biological fields is another primary objective in nanotechnology.

Despite the fact that biological systems are inherently nano in scale, nanoscience must merge with biology in order to deliver biomacromolecules and molecular machines that are similar to the human body or organ. In the twenty-first century, scientists have developed the technology to artificially tap into nanobiology. This process is best described as organic merging with synthetic. Interestingly, colonies of live neurons can live together on a biochip device. The self-assembling nanotubes have the ability to be used as a structural system as they would be composed together with rhodopsins, which would help the optical computing process and also help with the storage of biological materials. The most fascinating aspects would be of DNA as the software for all living things, which can be used as a structural proteomics system—a logical component of molecular computing.

16-Chromosomal Abnormalities

All nucleated cells contain chromosomes that consist of DNA and proteins (histones) in a compact structure. Chromosomes carry all of our genes and therefore all of our genetic information. DNA with its associated packaging proteins is referred to as **chromatin**. Some regions of chromosomes are tightly packed and are called heterochromatin, while other regions are less condensed and are called euchromatin. Less condensed packing of chromatin generally increases the transcription of genes in the region. Each species has a characteristic number and form of chromosomes, referred to as the **karyotype**. A **karyogram** is a photographic representation of stained chromosomes arranged in order of size, i.e., decreasing length (Fig. 1).



Fig. 1 Normal human male 46,XY karyogram. Humans have a total of 46 chromosomes that consist of two identical sets of 22 chromosomes (autosomal chromosomes) and 2 sex chromosomes (XX for female and XY for male). Because of the two identical sets of chromosomes, human cells are called diploid.

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Each chromosome is paired with its matched or **homologous chromosome**. The matched chromosomes are identical in size and structure but may carry different versions (known as alleles) of the same gene. Humans have 46 chromosomes, or 23 chromosome pairs, to carry our approximately 25,000 genes. Cells in our body that contain 46 chromosomes are **diploid** ($n = 2$); 23 chromosomes are derived from the mother's egg cell, and the other 23 are from the father's sperm. Egg cells and spermatozoa each contain only 23 chromosomes (Fig. 2) and therefore are **haploid** ($n = 1$). In diploid cells, the 46 chromosomes appear as 22 homologous pairs of

autosomes (nonsex chromosomes) and one pair of sex chromosomes, XX in females and XY in males.

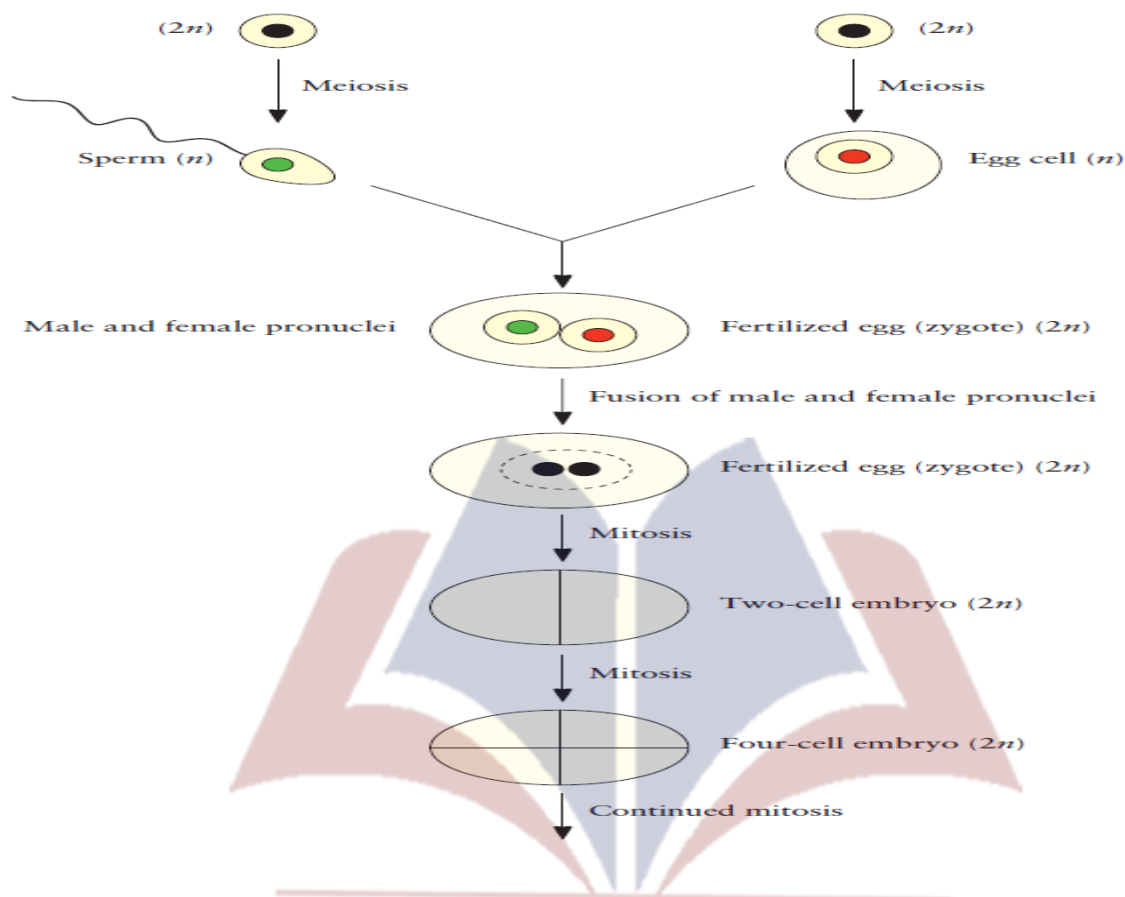


Fig.2 Generation of a diploid ($2n$) zygote. A diploid $2n$ fertilized egg (zygote) is produced by the fusion of a haploid (n) sperm and egg. Successive mitotic divisions generate many types of diploid cells in the body (somatic cells) during development and cell turnover (replacement of old cells with new ones) in an embryo.

Human chromosomes are usually studied in rapidly dividing cells, such as peripheral blood lymphocytes. Cell mitosis can be arrested in the metaphase stage of the cell cycle, and the chromosomes can be differentially stained to allow their identification microscopically. This microscopic analysis of chromosomes is known as **cytogenetics**. For routine karyotyping,

Giemsa staining is preferred, as this procedure produces alternating light and dark bands (**G banding**) that reflect differential chromosomal structures characteristic of each chromosomal pair (Fig. 1). These light and dark bands of chromosomes result from the specificity of binding of the Giemsa stain for the phosphate groups of DNA, as the stain attaches to regions of DNA where there are large amounts of A-T bonding. Thus, Giemsa staining can identify different types of changes in chromosomal structure as gene rearrangements.

Examination of a karyotype enables one to determine either if there is gain or loss of a chromosome(s) or if the structure of a given chromosome(s) is altered. The **centromere** of each chromosome separates the short arm (p) from the long arm (q). Most arms are divided into two or more regions by distinct bands, and each region is further subdivided into subbands (Fig. 3). For example, band Xp21.2 is found on the p arm of the X chromosome in region 2, band 1, subband 2. **Chromosome disorders** are caused by abnormalities in the number (increase or decrease of genes) or the structure of chromosomes. An individual's physical characteristics are called a **phenotype**, which is the combination of all of that individual's expressed traits, including morphology, development, behavior, and biochemical and physiological properties.

Phenotypes result from the expression of genes and environmental factors that interact with these genes. Thus, the phenotype of a person with a chromosomal disorder may vary with the type of chromosomal defect.

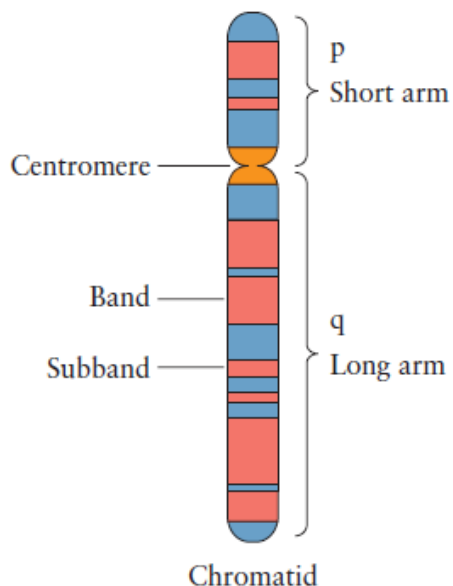


Fig.3 Schematic diagram of the structures of a chromosome. The p and q arms, centromere, chromatid, bands, and subbands (described in the text) are shown. A chromatid is an individual chromosome that is paired with a replicated copy of the identical chromosome. This pair of chromosomes is held together at the centromere for the process of cell division. The light and dark bands originate from the differential staining of the regions of the chromosome with the Giemsa stain, as explained in the text. The short arm of each chromosome is denoted with a “p” and the long arm with a “q.” Thus, 7q refers to the long arm of chromosome 7. Each arm may be further divided into regions, depending on the size.

Risk factors for chromosomal abnormalities:

- Radiation
- Smoking
- Fertility drugs
- Alcohol consumption
- Oral contraceptive

17-Chromosome Structural Abnormalities

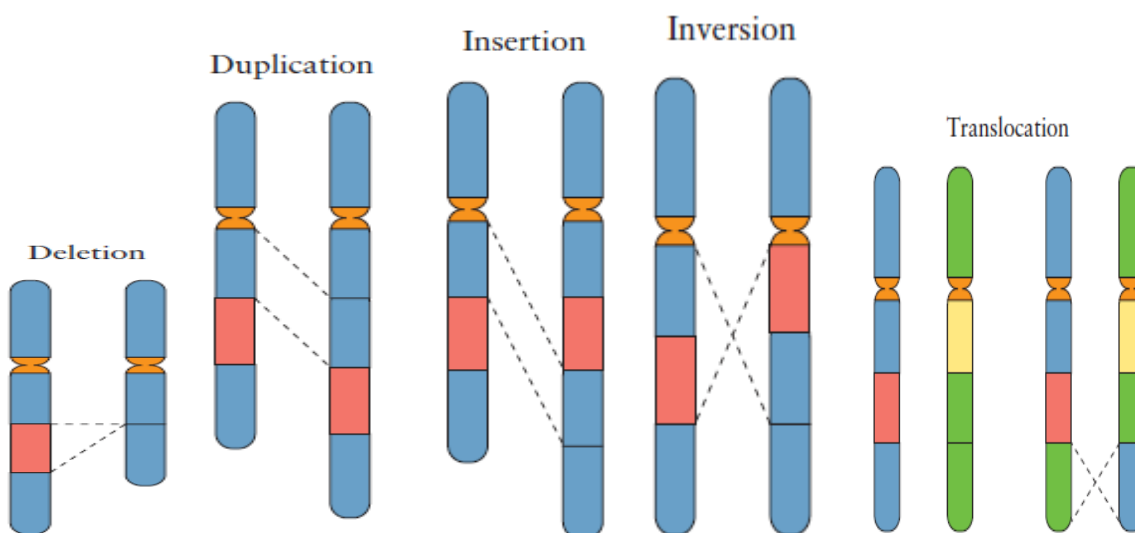
Genomic rearrangements may alter genome architecture and yield clinical consequences. Several genomic disorders caused by structural variation of chromosomes were discovered by cytogenetics. Recent advances in molecular cytogenetic techniques have enabled the rapid and precise detection of structural rearrangements on a whole-genome scale. This high resolution illustrates the role of **structural variants (SVs)** and **single-nucleotide polymorphisms (SNPs)** in normal genetic variation.

In analyzing the role of structural gene variants in cell function, it is important to consider the two types of such variants, i.e., **gain-of-function variants** and **loss-of-function variants**. A gain-of-function variant results from a mutation that confers new or enhanced activity on a protein. Most mutations of this type are not heritable (germ line) but, rather, are somatic mutations. A change in the structure of a gene that may arise during DNA replication and is not inherited from a parent, and also is not passed to offspring, is called a **somatic mutation**. Such a mutation that results in a single base substitution in DNA is known as a somatic point mutation.

A loss-of-function variant results from a point mutation that leads to reduced or abolished protein function. Most loss-of-function mutations are recessive, indicating that clinical signs are typically observed only when both chromosomal copies of a gene (one being inherited from each parent) carry such a mutation.

In a population, any two unrelated individuals are identical at about 99.5% of their DNA sequence. At a given chromosomal site, one individual may have the A nucleotide and the other individual may have the G nucleotide. This type of site in DNA is known as an SNP.

Each of the two different DNA sequences at this site (or gene) is called an **allele**. Molecular techniques such as array-based **comparative genomic hybridization (CGH)**, SNP arrays, array painting, and **next-generation sequencing** have facilitated and expedited the characterization of chromosome rearrangements in human genomes. These various genomic rearrangements can arise by several mechanisms, including deletions, amplifications, translocations, and inversions of DNA fragment. **Deletions** occur when a portion of a chromosome is missing or removed. **Duplications** result from the copying of a portion of a chromosome that results in extra genetic material. During a **translocation**, a portion of one chromosome is transferred to another chromosome. In a reciprocal translocation, DNA segments from two different chromosomes are exchanged. A DNA **inversion** results when a portion of a chromosome is broken off, turned upside down, and reattached. When a portion of a chromosome is disrupted, the chromosome may form a circle, or ring, without any loss of DNA.



18-Chromosome Numerical Abnormalities

An abnormality in which the chromosome number is an exact multiple of the haploid number ($n = 23$) and is larger than the diploid number ($n = 46$) is called **polyploidy**. Polyploidy arises from fertilization of an egg by two sperm (total number of chromosomes increases to 69) or the failure in one of the divisions of either the egg or the sperm so that a diploid gamete is produced. The survival of a fetus to full term of pregnancy is rare in the instance of polyploidy.

Aneuploidy occurs when the chromosome number is not an exact multiple of the haploid number and results from the failure of paired chromosomes (at first meiosis) or sister chromatids (at second meiosis) to separate at anaphase. Thus, two cells are produced, one with a missing copy of a chromosome and the other with an extra copy of that chromosome (Fig. 1). Examples of numerical chromosomal abnormalities are listed in Table 1.

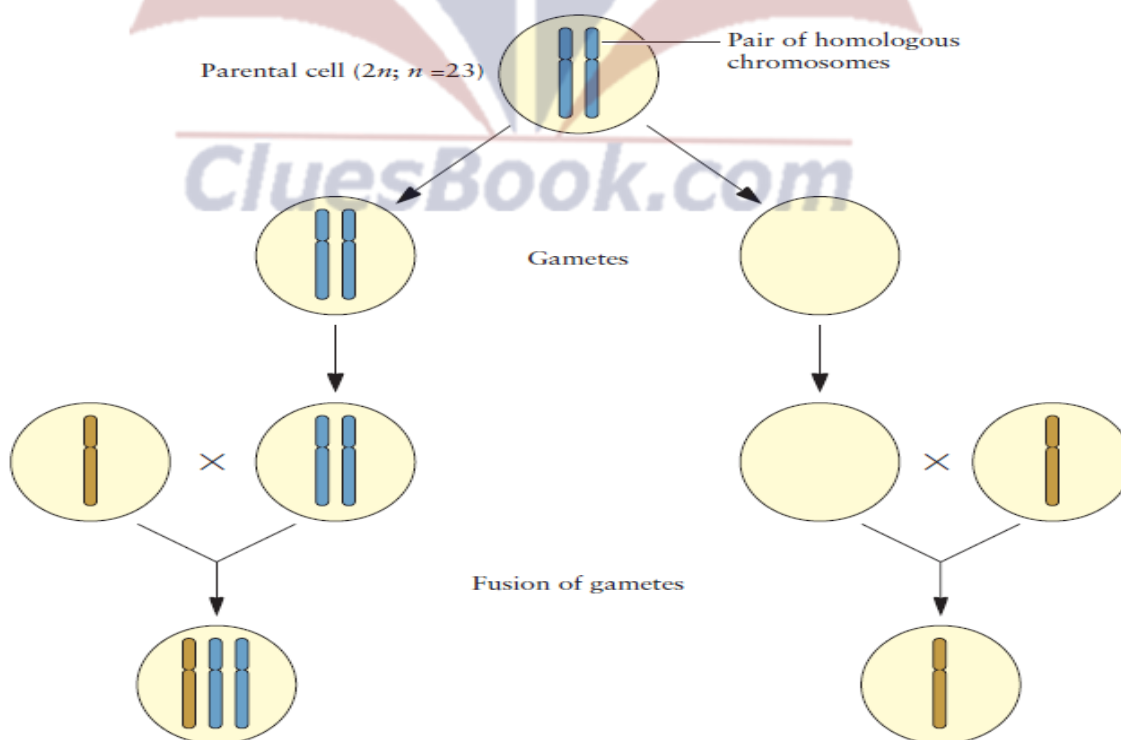


Fig.1 Nondisjunction in gamete cell formation and fusion of abnormal gametes with a normal haploid gamete. Aneuploidy occurs when paired chromosomes (at first meiosis) or sister chromatids (at second meiosis) do not separate from each other at anaphase, a stage of meiosis at which sister chromosomes move to opposite sides of the cell. This failure of paired chromosomes to separate, with the chromosomes instead moving to the same side of the cell, is termed nondisjunction. As a result, two cells are produced, one with a missing copy of a chromosome and one with an extra copy of that chromosome, as shown.

Table-1 Numerical chromosomal aberration syndromes

Aneuploidy condition	No. of chromosomes	Karyotype
Tetraploidy	92	XXYY
Triploidy	69	XXY
Trisomy 21 (Down syndrome)	47	XX+21
Trisomy 18 (Edward syndrome)	47	XY+18
Trisomy 13 (Patau syndrome)	47	XX+13
Klinefelter syndrome	47	XXY
Trisomy X	47	XXX
Turner syndrome	45	X

Trisomy 21, the first human chromosomal disorder discovered (in 1959), is an abnormality that displays an extra copy (total of 3 copies) of chromosome 21 (Fig. 2) and causes **Down syndrome**. The genes on all three copies of chromosome 21 are normal. However, not all individuals with Down syndrome show the same physical characteristics, indicating that their phenotypes can vary. People with Down syndrome have a typical facial appearance (the face is flat and broad) that includes an abnormally small chin, skin folds on the inner corners of the eyes, poor muscle tone, a flat nasal bridge, a protruding tongue due to a small oral cavity, an enlarged

tongue near the tonsils, a short neck, and white spots on the iris. Growth parameters such as height, weight, and head circumference are smaller in children with Down syndrome than in typical individuals of the same age.

All Down syndrome patients have some degree of mental retardation, albeit moderate. Despite this condition, many persons with Down syndrome can be educated and live with minimal daily assistance, while others require much attention and care. There are several possible health concerns, including cardiac failure and hearing loss. Individuals with Down syndrome are now living longer than they used to and can survive into their 50s and 60s. However, those individuals who survive to that age are at very high risk of developing Alzheimer disease.

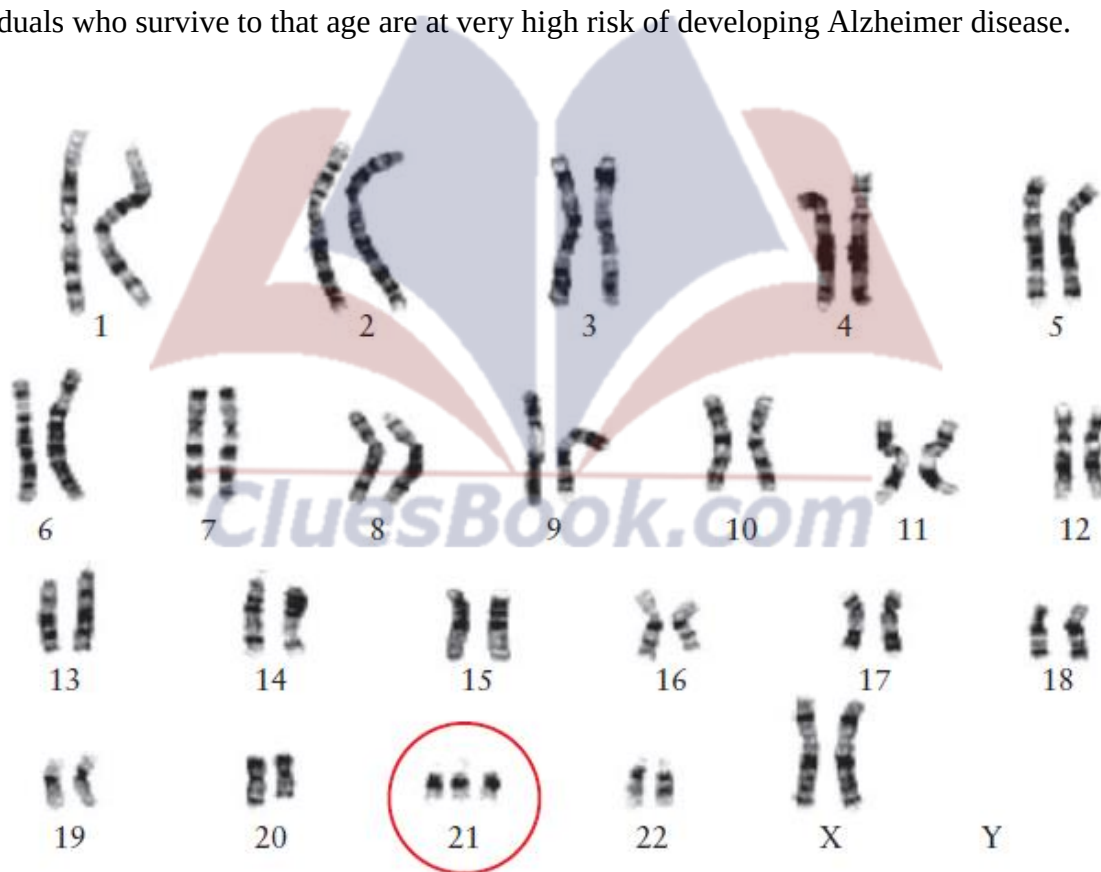


Fig-2 Trisomy 21. This syndrome appears when an individual inherits three copies of chromosome 21. The extra copy of chromosome 21 results in Down syndrome.

Trisomy 13, the presence of three copies of chromosome 13, causes **Patau syndrome**. Only about 5% of infants with this disorder survive past their first year, and most pregnancies involving trisomy 13 end in miscarriage. Children with trisomy 13 usually have a lot of trouble breathing, especially when they sleep, and many have seizures. All individuals with Patau syndrome have severe mental retardation, and other common characteristics include a small head, extra fingers and/or toes, and a cleft lip or cleft palate.

Trisomy 18, the presence of 3 copies of chromosome 18, elicits **Edward syndrome**. Only about 10% of babies with this syndrome survive past their first year, and the majority of survivors are female, indicating a prenatal selection against males with trisomy 18 after the time of amniocentesis. Children with trisomy 18 usually have problems with breathing and eating, and many have seizures or serious heart conditions. All individuals with trisomy 18 have severe mental retardation. Most babies with trisomy 18 are very small and have certain recognizable facial features. They also tend to overlap their fingers in a very distinct pattern.

Prader–Willi syndrome results from the absence or nonexpression of a group of genes on chromosome 15. A specific form of blood cancer, **chronic myeloid leukemia**, may be caused by a chromosomal translocation, in which portions of two chromosomes (chromosomes 9 and 22) are exchanged. No chromosomal material is gained or lost, but a new, abnormal gene that leads to the development of cancer is formed.

19-Sex Chromosome Abnormalities

The phenotype of chromosomal disorders can vary depending on whether the chromosomal abnormality occurs on the maternally or paternally derived chromosomes (Table 1).

Turner syndrome (45,X) is an example of monosomy, in which a girl is born with only one sex chromosome, an X chromosome.

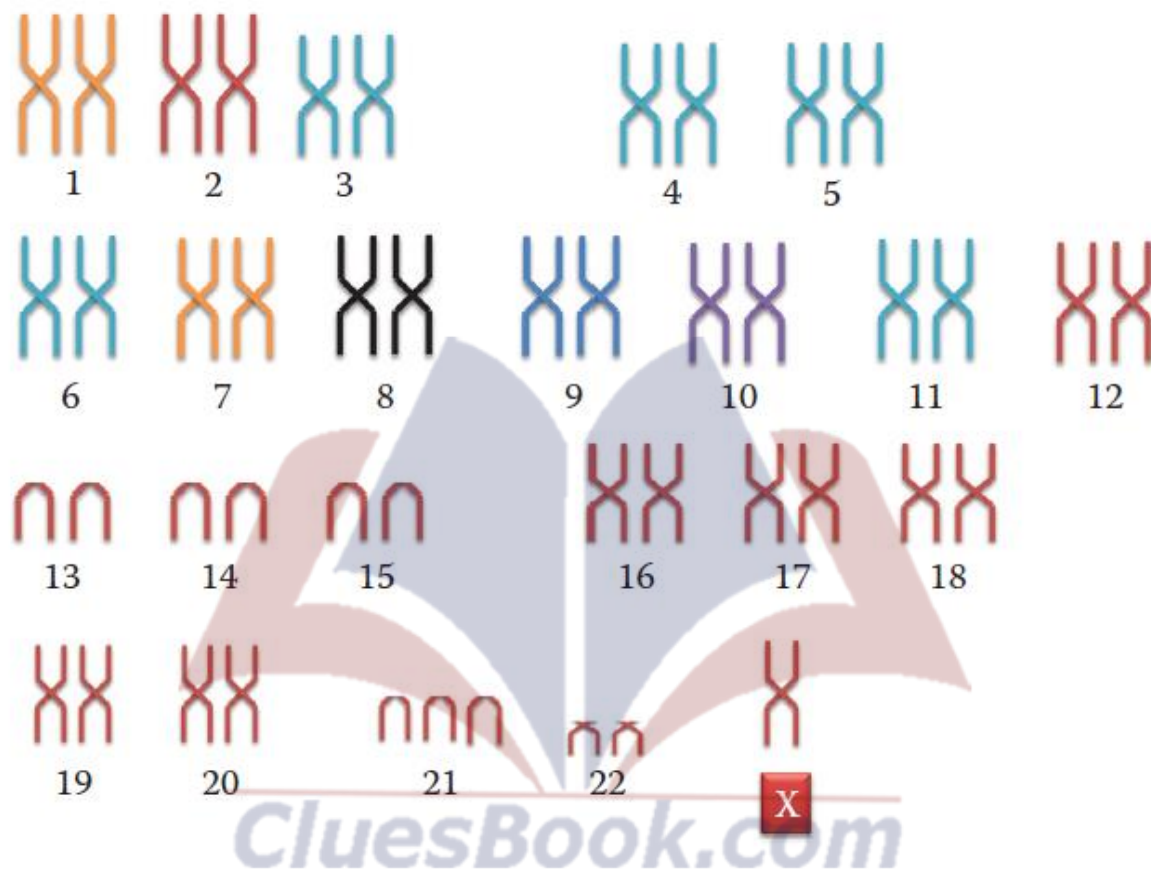


Fig- Turner syndrome (45 X).

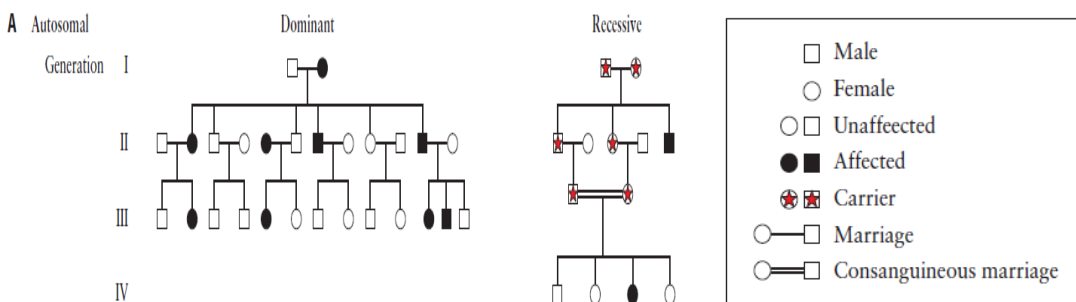
Klinefelter syndrome occurs in males with the genotype 47,XXY. Such males have 47 chromosomes and are classified as having a sex chromosome trisomy (three sex chromosomes), since they carry two X chromosomes and one Y chromosome. This syndrome affects about 1 in 1,000 males. Most affected males are taller than average, and they may have more body fat in the hips or chest as well as little facial and body hair. Some Klinefelter males are mentally retarded, while many others have normal intelligence. The most common feature of this syndrome is infertility; about 2% of infertile men have Klinefelter syndrome.

20-Single Gene Disorders

The genetic disorder in which a single gene gets mutated is called as a single-gene disorder. When a gene is mutated so that its protein product cannot carry out its normal body function, it leads to the development of a disorder.

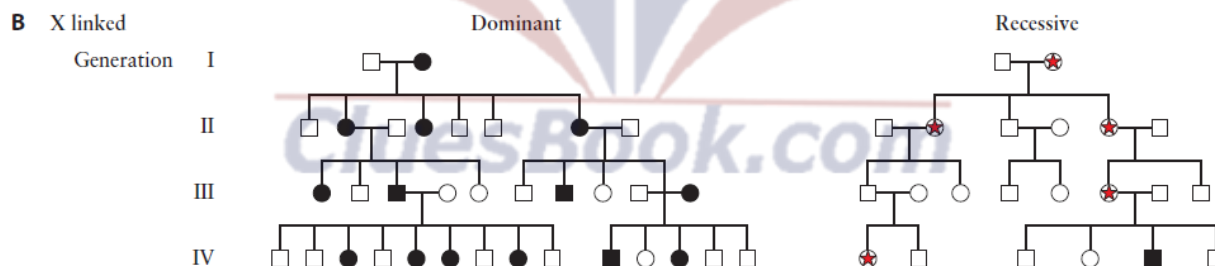
Single-gene disorders, or monogenic diseases, result from a mutation(s) in a single gene occurring in all cells of the body (somatic cells). Inheritance of these types of disorders follows a Mendelian segregation pattern. Although very rare, such disorders affect millions of people globally. Currently, it is estimated that more than 10,000 human diseases are monogenic. These diseases can occur in about 1 out of every 100 births and thus can cause a significant loss of life. So-called “pure genetic diseases” are caused by a single nucleotide change in a single gene in human DNA. The nature of disease depends on the functions performed by the modified gene. Monogenic diseases can be classified into three main categories: **autosomal dominant**, **autosomal recessive**, and **X linked** (Fig.).

All humans have two chromosomal copies of each gene or allele, one allele per each member of a chromosome pair. Dominant monogenic disorders involve a mutation in only one allele of a disease-related gene. In autosomal dominant inheritance, an affected person has at least one affected parent, and affected individuals have a 50% chance of passing the disorder on to their children. Huntington disease is an example of an autosomal dominant disorder.



Recessive monogenic disorders occur due to a mutation in both alleles of a disease-related gene. In autosomal recessive inheritance, affected children are usually born to unaffected parents. Parents of affected children usually do not have disease symptoms but carry a single copy of the mutated gene. There is an increased incidence of autosomal recessive disorders in families in which parents are related. Children of parents who are both heterozygous for the mutated gene have a 25% chance of inheriting the disorder, and the disorder affects either sex. Cystic fibrosis and sickle-cell anemia are examples of autosomal recessive disorders.

X-linked monogenic disorders are linked to mutations in genes on the X chromosome. The X-linked alleles can also be dominant or recessive. These alleles are expressed in both men and women, more so in men, as they carry only one copy of the X chromosome (XY), whereas women carry two (XX).



Y-linked inheritance would affect only males, the affected males would always have an affected father, and all sons of an affected man would have the disease. However, no Y-linked diseases have ever been discovered. Apart from male infertility, mutations in Y-linked genes may not give rise to any disorders.

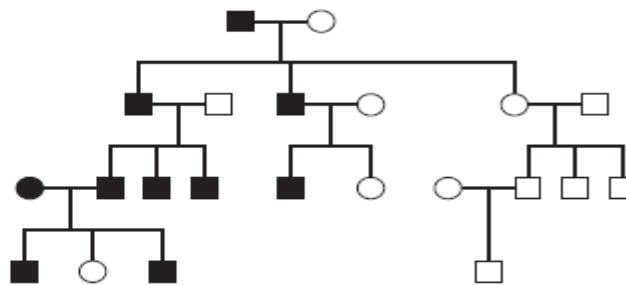
C Y linked

Generation I

II

III

IV



The prevalence of some monogenic inheritable disorders is shown in

Table

Disorder	Prevalence (approximate)
Autosomal dominant	
Familial hypercholesterolemia	1 in 500
Polycystic kidney disease	1 in 1,250
Neurofibromatosis type I	1 in 2,500
Hereditary spherocytosis	1 in 5,000
Marfan syndrome	1 in 4,000
Huntington disease	1 in 15,000
Autosomal recessive	
Sickle-cell anemia	1 in 625
Cystic fibrosis	1 in 2,000
Lysosomal acid lipase deficiency	1 in 40,000
Tay-Sachs disease	1 in 3,000
Phenylketonuria	1 in 12,000
Mucopolysaccharidoses	1 in 25,000
Glycogen storage diseases	1 in 50,000
Galactosemia	1 in 57,000
X linked	
Duchenne muscular dystrophy	1 in 7,000
Hemophilia	1 in 10,000

21-Cystic Fibrosis

Disorder and Genetics:

Cystic fibrosis is a genetic disorder that affects the respiratory, digestive, and reproductive systems. This disorder is mediated by the production of abnormally thick mucous linings in the lungs, can lead to fatal lung infections, and can give rise to obstruction of the pancreas and impair digestion. The severity of the condition varies from the mild forms (e.g., absence of the vas deferens in men) to the more common severe forms (epithelial gland dysfunction). Since an individual must inherit two defective cystic fibrosis genes, one from each parent, to get the disease, cystic fibrosis is classified as an autosomal recessive disorder.

In families in which both parents are carriers of the cystic fibrosis gene, there is a 25% chance that they will transmit cystic fibrosis to their child, a 50% chance that the child will carry the cystic fibrosis gene, and a 25% chance that the child will be a non carrier.

The cystic fibrosis gene encodes a protein known as the cystic fibrosis transmembrane regulator (CFTR), and the *CFTR* gene contains 27 exons within 250 kb of genomic DNA. The CFTR protein consists of 1,480 amino acids distributed in two membrane-spanning domains and two ATP-binding domains.

The membrane-spanning domains form a low-conductance cyclic adenosine monophosphate (cAMP)-dependent chloride channel. The ATP-binding domains control channel activity.

The most common mutation, a 3-bp deletion at codon 508 ($\Delta F508$), results in the loss of a phenylalanine residue from the first ATP-binding domain and blocks the transport of the CFTR protein to the epithelial cell membrane.

The next most common mutation ($\leq 5\%$ of *CFTR* mutations) is also present in the first ATP-binding domain and G551D (glycine at residue 551 is replaced by aspartic acid) in exon 11. Both common mutations are associated with a severe form of the disease in the homozygous state.

Prevalence

Cystic fibrosis is a common cause of death in childhood and the most common inherited disease in white populations. The incidence of cystic fibrosis varies significantly in different countries: 1 in 2,500 births in the United Kingdom, 1 in 2,000 to 3,000 births in Europe, and 1 in 3,500 births in the United States. Cystic fibrosis can arise in nonwhite populations, but only very rarely (1 in 100,000 births in African-American and East Asian populations).

Diagnosis and Prognosis

Cystic fibrosis patients have many symptoms (salty skin; persistent coughing, wheezing, or shortness of breath; and excessive appetite but poor weight gain), and their symptoms may vary due to the >200 mutations of the *CFTR* gene.

A diagnostic sweat test measures the amount of salt in sweat of patients with cystic fibrosis (a high salt level indicates cystic fibrosis). This test is usually performed in babies older than 3 to 4 weeks and can also confirm the diagnosis in older children and adults.

If pancreatic enzyme levels are reduced, stool analyses may reveal decreased or absent levels of the digestive enzymes (trypsin and chymotrypsin) or high levels of fat. If insulin secretion is reduced, blood sugar levels are high. Lung function tests may show that breathing is compromised.

Genetic testing on a small blood sample can help determine whether an individual has a defective *CFTR* gene. During pregnancy, an accurate diagnosis of cystic fibrosis in the fetus is possible.



22-Fragile X Syndrome

Disorder and Genetics:

Fragile X syndrome is caused by a “fragile” site at the end of the long arm of the X chromosome. This syndrome is manifested by many changes in behavior and cognitive recognition that vary widely in severity among patients.

Fragile X syndrome is the most common cause of inherited mental retardation. Although it is an X-chromosome-linked recessive trait with variable expression and incomplete penetrance, 30% of all carrier women are affected.

Penetrance is the proportion of individuals carrying a particular variant of a gene (allele or genotype) who also express an associated trait or phenotype. Full penetrance occurs when all individuals carrying a gene express the phenotype. Incomplete penetrance occurs when some individuals fail to express the phenotype, even though they carry the variant allele.

Fragile X syndrome is caused by loss-of-function mutations in the fragile X mental retardation 1 (*FMR1*) gene. *FMR1* encodes the FMRP protein found in many tissues and at particularly high levels in the brain and testes. In the brain, it may play a role in the development of neuronal synapses and cell communication. The synapses can change and adapt over time in response to experience, a characteristic called **synaptic plasticity**. The FMRP protein may help regulate synaptic plasticity and thereby control learning and memory.

Fragile X syndrome belongs to a growing class of **neurodegenerative disorders** known as **trinucleotide repeat disorders**. Among these disorders, 14 affect humans and elicit neurological dysfunction. Trinucleotide CGG repeat expansions (200 to more than 1,000 repeats) that inactivate the *FMR1* gene are the most common mutations observed at this locus.

Prevalence

Fragile X syndrome is the single most common inherited cause of mental impairment, affecting 1 in 3,600 males and 1 in 4,000 to 6,000 females worldwide. Approximately 1 in 259 women of all races carry the fragile X gene and may pass it to their children, whereas about 1 in 800 men of all races and ethnicities are carriers. Carrier females have a 30 to 40% chance of giving birth to a mentally retarded male child and a 15 to 20% chance of having a mentally retarded female child.

Diagnosis and Prognosis

The diagnosis of fragile X syndrome is made by the detection of mutations in the *FMR1* gene. Over 99% of individuals have an *FMR1* gene that expresses all of its known mutants. Tests used for diagnosis include chromosome analysis and various protein tests. Diagnosis is usually made when a child is young, and there is no current cure for this illness. Early diagnosis of the syndrome may allow for therapeutic interventions such as speech therapy, occupational therapy, psychotherapy, and special education, which can improve the quality of a patient's life considerably.



BT501-HEALTH BIOTECHNOLOGY

23-Hemophilia

It has been reported that the Arab physician Abu al-Qasim al-Zahrawi, commonly known as Abulcasis, wrote a book on hematology. His book *Al-Tasrif* written during the period 1000 AD provided the first description of hemophilia, a hereditary genetic disorder

Disorder and Genetics

Hemophilia is a hereditary bleeding disorder resulting from a partial or total lack of an essential blood clotting factor. It is a lifelong disorder that produces excessive bleeding, and spontaneous internal bleeding occurs very frequently.

Hemophilia A, the most common form, is caused by a deficiency in clotting factor 8 (factor VIII). Hemophilia B results from a deficiency in clotting factor 9(IX). This disorder is inherited in a sex linked recessive manner.

Prevalence

This disorder occurs more prominently in males than in females. About one-third of new diagnoses occur without a family history. It appears globally and occurs in all racial groups.

Diagnosis and Prognosis

Hemophilia can be diagnosed as type A or type B by blood tests in infants after 9 months of age. Administration of clotting factors helps affected individuals to live with the disease.

Hemophilia, though a serious disease, can be tolerable with proper precautions and therapy, and the prospects for children with hemophilia are excellent. Recent studies have documented a

greatly increased quality of life and life expectancy among hemophiliac patients in developed countries.

However, it is predicted that the number of people with hemophilia in developed countries will increase gradually during the next few decades.

24-Huntington Disease

Disorder and Genetics

Huntington disease is an autosomal dominant genetic disorder; if one parent carries the defective Huntington disease gene, his or her offspring have a 50-50 chance of inheriting the disease. Huntington disease is a neurodegenerative brain disorder in which afflicted individuals lose their ability to walk, talk, think, and reason. They easily become depressed, lose their short-term memory, and may experience a lack of concentration and focus. Every individual with the gene for the disease eventually develops the disease.

The Huntington disease gene is located on the long arm of human chromosome 4 and encodes a protein called **huntingtin** that is quite variable in its structure. The 5' end of the Huntington disease gene contains many repeats of the CAG trinucleotide (encodes glutamine). A highly variable number of **CAG trinucleotide repeats** accounts for the Huntington disease gene mutation, which leads to the expression of an abnormally long **polyglutamine tract** at the N terminus of the huntingtin protein beginning at residue 18.

Such polyglutamine tracts increase protein aggregation, which may alter cell function. Thus, like fragile X syndrome, Huntington disease is one of 14 trinucleotide repeat disorders that cause neurological dysfunction in humans. More specifically, Huntington disease is classified as a **polyglutamine disorder**.

Healthy unaffected persons have a CAG repeat count of 9 to 35 (Table). However, alleles with more than 36 CAG repeats give rise to Huntington disease (the highest reported repeat length is 250). Incomplete penetrance is found in alleles with 36 to 39 CAG repeats. People with 36 to 40 CAG repeats may or may not develop Huntington disease, while people with more than 40 CAG repeats are rather likely to develop the disorder. Alleles with more than 60 CAG repeats result in a severe form of Huntington disease known as juvenile Huntington disease, and children who get Huntington disease may range from 2 to 20 years of age.

Table- Variation in status of Huntington disease is controlled by number of CAG repeats in the *huntingtin* gene

CAG repeat count	Disease classification	Disease status
<28	Normal	Unaffected
28–35	Intermediate	Unaffected
36–40	Incomplete penetrance	Weakly affected with Huntington disease
>40	Full penetrance	All affected with Huntington disease
>60	Full penetrance	Affected with juvenile Huntington disease

The mass of the huntingtin protein depends mainly on the number of its glutamine residues. Wild-type (normal) huntingtin consists of 3,144 amino acids, contains 6 to 35 glutamines, and has a mass of ~350 kilodaltons (kDa). In Huntington disease patients, huntingtin contains more than 36 glutamines and has an overall higher molecular mass than wild-type huntingtin.

Prevalence

Huntington disease affects males and females equally and crosses all ethnic and racial boundaries. Usually, Huntington disease begins at age 30 to 45, but it may occur as early as the age of 2. Children who develop juvenile Huntington disease rarely live to adulthood. Everyone

who carries the gene develops the disease. In the United States, Canada, and western Europe, Huntington disease affects about 1 in 20,000 people.

Diagnosis and Prognosis

Currently, there is no treatment or cure for Huntington disease. The discovery of the Huntington disease gene in 1993 facilitated the development of specialized testing that may help to confirm the diagnosis of the disease in patients with an affected parent or characteristic symptoms of the disease. Blood samples are taken from patients, and DNA is directly analyzed by PCR for Huntington disease gene mutations to determine the number of CAG repeats in the Huntington disease gene region. Additional blood samples may be obtained from close or first-degree relatives (e.g., the mother or father) with Huntington disease to help confirm the results.

25-Sickle-Cell Anemia

Sickle-cell disease, or sickle-cell anemia or drepanocytosis, is a hereditary blood disorder, characterized by red blood cells that assume an abnormal, rigid, sickle shape. The sickling occurs because of a mutation in the hemoglobin gene.

Disorder

Sickle-cell disease is a blood-related disorder resulting from the substitution of a valine for glutamic acid in the β -globin chain of adult hemoglobin. The mutated sickle hemoglobin undergoes conformational change and polymerization upon deoxygenation, leading to red blood cell hemolysis and deformation (not doughnut shaped) and to pathology due to blockage of

capillaries. Sickled red blood cells cannot migrate through small blood vessels; rather, they cluster and block these vessels, depriving organs and tissues of oxygen-carrying blood. This process leads to periodic episodes of pain and ultimately can damage tissues and vital organs and lead to other serious medical problems. Normal red blood cells live about 120 days in the bloodstream, but sickled red blood cells die much more rapidly, after about 10 to 20 days. Since they cannot be replaced fast enough, the blood is chronically short of red blood cells, leading to the condition of sickle-cell anemia.

Genetics

Hereditary persistence of fetal hemoglobin decreases the severity of sickle-cell disease. Thus, induction of fetal hemoglobin in adults has been a long-standing goal of therapies for sickle-cell disease. Recently, a GWAS for the fetal hemoglobin phenotype revealed a single strong locus, with sequence variants in the intron of a transcription factor, B-cell chronic lymphocytic leukemia/lymphoma 11A (BCL11A). BCL11A represses fetal hemoglobin expression in red blood cells.

Inactivation of BCL11A in a mouse model of sickle-cell disease not only leads to the induction of fetal hemoglobin expression but also corrects the hematologic and pathological defects of the disease. Thus, BCL11A is a potential therapeutic target for sickle-cell disease.

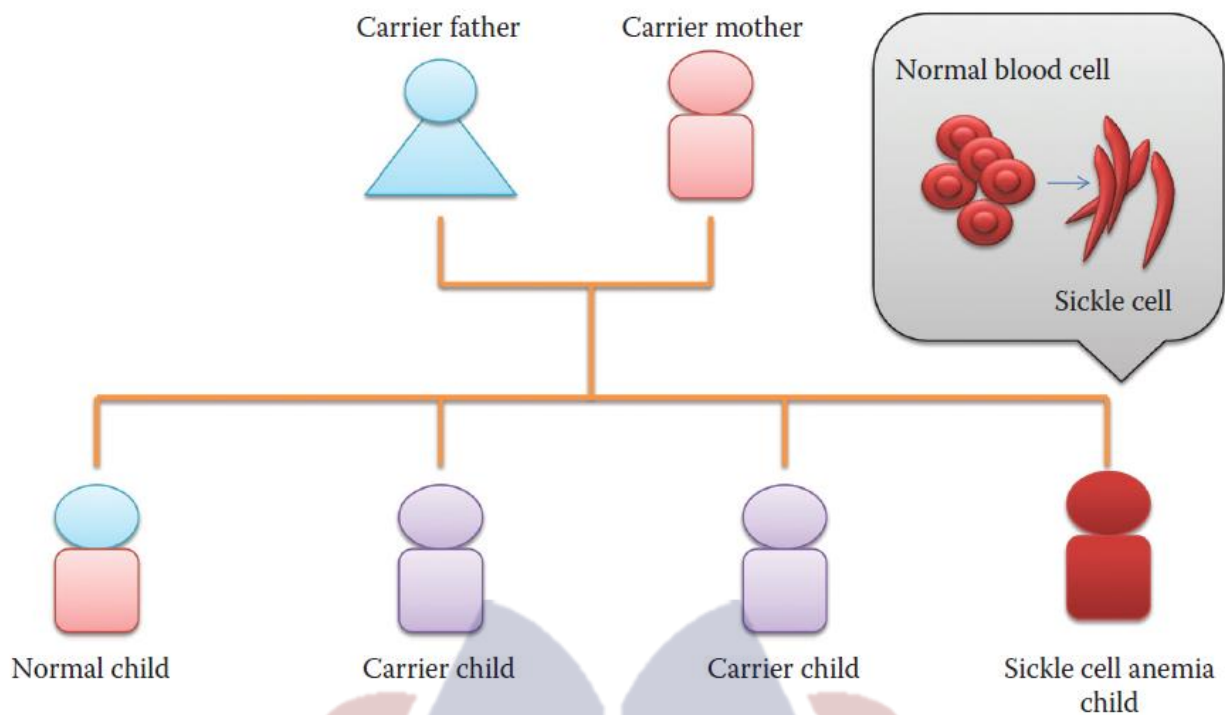


Figure- Single gene inheritance which causes sickle cell anemia

Prevalence

Sickle-cell anemia affects millions of people throughout the world. It is particularly common among people whose ancestors originate from sub-Saharan Africa, South America, Cuba, Central America, Saudi Arabia, India, and Mediterranean countries (e.g., Turkey, Greece, and Italy).

In the United States, it affects around 72,000 people, most of whose ancestors come from Africa. The disease occurs in about 1 in every 500 African-Americans and 1 in every 1,000 to 1,400 Hispanic Americans. About 2 million Americans, or 1 in 12 African-Americans, carry the sickle-cell allele.

Diagnosis and Prognosis

Sickle-cell disease can be diagnosed by a simple blood test. In many cases, sickle-cell anemia is diagnosed when newborns are screened. Vaccines, antibiotics, and folic acid supplements are administered in addition to painkillers. Blood transfusions and surgery are used in severe cases. The only known cure at present is a bone marrow transplant.

26-Tay–Sachs Disease

Disorder and Genetics

Tay–Sachs disease is an autosomal recessive fatal genetic disorder caused by a genetic mutation in the *HEXA* gene on human chromosome 15, which encodes the hexosaminidase A enzyme. This mutation leads to a decrease in function of hexosaminidase A, and as a result, harmful quantities of a sphingoglycolipid, termed ganglioside GM2, accumulate in brain neurons.

HEXA gene mutations are rare and are most frequently detected in genetically isolated populations. Tay–Sachs disease can occur from the inheritance of either two similar or two unrelated mutations in *HEXA* that cause disease. Many *HEXA* mutations have been identified, and these mutations can reach significant frequencies in certain populations (see below).

In the most common form of the disease (infantile Tay–Sachs), abnormal hexosaminidase A enzyme activity and the accompanying harmful accumulation of cell membrane-associated gangliosides in neurons lead to premature neuronal death, paralysis, dementia, blindness, psychoses, and even death of the patient. Although the degeneration of the central nervous

system begins at the fetal stage, the loss of peripheral vision and motor coordination are not evident until ~6 months of age, and death usually results by 4 years of age.

Prevalence

The frequency of Tay–Sachs disease is much higher in Ashkenazi Jews (Jews of eastern European origin) than in others. Approximately 1 in 27 Jews in the United States is a carrier of the Tay–Sachs disease gene. There is also a noticeable incidence of Tay–Sachs disease in non-Jewish French Canadians, known as Acadians, who originated from France and settled in southeastern Quebec.

Interestingly, while the French Canadians and Ashkenazi Jews carry different *HEXA* mutations, the Cajuns (people in southern Louisiana descended from the Acadians) carry the mutation found most frequently in Ashkenazi Jews. The ancestry of carriers from Louisiana families traces back to a single non-Jewish founder couple that lived in France in the 18th century. The Irish are also at increased risk for the Tay–Sachs gene, and among Irish-Americans the carrier rate is currently about 1 in 50. By contrast, the carrier rate in the general non-Jewish population as well as in Jews of Sephardic (Iberian or Middle Eastern) origin in the United States is about 1 in 250.

Diagnosis and Prognosis

Tay–Sachs disease may be diagnosed by a blood test that measures levels of the hexosaminidase A enzyme in serum, lymphocytes, or skin fibroblasts. During the past 25 years, carrier screening and genetic counseling in high-risk populations have greatly reduced the number of children born with Tay–Sachs disease in these groups. Thus, a high percentage of babies born with Tay–

Sachs disease today are born to couples not previously considered to be at high risk. Prenatal tests of hexosaminidase A activity, such as amniocentesis (at 15 to 16 weeks of pregnancy) and chorionic villus sampling (at 10 to 12 weeks of pregnancy), can now diagnose Tay–Sachs disease in the fetus.

27-Thalassemia

Disorder and Genetics

Thalassemia is a blood-related genetic disorder that involves either the absence of or errors in genes responsible for the production of hemoglobin. Each red blood cell contains 240 million to 300 million molecules of hemoglobin. The severity of the disease depends on the gene mutations that arise and the manner in which they influence each other.

Both of the hemoglobin α and β subunits are required to bind oxygen in the lungs and deliver it to other tissues. Genes on human chromosome 16 encode the α subunits, while genes on chromosome 11 encode the β subunits. A lack of expression of a particular subunit determines the type of thalassemia; e.g., a lack of an α subunit results in α -thalassemia. The lack of subunits corresponds to mutations in the genes on the appropriate chromosomes. There can be various grades of the disease depending on the gene and the type of mutations.

Prevalence

The α - and β -thalassemias are the most common inherited single-gene disorders, with the highest prevalence in areas where malaria was or still is endemic. The burden of this disorder in many regions is of such a magnitude that it represents a major public health concern. In Iran, about 8,000 fetuses may be affected by thalassemia each year. In some Mediterranean countries, control programs have achieved 80 to 100% prevention of disease in newborns.

Diagnosis and Prognosis

Diagnosis of thalassemia can be made as early as 10 to 11 weeks in pregnancy using procedures such as amniocentesis and chorionic villus sampling. Individuals can also be tested for thalassemia through routine blood counts. Thalassemic patients may have reduced fertility or even infertility.

Early treatment of thalassemia has been very effective in improving the quality of life of patients. Treatments for thalassemias depend on the type and severity of the disorder. Carriers or individuals who have mild or no symptoms of α - or β -thalassemia need little or no treatment.

Three types of treatments are used for moderate and severe forms of thalassemia. These include blood transfusions, iron chelation therapy (removes excess iron in the blood that increases after transfusions), and folic acid supplements (vitamin B supports red blood cell growth).

Currently, genetic testing and counseling, and prenatal diagnosis, play an increasingly important role in informing individual as well as professional decisions related to the prevention, management, and treatment of this disease.

28-Polygenic disorders- Diabetes

Genetic disorders are **polygenic** if they are causally associated with the effects of many genes.

Such disorders may also be **multifactorial**, if they are influenced by several different lifestyles and environmental factors.

Examples of multifactorial disorders include cancer, heart disease, and diabetes. Although such disorders frequently cluster in families, these disorders do not have a simple Mendelian pattern of inheritance. Moreover, as many of the factors that cause these disorders have not yet been identified, it is rather difficult to determine the risk of inheritance or transmission of these disorders.

Some principal features of the pattern of multifactorial inheritance of a polygenic diseases are summarized in Table.

No.	Characteristic
1	A simple pattern of inheritance of a polygenic disease does not exist within a family.
2	A lower incidence of a polygenic disease within a population is associated with a higher risk of disease in first-degree relatives.
3	The relative risk of a polygenic disease is much lower in second-degree and more distantly related family members.
4	The risk of recurrence of a polygenic disease in a family is higher when more than one family member is affected.
5	The more severe the malformation associated with a polygenic disease, the greater is the risk of disease recurrence.
6	If a polygenic disease is more frequent in one sex than the other, the risk is higher for relatives of patients of the less susceptible sex.
7	If an increased risk of recurrence of disease occurs when the parents are consanguineous (e.g., first-degree relatives), many factors with additive effects may be involved.

An individual may not be born with a disease but may be at high risk of acquiring it, a condition referred to as **genetic predisposition** or **genetic susceptibility**. The genetic susceptibility to a particular disease due to the presence of a gene mutation(s) in an allele(s) need not lead to disease.

Diabetes is a disease in which the body does not produce or properly use insulin. Insulin is a hormone that is needed to convert sugar, starches, and other food into energy needed for daily life. The cause of diabetes continues to be a mystery, although both genetics and environmental factors such as obesity and lack of exercise appear to play roles. There are three major classes of diabetes: type 1 diabetes, type 2 diabetes, and gestational diabetes. Only type 1 diabetes, a representative polygenic autoimmune disease.

29-Diabetes mellitus type 1

Disorder

Type 1 diabetes is an autoimmune disease that presents clinically with hyperglycemia resulting from the immune-mediated progressive destruction of insulin-producing pancreatic islet β -cells and associated metabolic dysfunction.

The resulting insulin deficiency requires lifelong exogenous insulin treatment for survival, and long-term complications can cause substantial disability and shorten life span. About 90 to 95% of the β -cell mass is already destroyed when symptoms of type 1 diabetes first appear, demonstrating that prediction and prevention are high priorities.

This disease affects about 1 in 300 individuals in North America. More than 20 million people worldwide (mostly children and young adults) are estimated to have type 1 diabetes.

Genetics

The etiology and pathogenesis of type 1 diabetes are largely accounted for by genetic predisposition. Within a pair of monozygotic twins (originating from one placenta), if one twin has type 1 diabetes, the probability of the other twin becoming a type 1 diabetic is about 40 to 50%. In pairs of dizygotic twins (originating from two placentas) in which one twin is type 1 diabetic, the probability of the other twin becoming a type 1 diabetic is only about 5 to 6%. This lower probability of susceptibility to type 1 diabetes is the same as that for unrelated children in the general population. These findings are consistent with the observed overall global increase in type 1 diabetes incidence during the last 20 to 30 years and strongly support the idea that environmental factors are also important in the control of susceptibility to type 1 diabetes.

The majority of genetic research to date has focused on the heritability that predisposes to islet autoimmunity and type 1 diabetes. The evidence indicates that type 1 diabetes is a polygenic, common, complex disease, with major susceptibility genes located in the *HLA* complex on human chromosome 6 and with other smaller effects found in non-*HLA* loci on other chromosomes. Recent advances in DNA technology, including high-throughput SNP typing and sequencing and GWASs, have advanced our understanding of the immune pathogenesis of type 1 diabetes.

Diagnosis

Type 1 diabetes is checked by analyzing the plasma glucose level and is diagnosed by fasting plasma glucose level at or above 126 mg/dL (considered as prediabetic conditions) and plasma glucose at or above 200 mg/dL (considered as diabetic condition).

To confirm type 1 diabetes, most physicians prefer to measure a fasting glucose level and according to the current definition, two fasting glucose measurements above 126 mg/dL are considered diagnostic for diabetes mellitus. Moreover, patients with fasting glucose levels from 100 to 125 mg/dL are considered to have impaired fasting glucose.

Furthermore, patients with plasma glucose at or above 140 mg/dL but not over 200 mg/dL, 2 h after a 75 g oral glucose load are considered to have impaired glucose tolerance. Interestingly, among these two prediabetic states, the latter in particular is a major risk factor for progression to full-blown diabetes mellitus as well as cardiovascular disease.

30-Diabetes mellitus type 2

It is known as Non Insulin dependent diabetes (NIDD) and is characterized by reducing response of body against insulin or relatively reduced insulin secretion by the beta cells. Type 2 diabetes is the most common type and starts appearing after the age of 30 years. It is due to the combination of sedentary lifestyle and genetic factors.

Obesity has been found to contribute to approximately 55% cases of the type 2 diabetes. However, because type 2 diabetes develops slowly, some people with high blood sugar experience no symptoms at all.

Treatment: Onset of type 2 diabetes can be delayed or prevented through proper nutrition and regular exercise. If the condition progresses, medications may be needed.

The synthesized insulins are used medically to treat patients with Type 1 diabetes mellitus, whereas patients with Type 2 diabetes mellitus are insulin resistant.

31-Alzheimer-diseases

Alzheimer's disease (AD), also known in the medical literature as Alzheimer disease, is the most common form of dementia. There is no cure for the disease, which worsens as it progresses, and eventually leads to death. It was first described by a German psychiatrist and neuropathologist Alois Alzheimer in 1906 and was named after him. Many different clinical features are associated with Alzheimer disease, including the inability to create new memories, the loss of short-term memory, and the inability to concentrate.

Disorder

Alzheimer disease is a devastating neurodegenerative disorder with a relentless progression. The pathogenesis of Alzheimer disease may be triggered by the accumulation of the amyloid β peptide ($A\beta$), which is due to overproduction of $A\beta$ and/or the failure of mechanisms to clear the peptide. $A\beta$ self-aggregates into oligomers of various sizes and forms diffuse and neuritic plaques in the parenchyma and blood vessels. $A\beta$ oligomers and plaques are potent mediators of

A β toxicity, block proteasome function, inhibit mitochondrial activity, alter intracellular Ca²⁺ levels, and stimulate inflammatory processes. Loss of normal A β function also contributes to neuronal dysfunction.

A β interacts with the signaling pathways that regulate the phosphorylation of the microtubule-associated protein tau. Hyperphosphorylation of tau disrupts its normal function in regulating axonal transport and leads to the accumulation of neurofibrillary tangles and toxic species of soluble tau. Degradation of hyperphosphorylated tau by the proteasome is inhibited by the actions of A β . These two proteins and their associated signaling pathways therefore represent important therapeutic targets for Alzheimer disease. Abnormalities in A β and tau can be measured upon neuropathological examination, in cerebrospinal fluid or by positive emission tomography scans.

Genetics

Susceptibility to Alzheimer disease is genetically controlled, and about 13% of Alzheimer disease is inherited as an autosomal dominant disorder. **Late-onset familial Alzheimer disease** is a complex polygenic disorder controlled by multiple susceptibility genes, and the strongest association is with the *APOE* e4 allele at locus *AD2*.

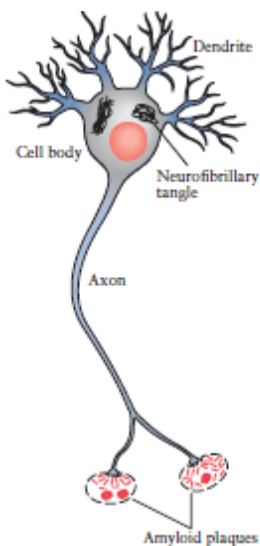
Mutations in three genes—those for **amyloid** precursor protein (*APP*), presenilin-1 (*PSEN1*), and presenilin-2 (*PSEN2*)—account for less than 5% of all cases of Alzheimer disease and often cause early onset of the disease in individuals who are 60 years old or younger. Thus, despite considerable research effort, the root cause of Alzheimer disease in the vast majority of cases is unknown.

Prevalence

Alzheimer disease is the most common cause of dementia in North America and Europe, with an estimate of four million affected individuals in the United States. The prevalence of Alzheimer disease increases with age: about 10% of persons older than 70 have significant memory loss, and more than half of these individuals have Alzheimer disease. An estimated 25 to 45% of persons over age 85 have dementia.

Diagnosis and Prognosis

A proper diagnosis of Alzheimer disease relies on a clinical neuropathological assessment. The formation of A β plaques and neurofibrillary tangles is thought to contribute to the degradation of the neurons in the brain and the subsequent symptoms of Alzheimer disease. A hallmark of Alzheimer disease is the accumulation of amyloid plaques between neurons in the brain. Amyloid is a general term for protein fragments that the body produces normally. A β is a protein fragment cleaved from a larger amyloid precursor protein. In a healthy brain, these A β protein fragments are broken down and eliminated. In Alzheimer disease, the fragments accumulate to form hard, insoluble plaques (deposits). The plaques are composed of a tangle of protein aggregates that have a fiber appearance and are called amyloid fibers. Neurofibrillary tangles consist of these insoluble twisted amyloid fibers found inside neurons. These tangles are formed primarily by a protein called tau, which forms part of a microtubule that helps to transport nutrients and other substances from one part of a neuron to another. In Alzheimer disease, however, the tau protein is abnormal and the microtubule structures collapse. As a result, neurons are deprived of certain nutrients and undergo cell death. In Alzheimer disease, however, the tau protein is abnormal and the microtubule structures collapse. As a result, neurons are deprived of certain nutrients and undergo cell death.



Neurofibrillary tangles and amyloid plaques associated with neurons in the brain are hallmarks of Alzheimer disease.

Three forms of early-onset familial Alzheimer disease caused by mutations in one of three genes—A β (A4) precursor protein (*APP*), presenilin 1 (*PSEN1*), and presenilin 2 (*PSEN2*)—are recognized. Molecular genetic testing of the three genes is available in clinical laboratories.

Early- onset familial Alzheimer disease A form of Alzheimer disease that arises at a younger age (<65 years) in families. About 1 to 6% of all Alzheimer disease is early onset, and about 60% of early- onset Alzheimer disease runs in families.

Late- onset familial Alzheimer disease A complex polygenic disorder controlled by multiple susceptibility genes. The strongest association is with the *APOE* e4 allele at locus *AD2*.

Senile plaque A dense spheroid amyloid body found outside of neurons and often associated with Alzheimer disease

32-Breast Cancer

Disorder

Cancer generally results from the sequential acquisition of mutations in genes that regulate cell multiplication, cell repair, and the ability of a cell to undergo malignant transformation. This multistep process is not an abrupt transition from normal to malignant but may take 20 years or more.

Genetics

This malignancy is one of the most commonly inherited cancers based on observations that

- (i) 20 to 30% of all patients with breast cancer have a family history of the disease and
- (ii) Twin studies show that 25% of breast cancer cases are heritable.

Discovery of the breast cancer type 1 susceptibility (*BRCA1*) and *BRCA2* genes more than 10 years ago has had a high impact on patient care, allowing for early detection and prevention of breast cancer. However, deleterious mutations in the *BRCA1* and *BRCA2* genes cause at most 3 to 8% of all breast cancer cases.

Prevalence

Lung, stomach, liver, colon, and breast cancers cause the most cancer deaths each year. Combined, these cancers are responsible for more than 4.2 million deaths annually.

Current epidemiological evidence predicts that 1 in 8 women will be diagnosed with breast cancer in her lifetime. Deaths from cancer worldwide are predicted to continue rising, with a

projected 13.1 million deaths in 2030. Breast cancer is the second most common malignancy diagnosed in women and the second leading cause of cancer-related deaths.

Diagnosis and Prognosis

The roles that genes play differ greatly, ranging from completely determining the disease state (disease genes) to interacting with other genes and environmental factors in causing cancer (susceptibility genes). The primary determinants of most cancers are lifestyle factors (e.g., smoking, dietary and exercise habits, environmental carcinogens, and infectious agents) rather than inherited genetic factors.

Breast cancer is a heterogeneous disease often characterized by the presence or absence of expression of estrogen receptors and human epidermal growth factor receptor 2 (HER2) on tumor cells. These molecular markers used to classify breast cancer subtypes may also predict typical responses to targeted therapies.

Genetic tests will be useful in determining the best course of treatment for cancers, which are presently classified as a single disease but may ultimately be classified into different types, each best managed by a different therapeutic strategy.

Thus, genetics will continue to play an important role in the control of cancer, including

- (i) The identification of individuals at risk for a specific cancer, leading to preventative or screening strategies for an individual or family members, and

- (ii) The identification of the subtype of a cancer so that treatment can be tailored to target that specific disease.

Women with a type of breast cancer that overproduces the human epidermal growth factor receptor 2 (HER2) protein on the surface of the tumor may be effectively treated with the humanized anti-HER2 monoclonal antibody trastuzumab (Herceptin).

Trastuzumab: The First Humanized Monoclonal Antibody Approved for the Treatment of Breast Cancer.

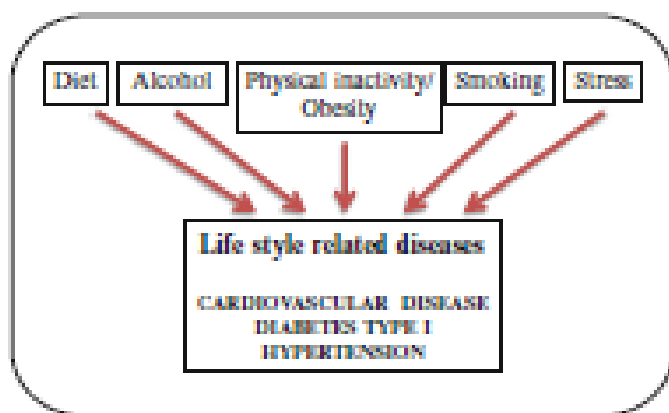
33-Cardiovascular Disease

Disorder

Major cardiovascular diseases include several subtypes, such as coronary heart disease, cerebrovascular disease, heart failure, rheumatic heart disease, and congenital heart disease. Cardiovascular disease is a leading health problem, affecting more than 80 million individuals in the United States. In Canada, 1 out of 3 deaths is attributable to cardiovascular disease.

The major risk factors associated with cardiovascular diseases are cigarette smoking, unhealthy diet, physical inactivity, hypertension, diabetes, and high blood cholesterol. Cardiovascular diseases may also result from a variety of genetic causes, including single-gene mutations, interactions between multiple genes, and gene–environment interactions.

Economic shifts, urbanization, industrialization, and globalization bring about lifestyle changes that promote heart disease. In developing countries, life expectancy is increasing rapidly and is expected to expose people to these environmental risk factors for longer periods.



Genetics

For most cardiovascular disease disorders, the relative risk of disease is attributable to inherited DNA sequence variants. However, the role of inheritance and the magnitude of its effect vary by disease and other factors such as age of disease onset and subtype of cardiovascular disease. Most cardiovascular disease traits, such as myocardial infarction and concentrations of plasma LDL cholesterol, show complex inheritance, suggestive of an interaction between multiple genes and environmental factors.

Disease	Possible causes	Affects	Symptoms	Treatment
Cardiovascular disease	Inherited and acquired, predominant lifestyle disease caused by smoking High blood pressure (hypertension) High cholesterol Diabetes Family history Peripheral artery disease Obesity	Coronary arteries are at risk for narrowing as cholesterol deposits, called plaques, build up inside the artery, affect the heart muscle or the blood vessels of the heart which can lead to congestive heart failure, and may also cause disturbances in normal heartbeat, called arrhythmia; clot in artery due to rupture of plaque can lead to myocardial infarction	Chest pain, shortness of breath, sweating, nausea, and indigestion	Medical intervention with multistep approach, lifestyle modification, regular exercise, cholesterol reduction, quitting smoking and alcohol consumption, and stress-free life

34-Mitochondrial Disorders

Disorders

For cells to function normally, they must generate energy in the form of ATP. In many types of cells, mitochondrial activity is the prime source of ATP. Mitochondria also regulate several other cellular processes, including production of heat in response to changes in temperature and diet, ion homeostasis (maintenance of an internal steady state of ions), innate immune response, production of reactive oxygen species, and programmed cell death (apoptosis).

It follows that if mitochondrial malfunction occurs and leads to deficient energy production and the impairment of other cellular functions, this may result in a mitochondrial disease(s). The knowledge that mitochondrial impairment may be involved in diseases is relatively new; it was first recognized in an adult in the late 1960s and then in children in the late 1980s.

Mitochondrial dysfunction can occur in the cells of many organs and systems in the body; more than 200 inherited diseases of metabolism can affect mitochondria, and more than 40 types of mitochondrial disorders have been reported. The brain, heart, muscles, and lungs are the organs that require the most energy, and when deprived of energy the functions of these organs begin to fail.

The most severe effects of mitochondrial disease occur in the brain and muscles because they are heaviest users of energy. Other commonly affected organs include the liver, nervous system, eyes, ears, and kidneys. Mitochondrial dysfunction is observed in single-gene mitochondrial disorders and may also be associated with the pathogenesis of polygenic diseases such as

Alzheimer disease, Parkinson disease, cancer, cardiac disease, diabetes, epilepsy, Huntington disease, and obesity. In addition, a progressive decline in the expression of mitochondrial genes is a main feature of human aging, raising the possibility that an increased understanding of the aging process may provide more insight into one or more of the above-mentioned polygenic diseases.

Genetics

Mitochondrial dysfunction generally results from a genetic mutation(s). Interestingly, mitochondria have their own DNA for 37 genes, 13 mitochondrial proteins, 2 rRNAs, and 22 tRNAs. Mitochondrial DNA is less complex than nuclear DNA and is not inherited according to Mendelian genetics (inheritance of DNA from both parents). Rather, because egg cells but not sperm contribute mitochondria to a developing embryo, only mothers can transmit mitochondrial traits to their children. Thus, mitochondrial DNA is inherited only maternally.

Mutations in mitochondrial DNA differ from those in nuclear DNA, and many more mutations occur in mitochondrial DNA than nuclear DNA. The mutation rate in mitochondrial DNA is 10 times higher than in nuclear DNA because mitochondrial DNA is subject to damage from reactive oxygen molecules released as a by-product during oxidative phosphorylation in cells. Mitochondrial dysfunction may also be caused by environmental factors (e.g., viral infection or drug treatment) that interfere with mitochondrial activity.

35-Mitochondrial homeostasis

Because neurons require considerable energy to maintain their metabolically active state, they rely heavily on the maintenance of mitochondria in a functional state, a property known as **mitochondrial homeostasis**.

Mitochondria constitute a population of organelles that require a careful balance and integration of many cellular processes, including the regulation of biogenesis, migration throughout the cell, shape remodeling, and autophagy (catabolic degradation of unnecessary cellular components by lysosomes).

These dynamic processes prompt mitochondrial recruitment to critical subcellular compartments, content exchange between mitochondria, mitochondrial shape control, and mitochondrial communication with the cytosol. Mitochondrial homeostasis is the term used to refer to a well maintained balance of these processes. The structure and function of the mitochondrial network are dependent on mitochondrial homeostasis, which is essential for maintaining the signaling, plasticity, and transmitter release of neurons. There are different pathways to maintain mitochondrial homeostasis.

36-Parkinson Disease

After Alzheimer disease, Parkinson disease is the second most common neurodegenerative disease. Parkinson disease affects about 1% of the population that is 60 years of age or older and is characterized by the progressive reduced capacity to initiate voluntary movements arising mainly from the loss of neurons that synthesize the neurotransmitter dopamine.

Among the many factors that mediate the development of Parkinson disease, mitochondrial dysfunction is considered to be a major factor in its etiology and pathogenesis. The dysfunction of mitochondria in Parkinson disease patients may result from one or more of the following conditions:

- Deletion of mitochondrial DNA,
- Accumulation of mitochondrial DNA
- Mutations, increase in oxidative stress from reactive oxygen species
- Deficient expression and function of the mitochondrial respiratory chain
- Abnormal morphology of mitochondria.

Neurodegenerative diseases such as Parkinson disease are often closely associated with an imbalance of mitochondrial homeostasis (Fig.). The levels of many mitochondrial proteins are altered in postmortem samples of brains from persons with Parkinson disease.

Genetic mutations in the *PTEN-induced putative kinase 1 (PINK1)*, *parkin (PARK2)*, *DJ-1*, *alpha-synuclein (SNCA)*, and *leucine-rich repeat kinase 2 (LRRK2)* genes are closely linked to

the recessive Parkinson disease genes. Moreover, these mutations are associated with important functions that maintain mitochondrial homeostasis, which include membrane potential, calcium homeostasis, structure of cristae (internal compartments formed by the inner membrane of mitochondria), respiratory activity, mitochondrial DNA integrity, and clearance of dysfunctional mitochondria from cells.

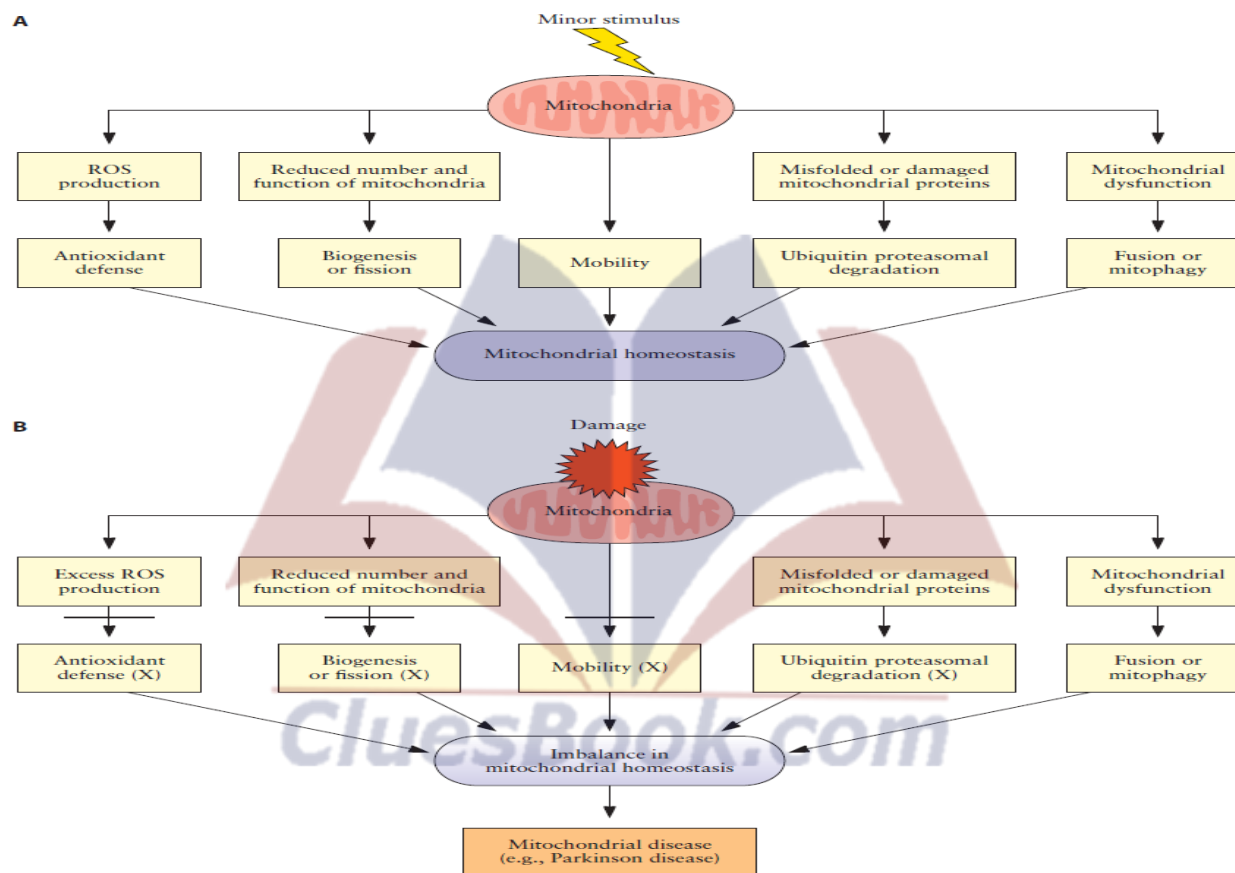


Fig-Parkinson disease is associated with an imbalance in mitochondrial homeostasis. Since mitochondria regulate energy metabolism, they are essential for the control of neuronal function. Mitochondria respond to exogenous and endogenous stimuli and maintain their homeostasis by undergoing continuous fusion, fission, mobility, mitophagy, and biogenesis. **(A)** Normally, upon exposure of mitochondria to a minor stimulus, changes in mitochondrial function and morphology occur to activate different cellular pathways that help to maintain homeostasis. **(B)** After mitochondria incur damage, changes in mitochondria occur that lead to excess reactive oxygen species (ROS) production, reduced number and mass, misfolding

and/or aggregation of damaged proteins, and mitochondrial dysfunction. As a result, an imbalance in mitochondrial homeostasis ensues that may lead to the onset of Parkinson disease. Mitophagy is a catabolic process of self-degradation of mitochondria in a cell. X indicates a mitochondrial dysfunction.

37-Immunological Approaches To Detect Protein Biomarkers of Disease

Proteins play a critical role in all cellular processes, including metabolism, communication, defense, reproduction, transport, and motility. Regulatory proteins maintain tight control over protein production to ensure that cells function normally.

Characteristic changes in proteins have been used extensively to diagnose disease, either by detecting the presence or measuring levels of a specific protein biomarker or by determining protein profiles in polygenic diseases. There are several advantages to using proteins as diagnostic biomarkers of disease. Abnormal levels of gene expression as a consequence of disease-associated mutations are more accurately quantified by measuring the protein directly rather than by measuring mRNA levels, which often do not correlate with protein levels.

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Furthermore, proteins in diseased tissues may exhibit irregularities in posttranslational modification that cannot be detected using nucleic acids. In addition, many diseases often are a consequence of altered protein conformation (e.g., prion diseases and some neurodegenerative diseases) that may not be detected from nucleic acid sequences. On the other hand, the unique shape and often very low cellular levels of a protein are a challenge for development of a diagnostic assay.

Antibodies can be produced that bind to target proteins with high affinity and specificity, and therefore immunological approaches meet the criteria of sensitivity, specificity, and simplicity for diagnostic assays.

Agglutination is a simple, inexpensive, rapid, and highly specific immunological test that is widely performed in diagnostic laboratories. For example, it is often used for human blood typing based on the presence of specific antigens on the surface of red blood cells, which vary among individuals.

Antiserum containing antibodies against either A or B surface antigen is mixed with red blood cells. Clumping (agglutination) indicates the presence of the antigen (Fig. A). Some individuals produce only the A antigen (blood type A), which agglutinates with anti-A antiserum but not with anti-B antiserum (Fig. A).

Blood type B individuals produce only the B antigen, which reacts with anti-B antiserum and not with anti-A antiserum (Fig. B). Others carry both antigens (blood type AB) or neither (blood type O). Blood samples from individuals with blood type O, the most common blood type, do not produce a positive agglutination result with either antiserum. This test, referred to as hemagglutination, is important for determining which blood to use in transfusions, as antibodies naturally produced against nonself red blood cell antigens would destroy the introduced blood. In some agglutination tests, the antigen or antibody employed to detect either a specific antibody or antigen, respectively, in patient samples is used to coat the surface of small latex beads.

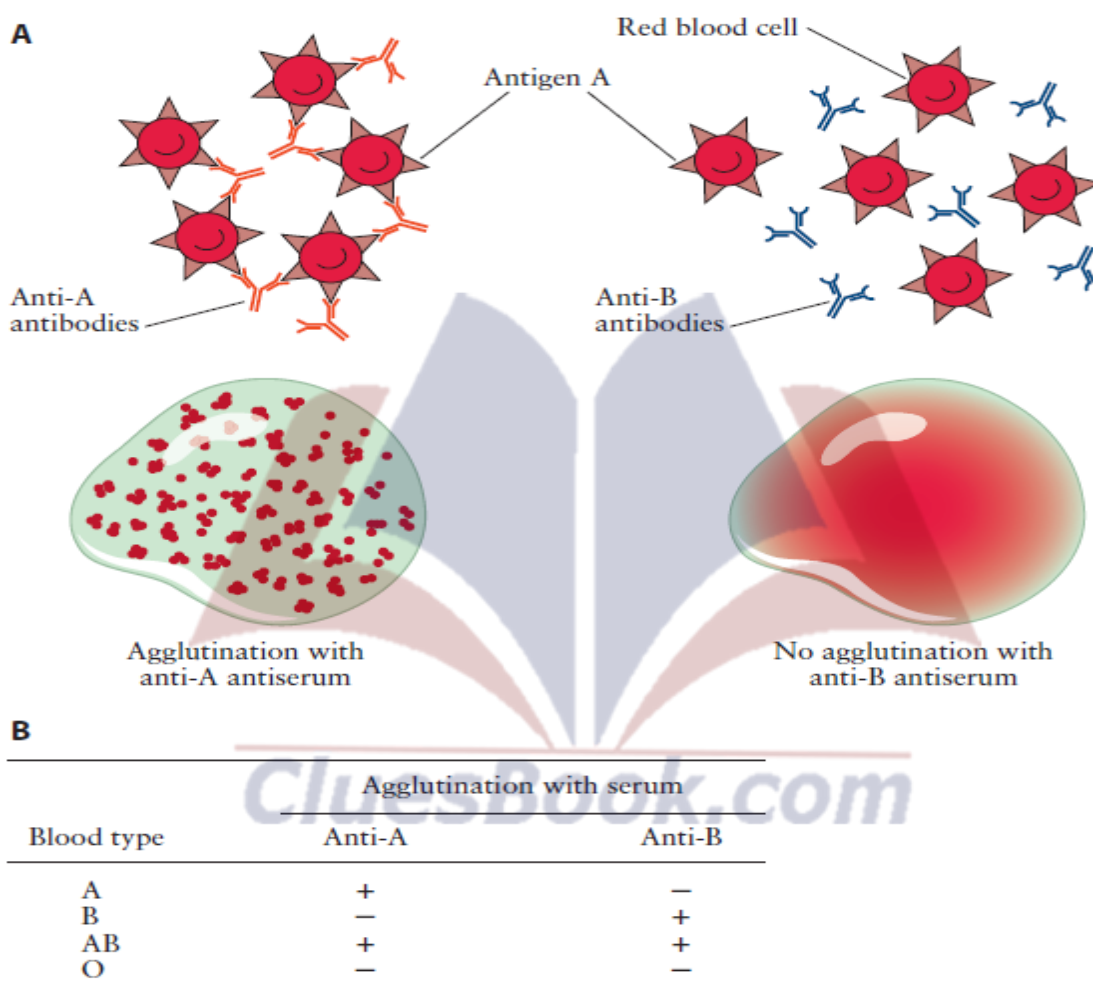


Fig-Hemagglutination test to determine blood type. **(A)** Antiserum containing antibodies against A or B surface antigens is mixed with red blood cells carrying A surface antigens. Clumping (agglutination) in the presence of anti-A antiserum but not with anti-B antiserum indicates blood type A. **(B)** Agglutination responses for different blood types.

38-Enzyme-Linked Immunosorbent Assays

Enzyme-linked immunosorbent assays (ELISAs) are widely used for diagnosing human diseases, including various cancers, autoimmune diseases, allergies, and infectious diseases. An ELISA measures antigens or antibodies produced against an antigen in a clinical sample from blood, urine, or tissues. It is based on the specific and high-affinity interaction between an antibody and an antigen and is a sensitive assay that can be used for rapid detection on a large scale (i.e., high-throughput assays). Detection relies on the activity of an enzyme that is covalently bound to an antibody employed in the assay.

An **indirect ELISA** can detect the presence of specific antibodies in patient serum that indicates an immune response to the presence of a particular protein or pathogen. A **sandwich ELISA** detects the presence of a specific antigen in a patient's sample and hence is sometimes referred to as an antigen capture assay.

In an indirect ELISA used for diagnostic purposes, a standardized antigen is bound to a solid support, usually the surface of a well in a microtiter plate (Fig. A). A patient's serum sample is applied to the well, and specific **primary antibodies** in the serum bind to the immobilized antigens. A **secondary antibody** that binds specifically to the primary antibody is applied. Because the primary antibody is present in human serum, the secondary antibody is an anti-human immunoglobulin antibody that was raised in another animal, for example, a goat, by injecting the animal with human immunoglobulin. The secondary antibody is covalently bound (conjugated) to an enzyme such as alkaline phosphatase or horseradish peroxidase that catalyzes

the conversion of a colorless substrate into a colored product. A colorless substrate is applied, and the formation of a colored product by the enzyme indicates the presence of the secondary antibody and, hence, the primary antibody–antigen complex.

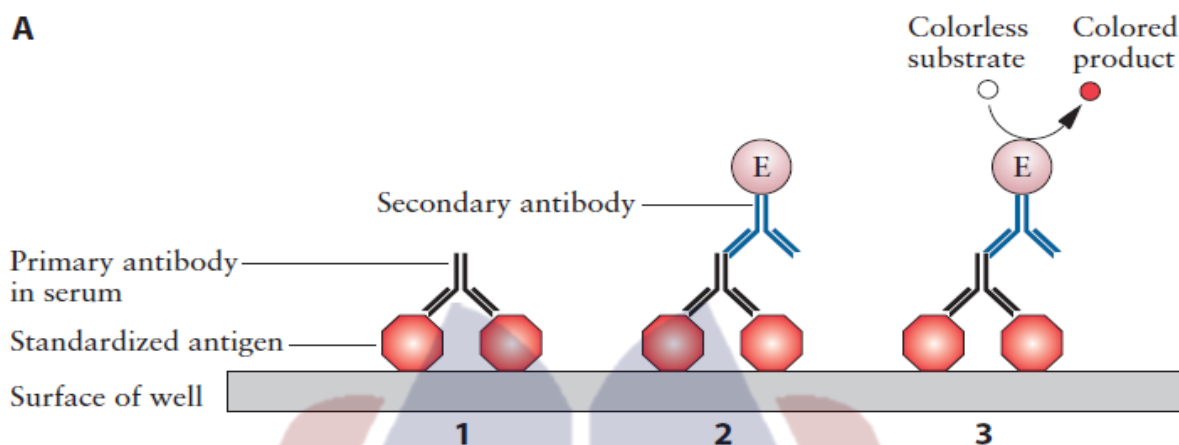


Fig. (A) Indirect ELISA. A specific antigen is immobilized on the surface of the wells in a microtiter plate. (1) Patient serum is applied, and primary antibodies present in the serum bind to the antigen. Unbound antibodies are removed by a washing step. (2) A secondary antibody that binds specifically to the primary antibody is applied. (3) The secondary antibody is conjugated to an enzyme (E) that catalyzes the conversion of a colorless substrate into a colored product. The formation of a colored product indicates that the serum sample contains antibodies directed against the antigen.

In contrast to an indirect ELISA, a sandwich ELISA directly detects a particular antigen in a complex clinical sample. To capture the antigen, a **monoclonal antibody** that is specific for the target antigen is first bound to the surface of a microtiter plate (Fig. B). The patient's sample is then added to a well; if the specific antigen is present in the sample, it binds to the immobilized antibody. A labeled primary antibody is added to detect the presence of bound antigen. The labeled antibody may detect a different **epitope** on the same antigen.

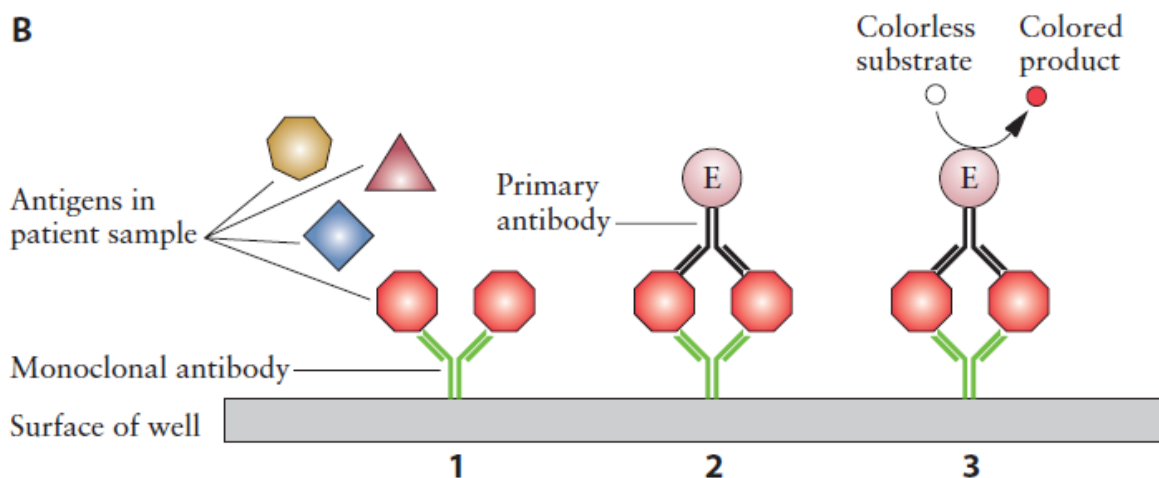


Fig. (B) Sandwich ELISA. Monoclonal antibodies specific for a target antigen are bound to the surface of a well in a microtiter plate. (1) A patient's sample is added to the wells, and specific antigens present in the sample bind to the immobilized antibody. A wash step removes any unbound molecules. (2) A primary antibody is applied to detect the presence of bound antigen. (3) The primary antibody is conjugated to an enzyme that catalyzes the formation of a colored product that indicates the presence of the target antigen.

39-Measuring Disease-Associated Proteins by Sandwich ELISA

Ovarian cancer is a devastating disease that kills over 15,000 women a year in the United States. More than 22,000 new cases are diagnosed each year, most at an advanced stage when the survival rate is less than 20%.

An immunoassay widely used to monitor progression and recurrence of ovarian carcinoma measures levels of the protein CA125 in serum. CA125 is a high-molecular-weight glycoprotein that is present at higher levels in 50% of women with ovarian cancer compared to healthy women.

Levels may also be elevated in women with lung, pancreatic, breast, cervical, and colorectal cancers and with noncancerous disorders such as pelvic inflammatory disease, hepatic disorders, and nonmalignant ovarian cysts. Because CA125 may indicate other disorders and is not elevated in all patients with ovarian cancer, especially at early stages of the disease when tumors are small and therefore secrete only low levels of the protein, it is not recommended as a screening test. Rather, it is commonly used to monitor a patient's response to treatment. Researchers have therefore sought to identify other biomarkers that are more specific for ovarian cancer.

A useful biomarker for ovarian cancer is a protein that is secreted specifically by ovarian tumors, and not by normal tissue or other tumor types, and that can be detected at low levels in the blood or urine, enabling early, and noninvasive, detection. Using cDNA microarrays and quantitative polymerase chain reaction (PCR), human epididymis protein 4 (HE4) was identified as a promising new biomarker that was expressed in ovarian carcinomas but not normal tissue or benign ovarian tumors. Mouse monoclonal antibodies were generated against two different HE4 epitopes and used to develop an ELISA. In an initial blinded study (researchers were not informed of the source of the patient samples), HE4 was found to be elevated in sera from patients with early and late-stage ovarian cancer compared to sera from healthy women and from women with benign ovarian tumors.

The U.S. Food and Drug Administration (FDA) has approved the use of a HE4 sandwich ELISA to monitor the recurrence or progression of ovarian cancer in women who are being treated for the disease.

Patient samples are incubated with biotinylated monoclonal antibodies generated against one HE4 epitope in streptavidin-coated wells (Fig.). A second antibody, conjugated to the enzyme horseradish peroxidase, that binds to a different HE4 epitope is added, and after generation of a colored product by the horseradish peroxidase, HE4 is quantified spectrophotometrically. Using this assay, HE4 was found to be present at elevated levels in more than 75% of patients with ovarian cancer, compared to 5% of healthy women and 13% of individuals with other nonmalignant conditions. An increase in HE4 levels of more than 25% is considered significant to suggest recurrence or disease progression, while a decrease of this magnitude suggests a positive response to treatment.

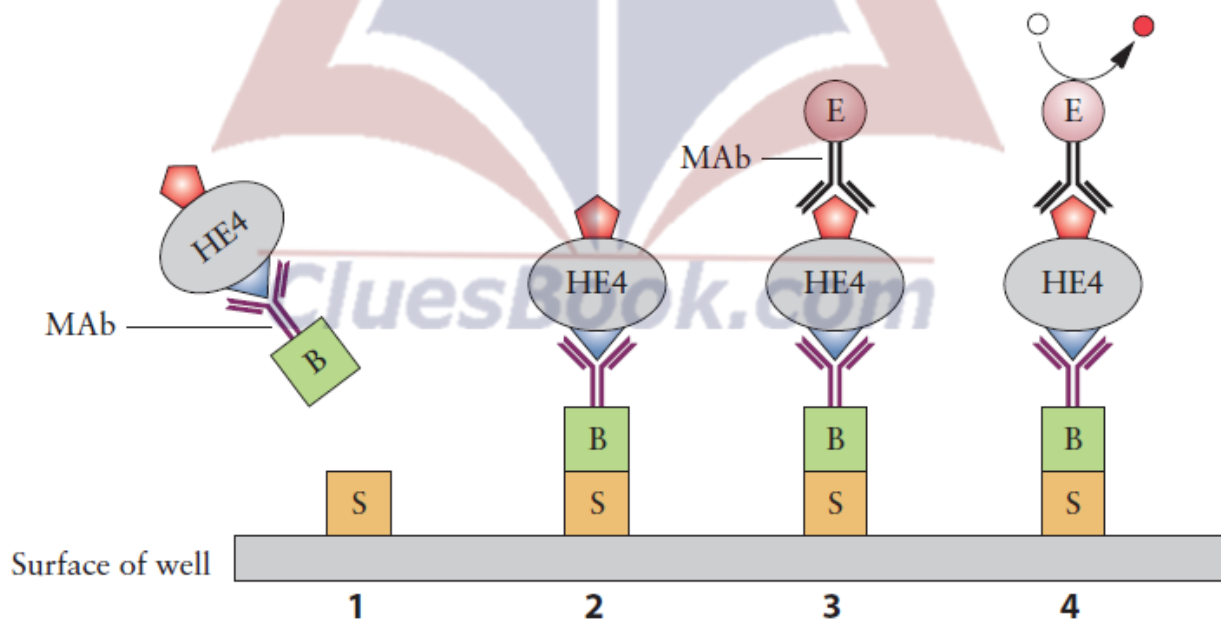


Fig- Sandwich ELISA to monitor progression or recurrence of ovarian cancer. **(1)** Patient samples are incubated with biotinylated (B) monoclonal antibodies (purple MAb) generated against one HE4 epitope (blue). **(2)** Biotin, together with the MAb–HE4 complex, binds with high affinity to streptavidin (S) that coats the wells of a microtiter plate. **(3)** A

second antibody (black MAb) that binds to a different HE4 epitope (red) is added, and after generation of a colored product by horseradish peroxidase (E) conjugated to the antibody **(4)**, HE4 is quantified spectrophotometrically.

40-Diagnosing Autoimmune Diseases by an Indirect ELISA

Autoimmune diseases occur when the body's immune system does not recognize normal cellular molecules and structures as "self" but rather produces antibodies (autoantibodies) that destroy those targets. Several autoimmune diseases have been identified, including celiac disease, type 1 diabetes, lupus erythematosus, and rheumatoid arthritis.

Indirect ELISAs have been developed to diagnose some autoimmune diseases. In these assays, autoantibodies produced against a self-protein are detected in patient blood samples. Rheumatoid arthritis is an autoimmune disease that results in chronic, systemic inflammation of the joints, mainly the synovial (flexible) joints. The synovial membrane that lines these joints secretes a fluid that lubricates the articulating bones. Initially, an inflammatory response occurs in the synovial membranes in the small joints of the hands and feet; it then occurs in larger joints, causing accumulation of excess fluid and damage to the joints. Systemic inflammation may damage organs such as the heart and lungs. The disease affects about 1% of the population of the United States.

Rheumatoid factor is an autoantibody that targets the Fc region of immunoglobulin G (IgG) antibodies and contributes to rheumatoid arthritis. It is commonly used as a diagnostic biomarker to differentiate rheumatoid arthritis from other forms of arthritis and other inflammatory

conditions. An indirect ELISA has been developed to detect the presence of rheumatoid factor in patient blood samples. In this test, IgG molecules, usually from a rabbit, are the standardized antigen bound to the surface of a multiwell plate, and diluted patient sera are applied to the wells (Fig.).

After incubation to allow binding of rheumatoid factor in the serum to the rabbit IgG molecules and washing to remove unbound molecules, an anti-human IgG antibody is added. The anti-human IgG antibody specifically targets rheumatoid factor (and not rabbit IgG) and is conjugated to an enzyme such as horseradish peroxidase for colorimetric detection. Other ELISAs have been developed that target rheumatoid factor IgM and IgA that can assist in accurate diagnosis of rheumatoid arthritis.

Early diagnosis is important to prevent irreversible joint damage, as rheumatoid arthritis may be managed in the early stages by administering antirheumatic drugs.

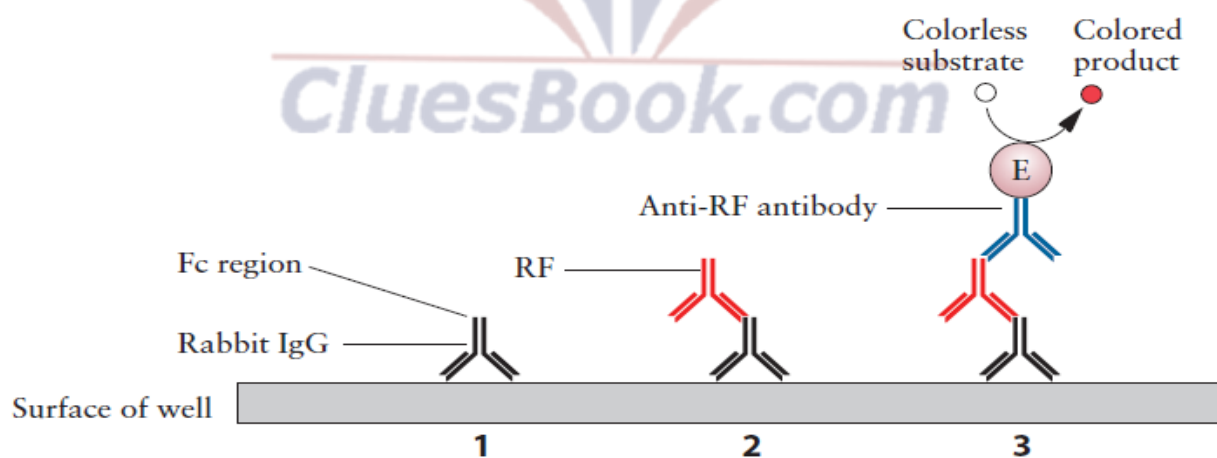


Fig- Indirect ELISA for diagnosis of the autoimmune disease rheumatoid arthritis. **(1)** Rabbit IgG molecules are bound to the surface of a microtiter plate, and diluted patient serum is applied. **(2)** If rheumatoid factor (RF) is present in the serum, it binds to the Fc region of the rabbit IgG molecules. **(3)** After a washing step to remove unbound molecules, an anti-RF antibody conjugated to an enzyme (E) is applied for colorimetric detection.

41-Immunoassay for infectious diseases

Clinical laboratories often identify pathogenic microbes in patient samples based on their physiological or biochemical characteristics. For example, a pathogenic bacterium may ferment specific carbohydrates or produce specific enzymes, and detection of the products of these reactions is the basis for some diagnostic assays. While these methods are effective, they require growth and isolation of the pathogen and therefore are slower than immunological and other molecular methods, typically requiring more than 48 h.

Immunological detection methods such as agglutination assays and ELISAs eliminate the need to grow the pathogen in culture and can be used to detect specific viral, bacterial, fungal, or protozoan pathogens in body fluids and tissues. Because antibody-based approaches detect a target antigen with high specificity and sensitivity, they are well suited to distinguish a specific pathogen from the other hundreds of microbes that are normally present in some human tissues.

Immunological assays for infectious disease may target proteins produced by a pathogen or may detect the presence of antibodies produced against the pathogen. The latter, however, does not differentiate between current and past infections.

To prevent transmission of infectious disease, donated blood is screened for several pathogens, including human immunodeficiency virus and hepatitis viruses. Although five hepatitis viruses can infect the liver and cause liver inflammation, cell death, and even liver failure, most cases of chronic viral hepatitis are caused by hepatitis B and C viruses. Both viruses may be transmitted via blood transfusion.

ELISAs have been developed for routine blood screening for hepatitis B surface antigen, a viral envelope lipoprotein, and for antibodies produced against both hepatitis B and C viruses. Viral surface antigens are the first markers to be detected in serum after infection, while antibodies appear up to several months later. Detection of the viral surface antigen with a sandwich ELISA utilizes specific monoclonal antibodies to capture the hepatitis B surface antigens present in blood in wells of an assay plate (Fig. A). The monoclonal antibodies were generated using recombinant hepatitis B surface protein expressed in yeast. An anti-hepatitis B surface antigen detection antibody with a conjugated enzyme and the enzyme substrate are added to measure virus titers.

An indirect ELISA that detects the presence of anti-hepatitis B virus antibodies in the blood sample is often used to confirm a positive result (Fig. B); however, blood from individuals immunized with a hepatitis B vaccine also produces a positive result with this test. Blood may additionally be screened for hepatitis B DNA and C virus RNA.

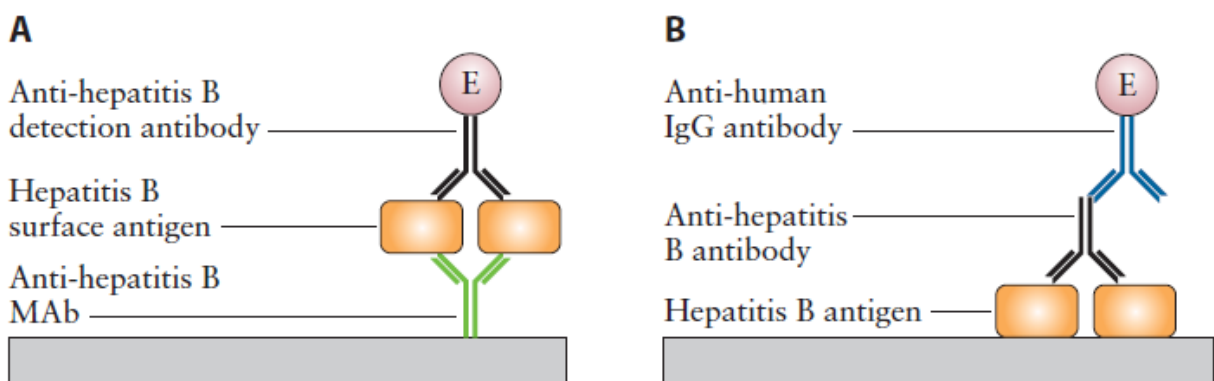


Fig- Diagnosis of hepatitis B virus infection by ELISA. **(A)** A sandwich ELISA detects the presence of hepatitis B surface antigen in blood. **(B)** An indirect ELISA detects the presence of anti-hepatitis B virus antibodies in blood.

42-Protein Array to Detect Polygenic Disease

An ELISA typically measures a single target protein; however, for some diagnoses, it may be more informative to measure multiple target proteins in a single assay. Analysis of **proteomes** is commonly used in research to identify and quantify protein changes in diseased tissue versus normal tissue and is useful for diagnosis of **polygenic diseases** (resulting from mutations in more than one gene) such as breast cancer, Alzheimer disease, type 1 diabetes, and cardiovascular disease.

Protein microarrays are multiplex immunoassays that can detect multiple biomarkers in a clinical sample. Biomarkers may be subsets of proteins in complex clinical samples such as biopsied tumor tissue. Currently, commercially available protein microarrays are most commonly used to detect antibodies in serum from patients suffering from allergies, autoimmune diseases, or infections.

Approximately 25% of the population in industrialized countries suffers from type I allergies, which are IgE-mediated immediate hypersensitivity reactions, such as asthma, hay fever, and eczema. In susceptible individuals, the first exposure to an allergen elicits the production of high levels of IgE antibodies that bind to mast cells. In this way, the individual becomes sensitized to the allergen. On subsequent exposures, the allergen reacts with mast cell-bound IgE and causes bridging between adjacent IgE molecules, thereby stimulating the release of inflammatory

mediators such as histamine, leukotrienes, prostaglandins, and tumor necrosis factor. These biochemicals stimulate smooth muscle contraction and dilation of capillaries that cause edema (swelling), itching, and development of a rash.

Diagnostic allergen microarrays detect IgEs produced against common allergens, for example, pollen, food proteins, and molds. Purified allergens are arrayed in triplicate on a solid support and probed with sera from allergic patients (Fig.). Specific IgEs present in the serum bind to an immobilized allergen and are retained on the surface of the array. Bound IgEs are detected using an anti-human IgE antibody, often labeled with a fluorescent dye. The array format enables detection of hundreds of allergens in a single assay.

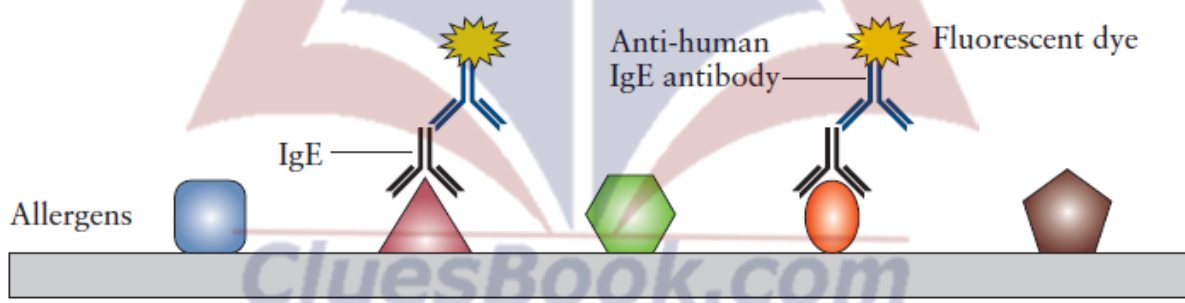


Fig- Diagnostic allergen microarray. Purified allergens are arrayed on a solid support and probed with serum from an allergic patient. Specific IgE present in the serum binds to an immobilized allergen and is detected using an anti human IgE antibody conjugated to a fluorescent dye

Another type of protein microarray measures levels of fifty different protein biomarkers of heart disease in a patient serum sample. Increased or decreased levels of these proteins have been associated with increased risk of heart disease. A liquid bead array has been developed to

quantify levels of the different protein biomarkers are detected in a single well of a multiwell plate (Fig.), which enables rapid analysis of many patient samples simultaneously.

Primary antibodies directed against each protein biomarker are attached to tiny capture beads that are distinguished by either size, color, or shape. A serum sample is applied to a well, and specific proteins in the sample bind to their cognate antibody (Fig. A). For detection of the bound proteins, secondary antibodies conjugated to a fluorescent molecule are added to the well (Fig. B). Each bead, and any captured proteins, is analyzed by passing by a laser that measures fluorescence emitted from bound secondary antibodies and distinguishes the different capture beads by their size, color, or shape.

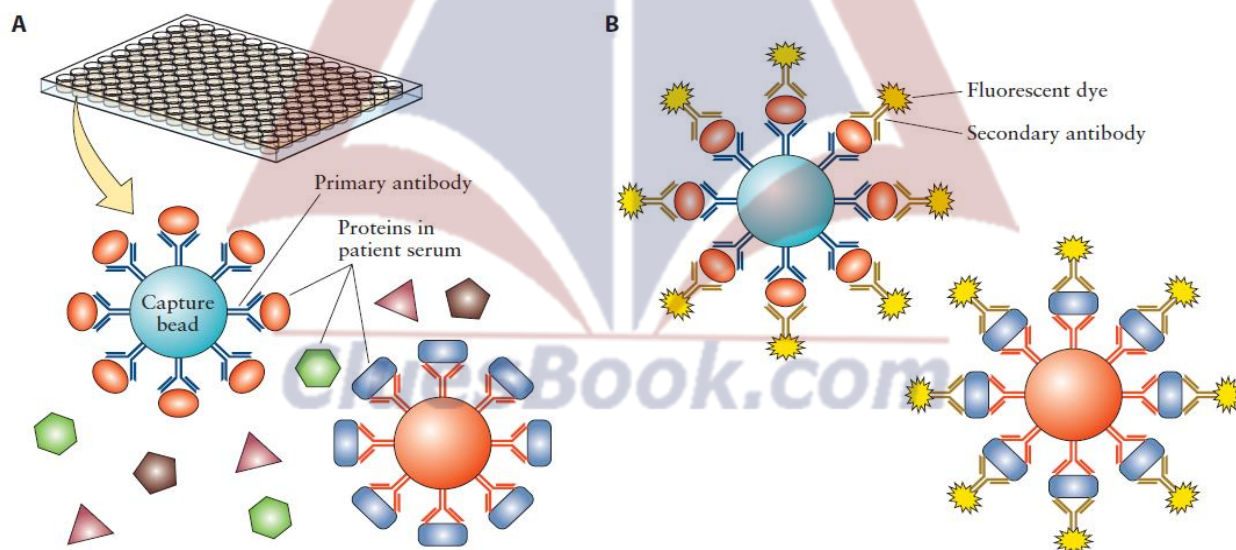


Fig- Liquid bead assay to detect protein biomarkers associated with increased risk of cardiovascular disease. **(A)**

Patient serum is applied to a well of a microtiter plate containing capture beads bound to primary antibodies directed against all of the protein biomarkers. Capture beads of different colors distinguish primary antibodies that bind to different protein biomarkers. Target proteins in the sample bind to their cognate antibodies. **(B)** For detection of the bound proteins, secondary antibodies conjugated to a fluorescent molecule are added to the well. Each bead, and any captured proteins, is analyzed by passing by a laser that measures fluorescence. The presence of specific protein biomarkers is determined by the color of the capture bead to which they are bound.

43-Immunoassay for Protein Conformation-specific Disorder

Several human neurological disorders arise as a consequence of protein misfolding that leads to protein aggregation and cell death. Parkinson disease, Alzheimer disease, and prion diseases are examples of protein conformational disorders.

Alzheimer disease is a degenerative brain disorder that is characterized by the progressive loss of abstract thinking and memory, personality change, language disturbances, and a slowing of physical capabilities. Clinical diagnosis of Alzheimer disease is poor, although 1% of the population between 60 and 65 years old and 30% of the population over 80 years old may develop it.

Two hallmarks of Alzheimer disease found in the brain are (i) neurofibrillary tangles of the cytoskeletal protein tau that accumulate within nerve cell bodies and (ii) dense extracellular aggregates of insoluble proteins called amyloid plaques that develop at the ends of inflamed nerves (Fig.).

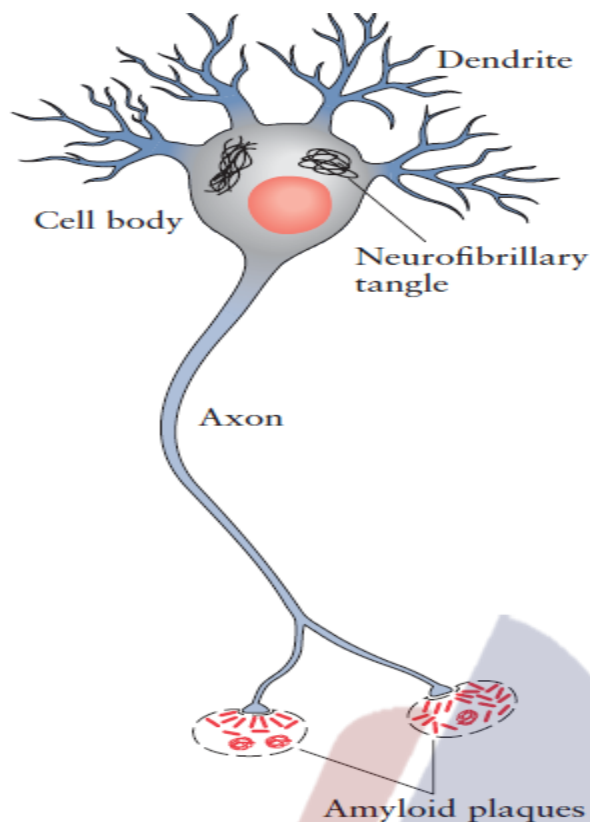


Fig- Neurofibrillary tangles and amyloid plaques associated with neurons in the brain are hallmarks of Alzheimer disease.

The principal protein of an amyloid plaque is a small protein called A β (β -amyloid protein). The A β protein ranges in length from 39 to 42 amino acid residues; the A β 40 and A β 42 forms are the main variants. All A β proteins are derived from the β -amyloid precursor protein (APP) by proteolytic cleavage. Abnormal cleavage of APP results in production of A β 40 and A β 42 and alters protein folding, causing exposed regions of the protein to self-interact; hence, the proteins aggregate in amyloid plaques.

Diagnosis of Alzheimer disease using immunological methods may exploit the development of antibodies that differentiate between the conformations of the disease-associated A β proteins and

normal APP. To develop conformation-specific antibodies, researchers used an approach that mimics protein aggregation during the disease process, that is, the self-interaction of the A β proteins. They grafted short motifs (10-mers) from the A β 42 protein into a complementarity-determining region (CDR3, VH domain) of an antibody to generate antibodies that would interact with aggregated A β 42 via the grafted motif (Fig. A).

Out of twelve antibodies, carrying different but overlapping A β 42 sequences, three bound specifically to insoluble A β 42 aggregates but not A β 42 monomers (Fig. 8.11B). A motif of four amino acids, common to all three positive antibodies, was found to be the minimal sequence required for binding. To determine whether binding of the grafted antibodies is mediated by interactions between the A β motif on the antibody and the same motif in the aggregated A β proteins, each grafted antibody was bound to immobilized A β 42 aggregates and then a second antibody was applied. Binding of the second antibody was reduced only when it recognized a motif that overlapped with that grafted on the first antibody. This relatively simple approach may be used to generate conformation-specific antibodies against other proteins that not only aid in accurate disease diagnosis but also may be used in therapeutic strategies to target these proteins.

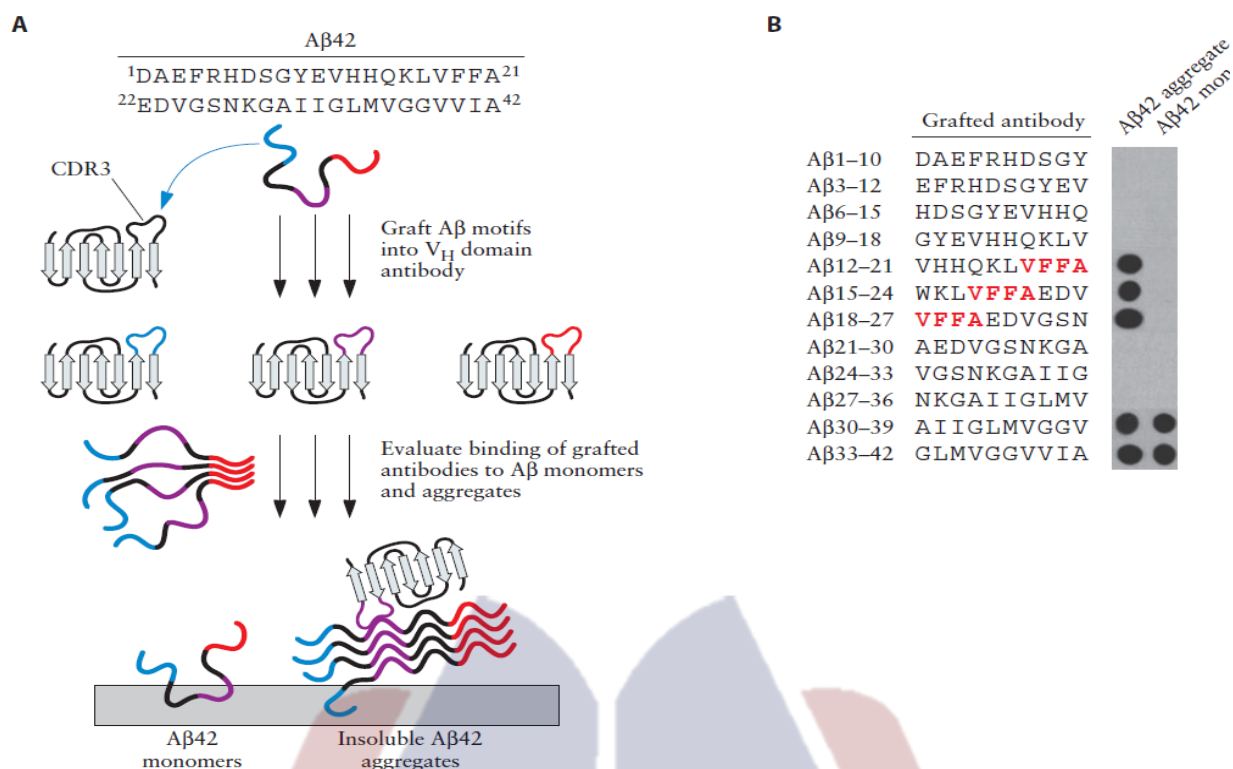


Fig- Development of conformation-specific antibodies to detect aggregated A β proteins indicative of Alzheimer disease. **(A)** Short (10-amino-acid) overlapping A β 42 peptide segments were grafted into a complementarity-determining region (CDR3, V_H domain) to generate antibodies that bind specifically to aggregated, insoluble A β 42 proteins. **(B)** Out of 12 grafted antibodies, three (A β 12-21, A β 15-24, and A β 18-27) bound to aggregated A β 42 proteins, and not A β 42 monomers, that were deposited on a membrane. A sequence of 4 amino acids (highlighted in red) was found to be the minimal requirement for binding.

44-DNA based approaches to diagnosis diseases

DNA-based diagnostic tests determine the existence of specific nucleotide sequences, including human genetic mutations and sequences present in human pathogens. They are highly sensitive and specific and can detect single nucleotide mutations or copy number variations. The ability to

diagnose diseases in humans at the genetic level makes it possible to determine the cause of an illness and to predict whether individuals or their offspring are predisposed to the disease.

Because a DNA-based test does not require expression of a gene, in contrast to diagnostic detection of proteins, DNA analysis can be used for the identification of asymptomatic carriers of hereditary disorders, for prenatal diagnosis of serious genetic conditions, and for early diagnosis before the onset of symptoms.

DNA sequence-specific diagnostic approaches include hybridization of a unique DNA probe to a complementary target sequence, target sequence amplification by PCR, microarray analysis to detect multiple sequences in a single sample, and mass spectrometry to identify single-nucleotide polymorphisms (SNPs).

Hybridization Probes:

Hybridization is the formation of hydrogen bonds between two complementary strands of nucleic acids. A diagnostic test involving DNA hybridization utilizes a DNA probe to detect a complementary target DNA sequence that is characteristic of the disease. The probe is labeled with a reporter molecule that indicates hybridization between the target and probe DNA.

Hybridization probes are often employed to detect the presence of microbial pathogens. Malaria, caused by the parasite *Plasmodium falciparum*, is one of the most common infectious diseases and is especially fatal in young children.

A DNA diagnostic test for active infections that measures the presence of the pathogen was developed by using highly repetitive DNA sequences (present in many copies) from *P. falciparum*. The DNA sequence that was chosen as a specific probe hybridized with *P. falciparum* but not with *Plasmodium vivax*, *Plasmodium cynomolgi*, or human DNA, despite the fact that *P. vivax* causes a less severe form of malaria. This probe can detect as little as 10 picograms (pg) of purified *P. falciparum* DNA or 1 nanogram (ng) of *P. falciparum* DNA in blood.

45-Allele specific Hybridization

In addition to infectious diseases, hybridization probes are widely used to detect specific disease-associated alleles. Monogenic diseases are caused by mutations in a single gene; however, any one of several alterations to the normal nucleotide sequence of a gene may be responsible.

This is exemplified by cystic fibrosis, a common lethal autosomal recessive disorder that affects approximately 1 in every 2,500 live births. Mutations in a single gene, the cystic fibrosis transmembrane conductance regulator (CFTR) gene, result in defects in chloride ion transport.

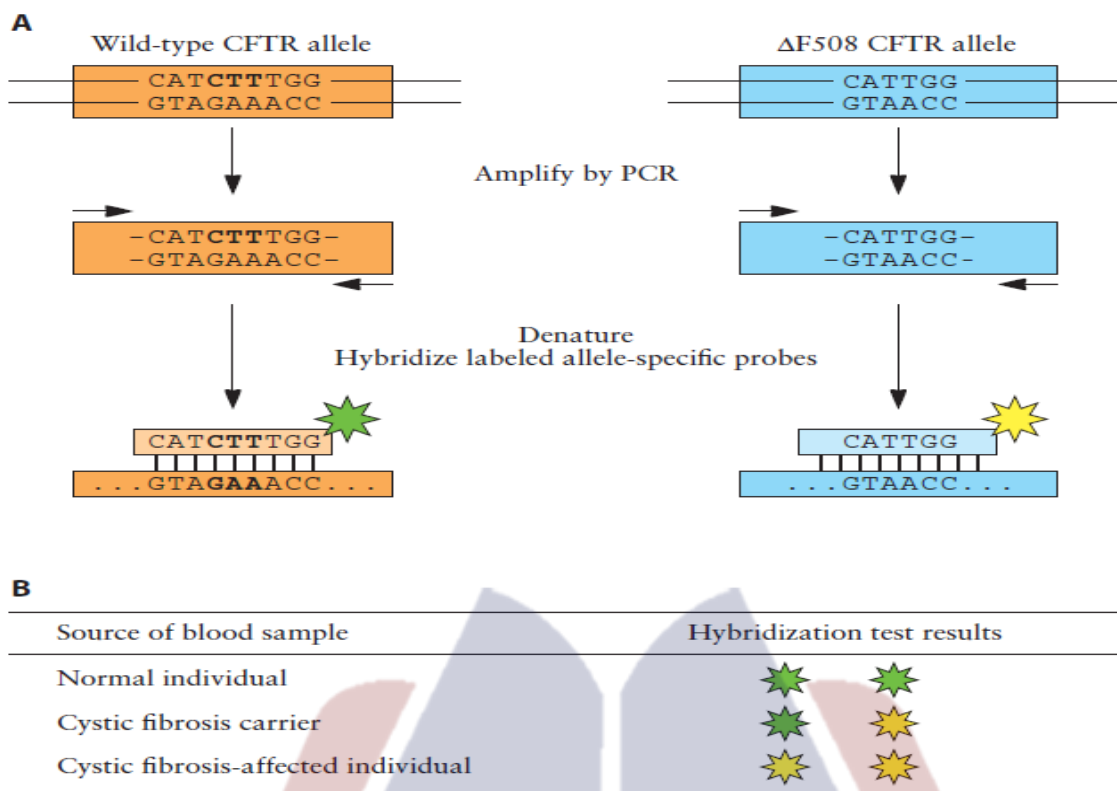
As a consequence, the mucus of lung and other mucosal tissues is thick and viscous, obstructing the respiratory, digestive, and reproduction system functions. Every state in the United States now routinely screens newborns for cystic fibrosis. An ELISA is used to detect higher-than-normal levels of immunoreactive trypsinogen in blood. This protein is produced by the pancreas and is linked to cystic fibrosis. Premature babies and babies from stressful births also may have

elevated levels of immunoreactive trypsinogen, and in these cases, genetic tests are employed for confirmation.

About 1,900 different mutations are reported to occur in the CFTR genes of patients with cystic fibrosis. Screening individuals who may be at risk for cystic fibrosis for such a large number of possible mutations is a daunting task. However, some of the mutations that cause cystic fibrosis are much more common than others. The most common mutation is an in-frame deletion of three nucleotides in exon 10 of the CFTR gene that leads to loss of the amino acid phenylalanine at codon 508 ($\Delta F508$).

Over 90% of cystic fibrosis patients carry at least one $\Delta F508$ allele, and nearly 50% of cystic fibrosis patients are individuals who are homozygous for $\Delta F508$. It is estimated that about 160 different mutations account for 96 to 97% of cystic fibrosis alleles.

Allele-specific hybridization is commonly used to screen for cystic fibrosis. With this technique, an individual's CFTR gene is amplified by PCR and then hybridized to labeled oligonucleotide probes for the mutant (e.g., $\Delta F508$) and wild-type genes, separately (Fig. A). In this way, it is possible to distinguish between healthy individuals with two wild-type alleles, cystic fibrosis carriers with one mutant and one wild-type allele, and cystic fibrosis-affected individuals with two mutant alleles (Fig. B).



(A) The CFTR gene is amplified by PCR, and then the PCR products are denatured and incubated with oligonucleotide probes that specifically hybridize to wild-type or mutant (usually $\Delta F508$) alleles (note that only part of the sequence of the CFTR gene and probe are shown). After a washing step to remove unbound probe, hybridization is detected by measuring fluorescence emitted by the fluorophore attached to the probe.

(B) Healthy individuals are homozygous for the wild-type allele, while those affected by the disease carry two mutant alleles. Cystic fibrosis carriers are heterozygous for the CFTR alleles.

Diagnostic kits are commercially available that test patient blood samples for the presence of a panel of common CFTR mutations as recommended by the American College of Medical Genetics.

46-Oligonucleotide ligation assay

The oligonucleotide ligation assay (OLA) is also commonly used to detect SNPs known to be associated with human diseases with a high degree of accuracy. In this diagnostic assay, two short oligonucleotide probes (~50 nucleotides) are designed to anneal to adjacent sequences within a gene that encompass the polymorphic nucleotide (Fig.).

Importantly, one of the probes (allele-specific probe) has as its last base at the 3' end the nucleotide that is complementary to the polymorphic nucleotide. The second probe (common probe) is complementary to the sequence immediately downstream of the polymorphic nucleotide.

When these two probes are hybridized with the target gene (which has been amplified by PCR), base-pairing occurs between the allele-specific probe and the target sequence, including the 3' nucleotide if the complementary polymorphic nucleotide is present, and the common probe binds immediately downstream. DNA ligase, added to the reaction, covalently joins the two probes.

However, if the nucleotide at the 3' end of the allele-specific probe is mismatched, it will not base-pair with the polymorphic nucleotide in the target DNA sequence, although the common probe will be perfectly aligned. As a consequence of the single-nucleotide misalignment, DNA ligase cannot join the two probes.

In short, OLA is designed to distinguish between two possibilities: ligation if the probes are perfectly matched and no ligation if the allele-specific probe carries a mismatched nucleotide.

Other oligonucleotide probes may be designed to detect different SNPs in the same gene.

To determine whether ligation has occurred, one of the two probes is labeled with a reporter molecule. For example, the allele-specific probe may be labeled at the 5' end with a fluorescent dye that can be detected by laser excitation and fluorescence emission. Probes may be labeled with different fluorescent molecules corresponding to different SNPs. If an individual is homozygous for a particular allele, either two copies of a normal gene or two copies of a disease-associated SNP, a positive signal will be detected only from one of the fluorescent molecules. Heterozygous individuals will yield positive signals from two different probes. Overall, the OLA system is rapid, sensitive, highly specific, and amenable to automation.

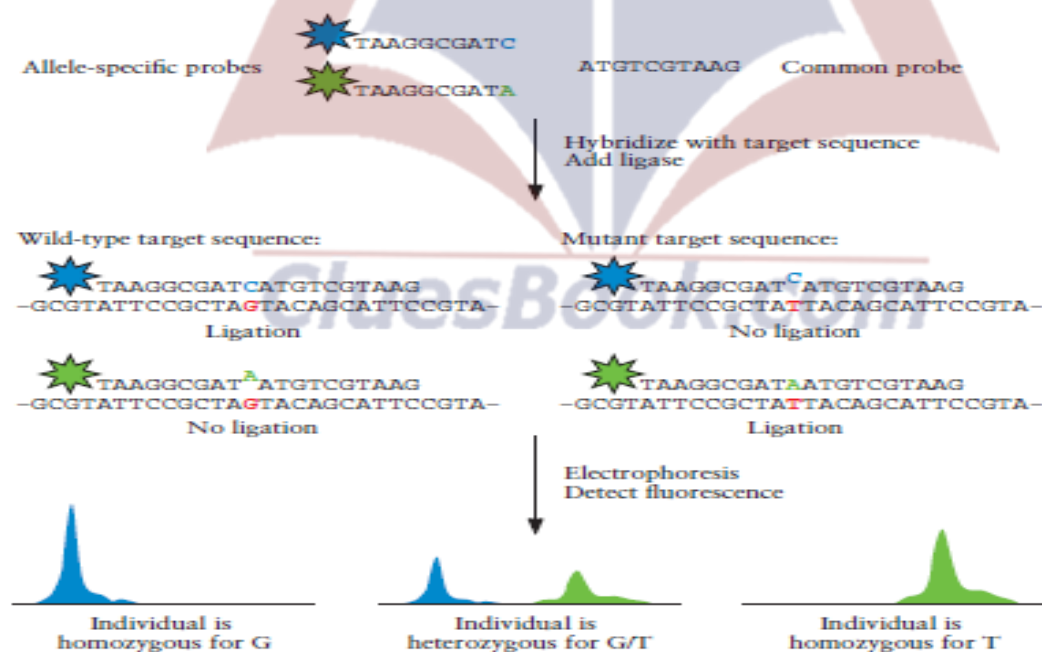


Fig-OLA. Ligation of the allele-specific and common probes occurs only when the allele-specific probe is perfectly complementary to the target sequence. Longer ligated probes may be separated from shorter unligated probes by electrophoresis, and then probe fluorescence can be detected to distinguish homozygous wild-type, homozygous mutant, and heterozygous individuals.

47-Padlock Probes

Diagnostic assays for SNPs that use padlock probes are very similar to those that use OLA probes, except that the former utilizes only one probe rather than two as used in the OLA procedure.

A padlock probe is an oligonucleotide that is complementary to a target sequence at its 5' and 3' ends but not in its middle region (Fig.). When a padlock probe hybridizes to its target sequence, the 5' and 3' ends of the probe come into close proximity to each other and the middle portion loops out. If the ends of the probe are exactly complementary to the target sequence, after hybridization they can be joined together by DNA ligase. If there is a mismatch between the target and probe, ligation does not occur.

The requirement for both ends of the probe to bind perfectly to the target sequence for ligation to occur ensures a high specificity of detection and therefore the ability to easily detect SNPs. Following the ligation reaction, the probe–target hybrid can be detected by the activity of reporter molecules attached to the middle (linker) portion of the probe. Padlock probes typically have sequences approximately 15 to 20 nucleotides in length at each end that are complementary to the target sequence and a middle region of approximately 50 nucleotides.

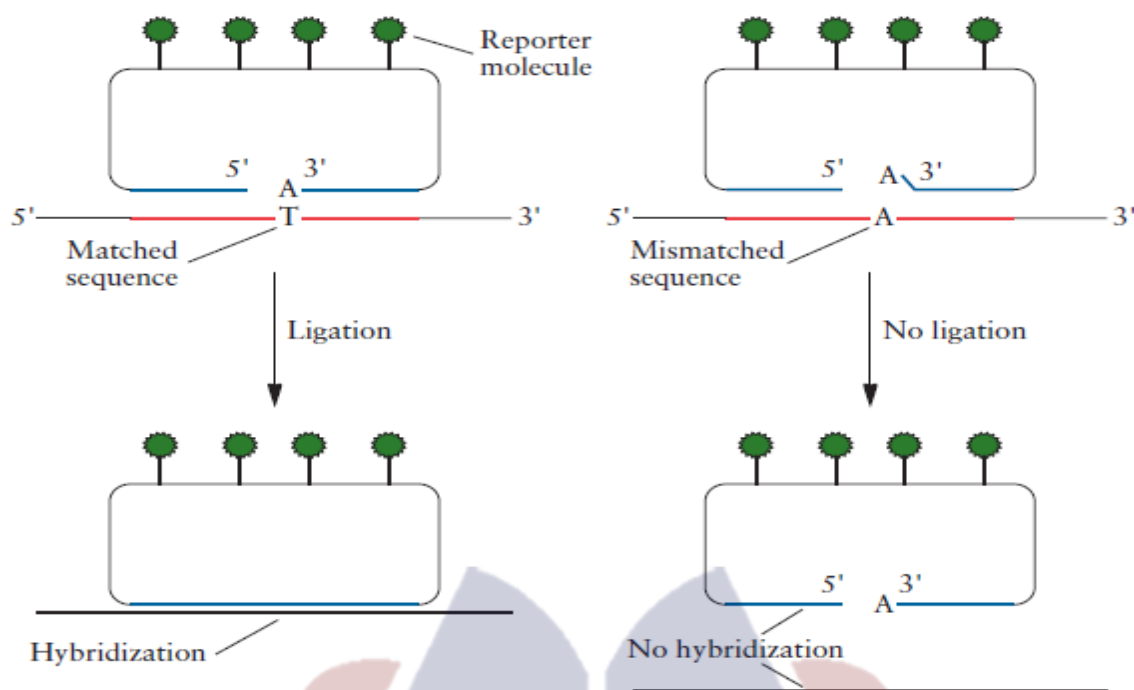


Fig-A padlock probe. Ligation (circularization) of the padlock probe occurs only when the 5' and 3' ends of the probe are perfectly complementary to the target sequence. When there is a single-base mismatch at the 3' end of the probe, ligation does not occur and the probe assumes a conformation that does not allow hybridization. Under stringent conditions, the ligated probe remains bound to the target DNA, while a nonligated probe is removed in a washing step. Hybridized probe can be detected by the activity of the reporter molecule, for example, by fluorescence emitted from a fluorescent dye.

48-Allele-Specific PCR

Many nucleic acid diagnostic tests are based on PCR. Advantages of PCR-based tests include

- (i) specificity that enables detection of a particular nucleotide sequence in complex samples,
- (ii) sensitivity that enables detection of low-abundance targets,
- (iii) an amplification step that generates substantial amounts of a target sequence for additional analyses such as hybridization or sequencing,
- (iv) rapid analysis (usually completed in 1 to 2 h or less),

(v) multiplexing that enables identification of multiple targets in a single sample, and
(vi) low cost. Conventional PCR has been used extensively in diagnostic laboratories over the last 15 years; however, more recently, it has become possible not only to detect but also to quantify the pathogen in a clinical sample using quantitative real-time PCR.

Allele-specific PCR (also referred to as PCR amplification refractory mutation system) screens for known SNPs. Different forward PCR primers are used to distinguish among alleles that differ by a single nucleotide. One primer is exactly complementary to the normal DNA sequence, and another primer anneals to a variant sequence containing the disease-associated SNP.

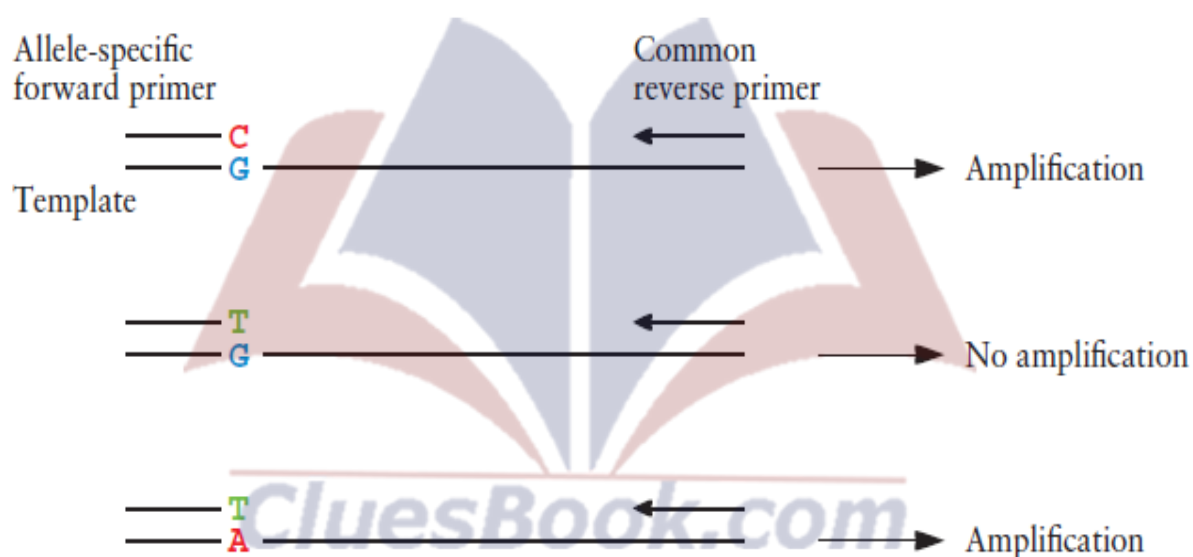


Fig- Allele-specific PCR to detect SNPs. Amplification occurs only when the forward and reverse PCR primers perfectly match the target sequence. Different allele-specific primers that carry an SNP-specific nucleotide at the 3' end are employed to distinguish among alleles that differ by a single nucleotide.

The primers are usually designed to place the polymorphic nucleotide at the 3'-terminal end of the primer because most polymerases used for PCR do not extend 3' mismatched primers efficiently. A third reverse primer is complementary to the opposite strand and is common to all

reactions. Each reaction contains only one of the allele-specific forward primers and the common reverse primer, together with the patient DNA sample, a thermostable DNA polymerase, and all four deoxyribonucleotides.

PCR amplification occurs only when a forward primer is present that is exactly complementary to the target sequence in the patient sample; mismatches between primer and template DNA prevent primer annealing and therefore primer extension during DNA synthesis. An advantage of this method, compared to allele-specific hybridization described above, is that the amplification and diagnostic steps are combined.

One variation of allele-specific PCR is known as competitive oligopriming, in which two different SNP-specific forward primers are included in a single reaction. To discriminate between the PCR products, each forward primer is labeled with a different fluorescent dye. For example, one forward primer that is exactly complementary to a normal allele may be labeled at its 5' end with rhodamine (which fluoresces red), while another forward primer that is complementary to the disease-associated allele may be labeled at its 5' end with fluorescein (which fluoresces green)

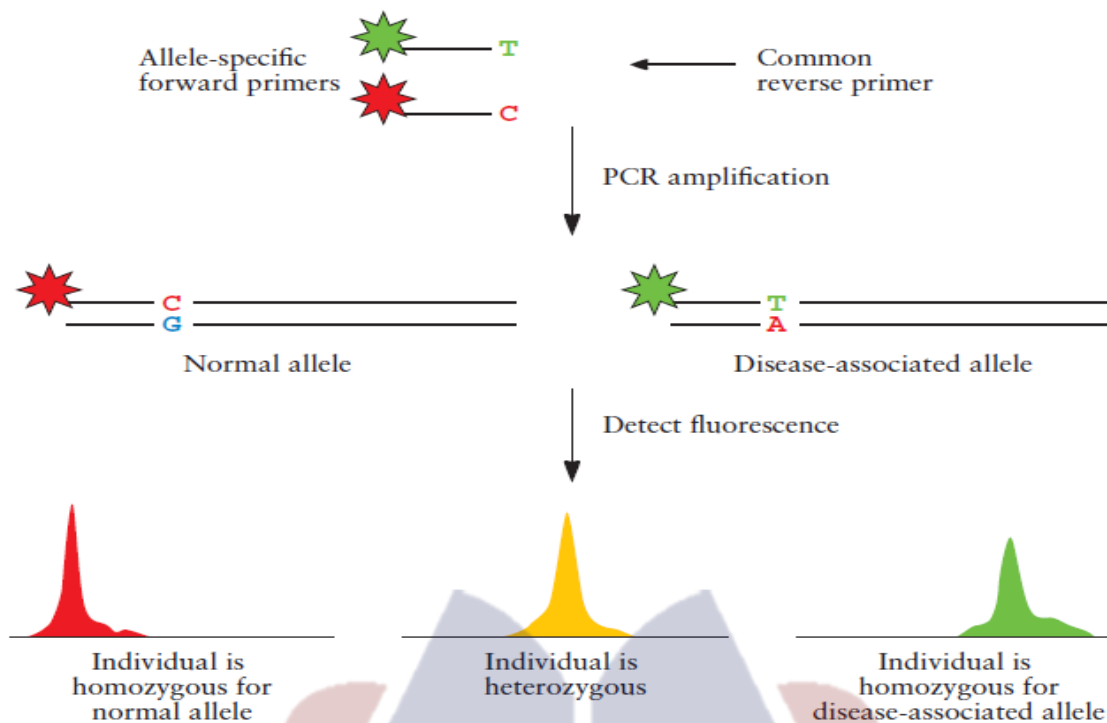


Fig- Competitive oligopriming to detect SNPs. Two allele-specific forward primers that differ in the nucleotide at the 3' end are included in a single reaction together with a reverse primer that anneals to a sequence that is common among variant alleles. PCR amplification occurs only when a forward primer is perfectly complementary to the target DNA. To discriminate between PCR products corresponding to normal and disease-associated alleles, each forward primer is labeled with a different fluorescent dye. In the example shown here, PCR products from individuals who are homozygous for the normal allele fluoresce red, those from individuals who are homozygous for the disease associated allele fluoresce green, and those from individuals who are heterozygous fluoresce yellow.

In both cases, amplification requires a third, unlabeled primer that is complementary to the opposite strand. PCR amplification occurs only when a forward primer is exactly complementary to the target DNA; therefore, the presence of these two forward primers in the same reaction mixture will result in the amplification of either the normal or disease-associated DNA sequence or both, depending on which alleles are present. If an individual is homozygous for the normal

allele, after PCR and removal of unincorporated primers, the reaction mixture will fluoresce red; if he or she is homozygous for the disease-associated allele, the reaction mixture will fluoresce green; and if he or she is heterozygous, the reaction mixture will fluoresce yellow. This assay can be automated and adapted for any single-nucleotide target site of any gene that has been sequenced.

49-TaqMan PCR

The TaqMan PCR protocol is used to screen individuals for the presence of SNPs that are indicative of any of a variety of genetic diseases. Made popular by one particular company, it is based on the 5' nuclease activity of *Taq* polymerase, which is commonly used to amplify DNA in PCR applications.

To simultaneously detect normal and disease-associated alleles, two TaqMan probes are utilized. Each probe is exactly complementary to either the normal or the disease-associated DNA sequence, and each probe has a different fluorescent dye attached to its 5' end (Fig.). Intact probes, whether unbound or bound to template DNA, do not fluoresce because of the presence of a quencher molecule at the 3' end of the probe.

PCR primers anneal to sequences that flank the probe hybridization site, and as PCR amplification proceeds, the TaqMan probe is displaced by the growing DNA strand. The 5' nuclease activity of the *Taq* polymerase degrades the 5' end of the TaqMan probe, thereby releasing the fluorescent dye and removing it from the proximity of the quencher molecule. Thus, only TaqMan probes that were previously bound to target DNA are degraded and subsequently fluoresce. Any mismatched probes that are not complementary to the SNP in the

region where the TaqMan probe binds will be displaced but not cleaved, so they will not fluoresce.

By monitoring the fluorescence at two different wavelengths (one for each TaqMan probe), it is possible to distinguish individuals that are homozygous for each SNP (either normal or disease associated) and heterozygous (carrying one normal and one disease-associated allele). This technique may be used to detect more than two different SNPs at the same time as long as the fluorescent dyes attached to each SNP-specific probe have well-separated, nonoverlapping fluorescence maxima.

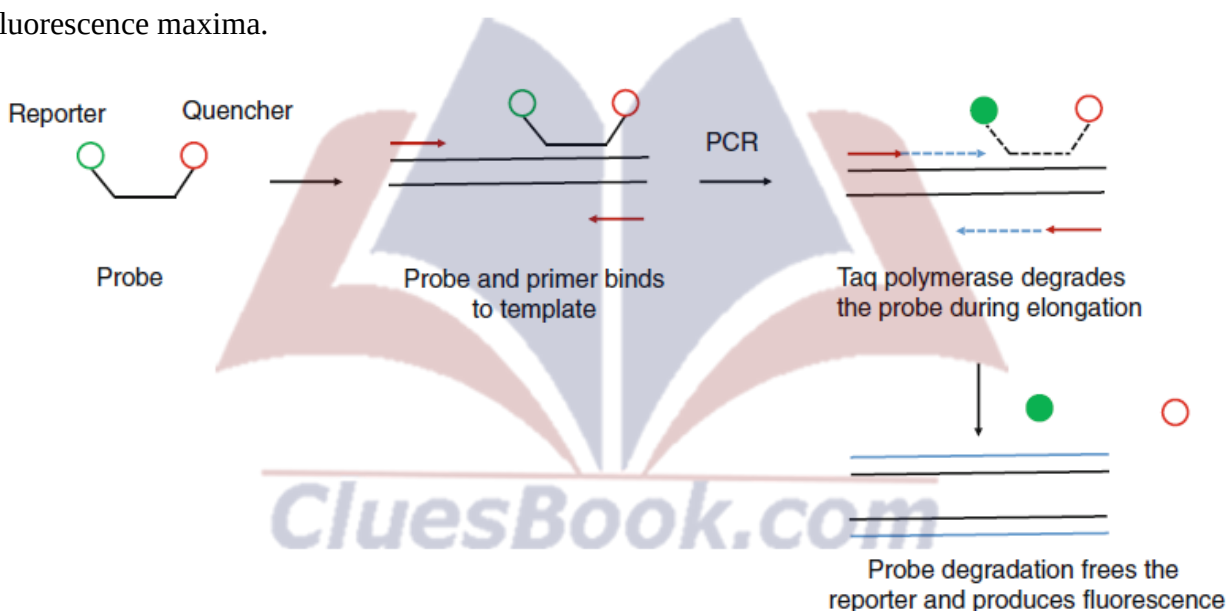


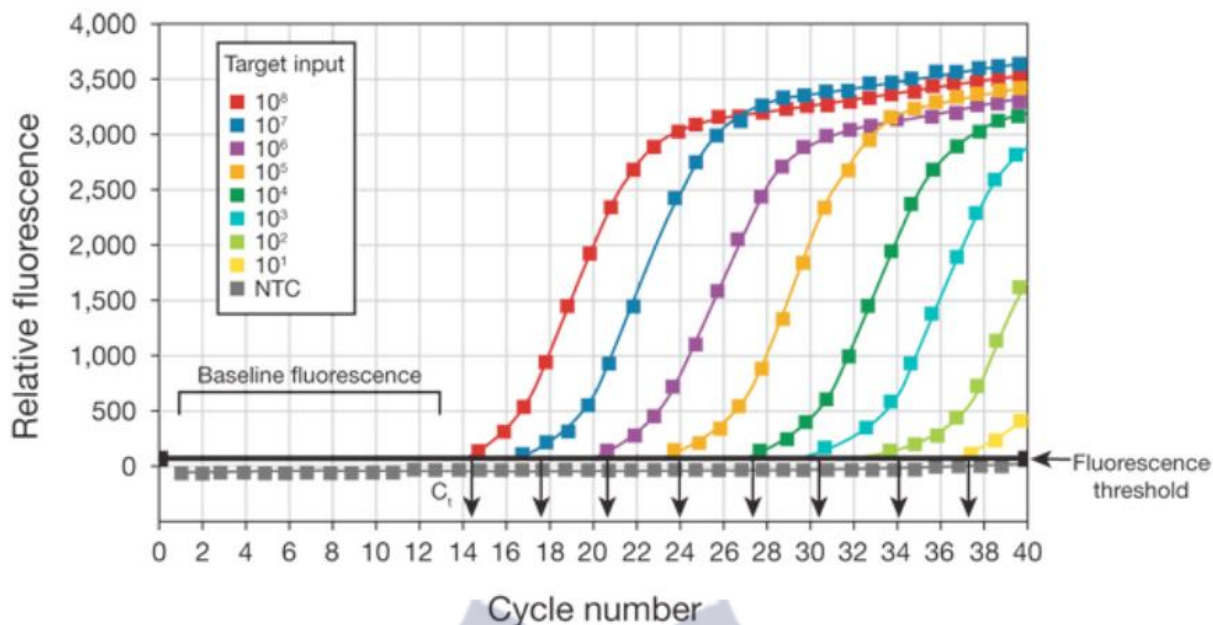
Fig-TaqMan q-PCR principle. A gene-specific probe with a reporter and quencher binds along with the primer. Since the reporter is close to the quencher, it does not produce fluorescence. During PCR reaction, Taq DNA polymerase degrades the probe due to 5' nuclease activity. The released reporter produces fluorescence signal that is captured after each cycle

50-Real-Time PCR To Detect Infectious Disease

Nucleic acid detection and accurate quantitation has become integral in all areas of scientific research. It has a wide range of applications in basic science, biotechnology, medicine, diagnostics, and forensic science. q-PCR is a valuable technique for detection and quantification of DNA and RNA. Unlike endpoint PCR where the amplicon is quantitated and detected at the end of the reaction, in real-time PCR, the amount of DNA is measured after each cycle based on increasing yield of fluorescent signal produced by the accumulation of fluorescent dye.

The signal is directly proportion to the number of molecules generated. Data collected in the exponential phase of the reaction yield quantitative information on the starting quantity of the target. The change in fluorescence is measured by a thermal cycler that has fluorescent dye scanning capability.

By plotting fluorescence against the cycle number, it generates an amplification plot that represents the accumulation of product over the duration of the entire PCR reaction.



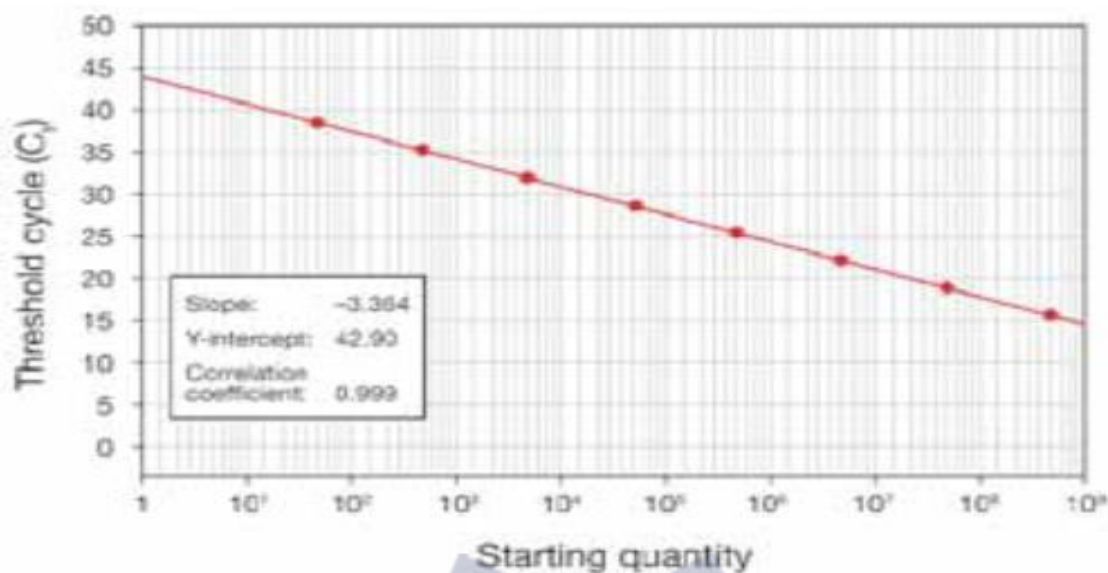
The steps in q-PCR are similar to that of conventional PCR . It includes denaturation of the DNA template, followed by primer annealing and extension. In case of RNA quantitation, a reverse transcription step is done before PCR where the RNA is reverse transcribed to complementary DNA (cDNA). The cDNA is used as a template for q-PCR. After the completion of every cycle, the fluorescence signal is measured and plotted against the cycle number to provide an amplification curve (Fig.).

Amplification curve has two phases: exponential and non-exponential phase. In exponential phase, the PCR product doubles after every cycle. Hence the fluorescence signal increases exponentially. The cycle number at which the fluorescence signal reaches detectable limit is called as threshold cycle or C_T . Since C_T is calculated at exponential phase when the reagents are not limiting, it can be used to calculate the initial amount of template accurately and reliably.

C T value is inversely related to the starting quantity of the template. The more the amount of the template, the earlier it will reach the C T . q-PCR can be used for exact quantitation or relative quantitation of a target. For exact quantitation, it is important to create a standard curve with known amount of the target. This can be done by performing q-PCR reaction with serial dilution of the purified target. Log of known concentration of the target template is plotted against the corresponding C T value to prepare the standard curve. This is used for determining the exact quantity of the target template in the experimental sample and also to calculate the efficiency of the reaction. Most of the time, relative quantitation is done where the expression of the target template is calculated as fold changes with regard to an internal control that could either be a single- copy gene or a housekeeping gene that is expressed steadily in all tissue in every condition.

Standard curve

A dilution series of known template concentrations can be used to establish a standard curve for determining the initial starting amount of the target template in experimental samples or for assessing the reaction efficiency (Fig.) The log of each known concentration in the dilution series (x-axis) is plotted against the Ct value for that concentration (y-axis). From this standard curve, information about the performance of the reaction as well as various reaction parameters (including slope, y-intercept, and correlation coefficient) can be derived.



Correlation coefficient (R²)

The correlation coefficient is a measure of how well the data fit the standard curve. The R² value reflects the linearity of the standard curve. Ideally, $R^2 = 1$, although 0.999 is generally the maximum value.

Y-intercept

The y-intercept corresponds to the theoretical limit of detection of the reaction, or the Ct value expected if the lowest copy number of target molecules denoted on the x-axis gave rise to statistically significant amplification. Though PCR is theoretically capable of detecting a single copy of a target, a copy number of 2–10 is commonly specified as the lowest target level that can be reliably quantified in real-time PCR applications. This limits the usefulness of the y-intercept value as a direct measure of sensitivity. However, the y-intercept value may be useful for comparing different amplification systems and targets.

Slope

The slope of the log-linear phase of the amplification reaction is a measure of reaction efficiency.

To obtain accurate and reproducible results, reactions should have an efficiency as close to 100% as possible, equivalent to a slope of -3.32

There are two types of PCR chemistry that is widely used for q-PCR: SYBR ® Green dye-based assay and 5' nuclease assay or TaqMan ® Assay.

Most nucleic acid diagnostic tests for etiological agents of infectious disease are based on PCR. As for other molecular approaches, PCR circumvents the requirement to grow the pathogens, which is time-consuming for viruses and slow-growing or fastidious bacteria. Not only does this technique enable identification of a viral, bacterial, or fungal pathogen, but also it can reveal specific characteristics of that pathogen, for example, the presence of antibiotic resistance genes that can be used to determine the best course of treatment.

In a real-time PCR, the double-stranded DNA product is bound by a fluorescent dye and the fluorescence is measured after each amplification cycle to quantify the amount of product in real time. Post-PCR processing is not required, and therefore, time to pathogen identification is shorter than for conventional PCR.